

Same Bug Twice:

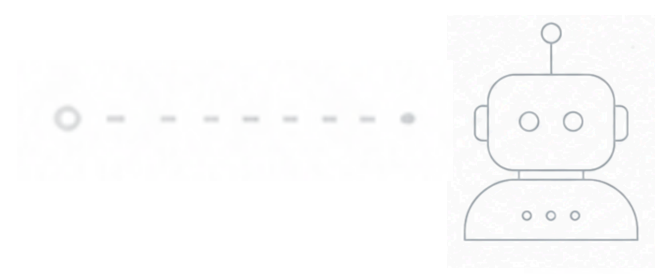
What Happens When AI Writes Your Code And Your Tests

Mourjo Sen

September 2026







\$ mvn clean test





```
$ mvn clean test
```

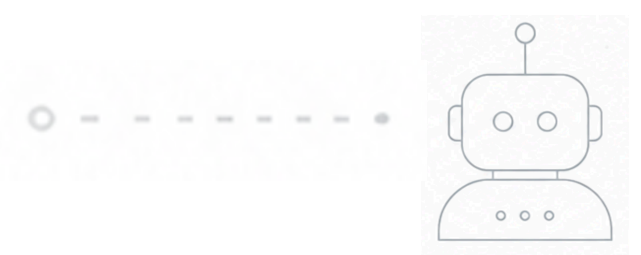
```
...  
tries = 187 | # of calls to property  
checks = 187 | # of not rejected calls  
generation = RANDOMIZED | parameters are randomly generated  
...
```

```
[ERROR] Tests run: 3, Failures: 1, Errors: 0, Skipped: 0
```





\$ mvn clean test



Invariant 'no-overlap' failed after the following actions: [

]

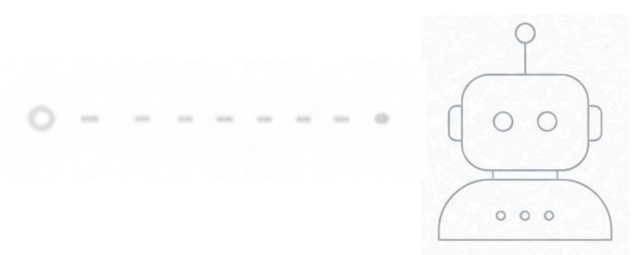
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```
$ mvn clean test
```

```
Invariant 'no-overlap' failed after the following actions: [  
  alice creates Meeting-1 in calendar default (UTC) from 10:00 to 10:01
```

```
]
```

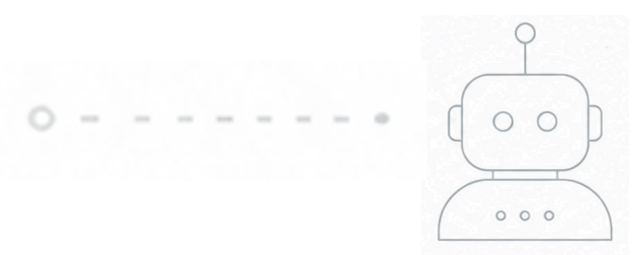
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```
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```
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  alice creates Meeting-1 in calendar default (UTC) from 10:00 to 10:01  
  bob creates Meeting-2 in calendar default (UTC) from 10:00 to 10:01  
  
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  alice creates Meeting-1 in calendar default (UTC) from 10:00 to 10:01  
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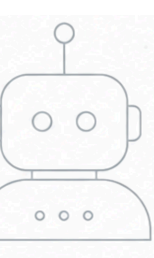


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Invariant 'no-overlap' failed after the following actions: [
 alice creates Meeting-1 in calendar default (UTC) from 10:00 to 10:01
 bob creates Meeting-2 in calendar default (UTC) from 10:00 to 10:01
 bob invites alice to Meeting-2
 alice accepts invitation to Meeting-2
]

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How did I miss that?

Invariant 'no-overlap' failed after the following actions: [
 alice creates **Meeting-1** in calendar default (UTC) from 10:00 to 10:01
 bob creates **Meeting-2** in calendar default (UTC) from 10:00 to 10:01
 bob invites **alice** to Meeting-2
 alice accepts invitation to Meeting-2
]

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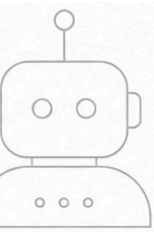
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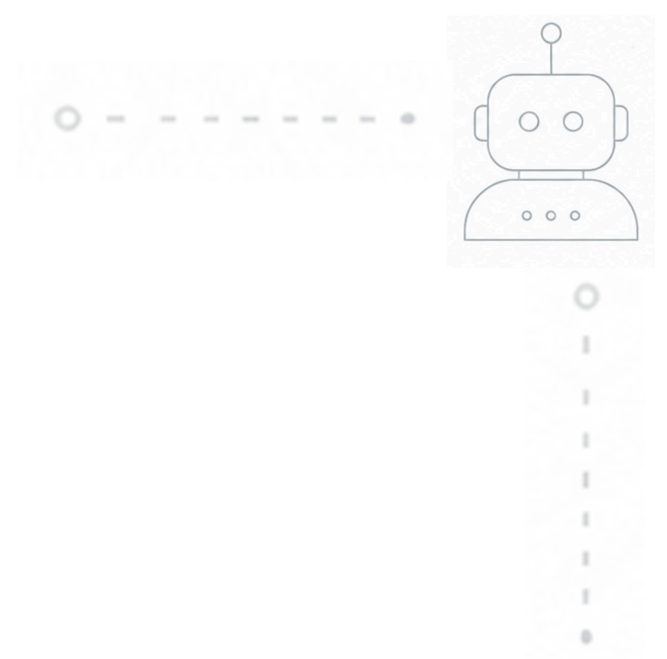




Namaste

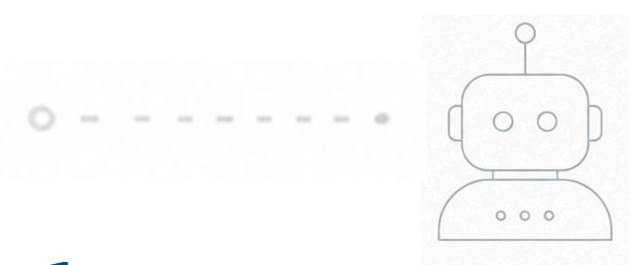
- India → Netherlands
 - 10 years in engineering
 - Staff Engineer @ booking.com
- 
- 
- 





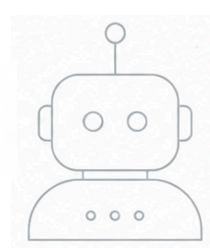
Everything's Changed





LLMs Write Faster Than We Can Read





LLMs Write Faster Than We Can Read

- Code generation is fast but probabilistic
 - Quality of software cannot be probabilistic





LLMs Write Faster Than We Can Read

- Code generation is fast but probabilistic
 - Quality of software cannot be probabilistic
 - Use LLMs for testing software?
 - Because we cannot read at that speed
- 
- 



Are LLMs Good At Testing Software?



Are LLMs Good At Testing Software?

Rethinking Verification for LLM Code Generation: From Generation to Testing

Zihan Ma^{1,2,3,*}, Taolin Zhang^{1,*}, Maosong Cao¹, Junnan Liu¹, Wenwei Zhang¹,
Minnan Luo^{2,3,†}, Songyang Zhang^{1,†,‡}, Kai Chen^{1,†}

¹Shanghai AI Laboratory

²School of Computer Science and Technology, Xi'an Jiaotong University, China

³MOE KLINNS Lab, Xi'an Jiaotong University, China

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{zhangtaolin, zhangsongyang}@pjlab.org.cn

{minnluo}@xjtu.edu.cn

Abstract

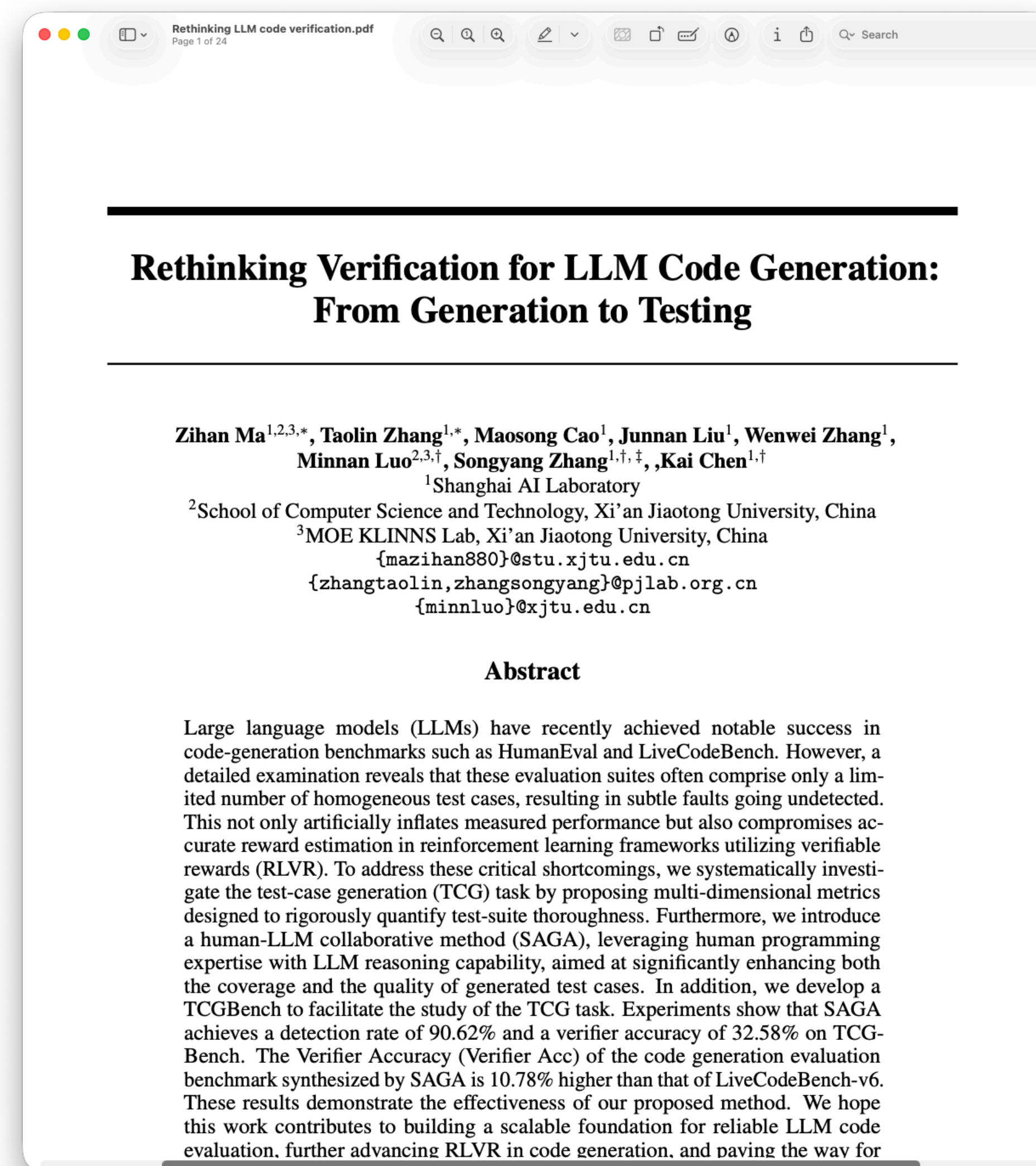
Large language models (LLMs) have recently achieved notable success in code-generation benchmarks such as HumanEval and LiveCodeBench. However, a detailed examination reveals that these evaluation suites often comprise only a limited number of homogeneous test cases, resulting in subtle faults going undetected. This not only artificially inflates measured performance but also compromises accurate reward estimation in reinforcement learning frameworks utilizing verifiable rewards (RLVR). To address these critical shortcomings, we systematically investigate the test-case generation (TCG) task by proposing multi-dimensional metrics designed to rigorously quantify test-suite thoroughness. Furthermore, we introduce a human-LLM collaborative method (SAGA), leveraging human programming expertise with LLM reasoning capability, aimed at significantly enhancing both the coverage and the quality of generated test cases. In addition, we develop a TCGBench to facilitate the study of the TCG task. Experiments show that SAGA achieves a detection rate of 90.62% and a verifier accuracy of 32.58% on TCGBench. The Verifier Accuracy (Verifier Acc) of the code generation evaluation benchmark synthesized by SAGA is 10.78% higher than that of LiveCodeBench-v6. These results demonstrate the effectiveness of our proposed method. We hope this work contributes to building a scalable foundation for reliable LLM code evaluation, further advancing RLVR in code generation, and paving the way for

Source: <https://arxiv.org/abs/2507.06920>, presented at NeurIPS 2025

Are LLMs Good At Testing Software?

- 1840 programming problems
 - From competitive coding platforms

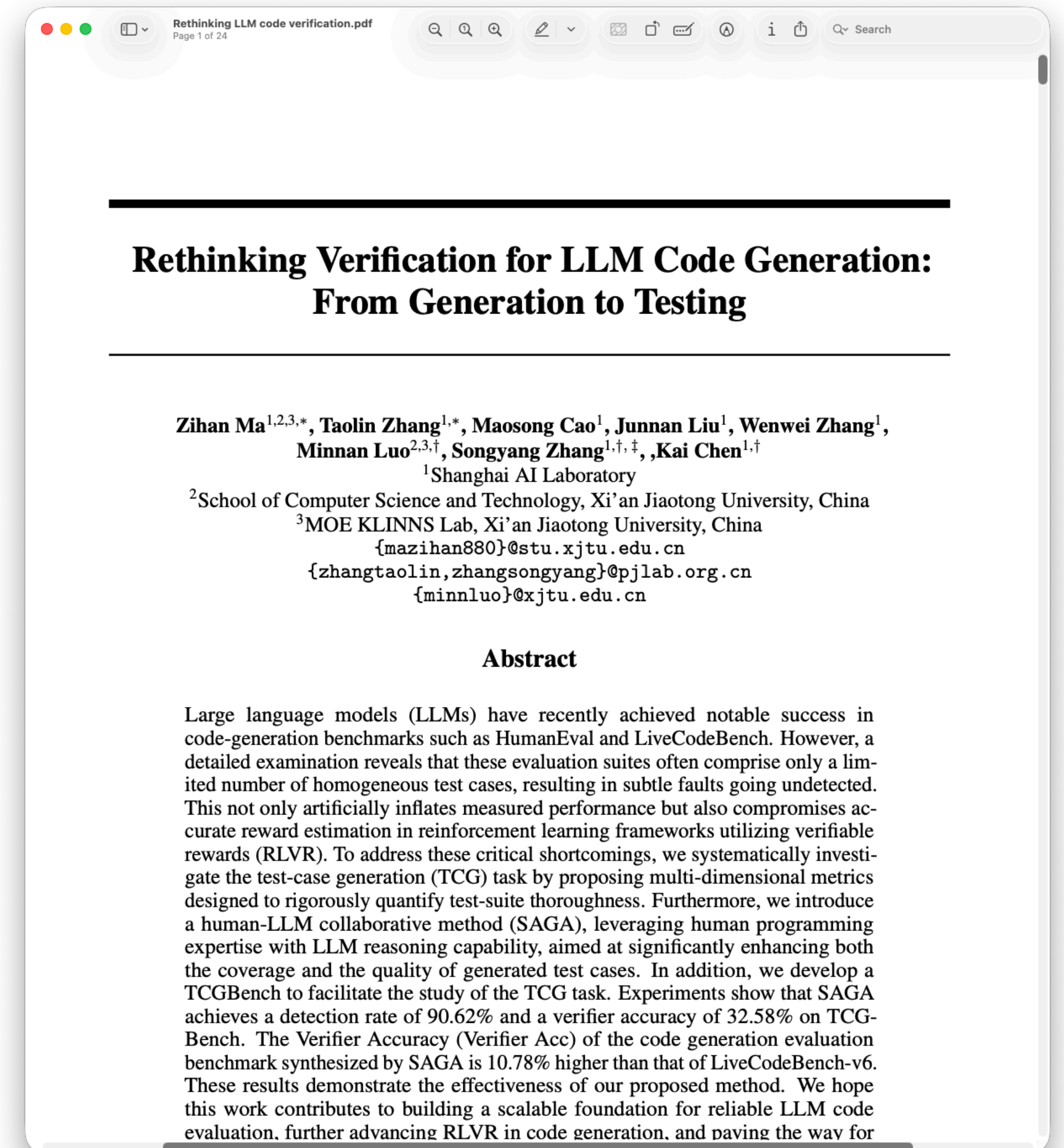
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Are LLMs Good At Testing Software?

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- 36.6 incorrect solutions per problem
 - ~67K solutions analyzed

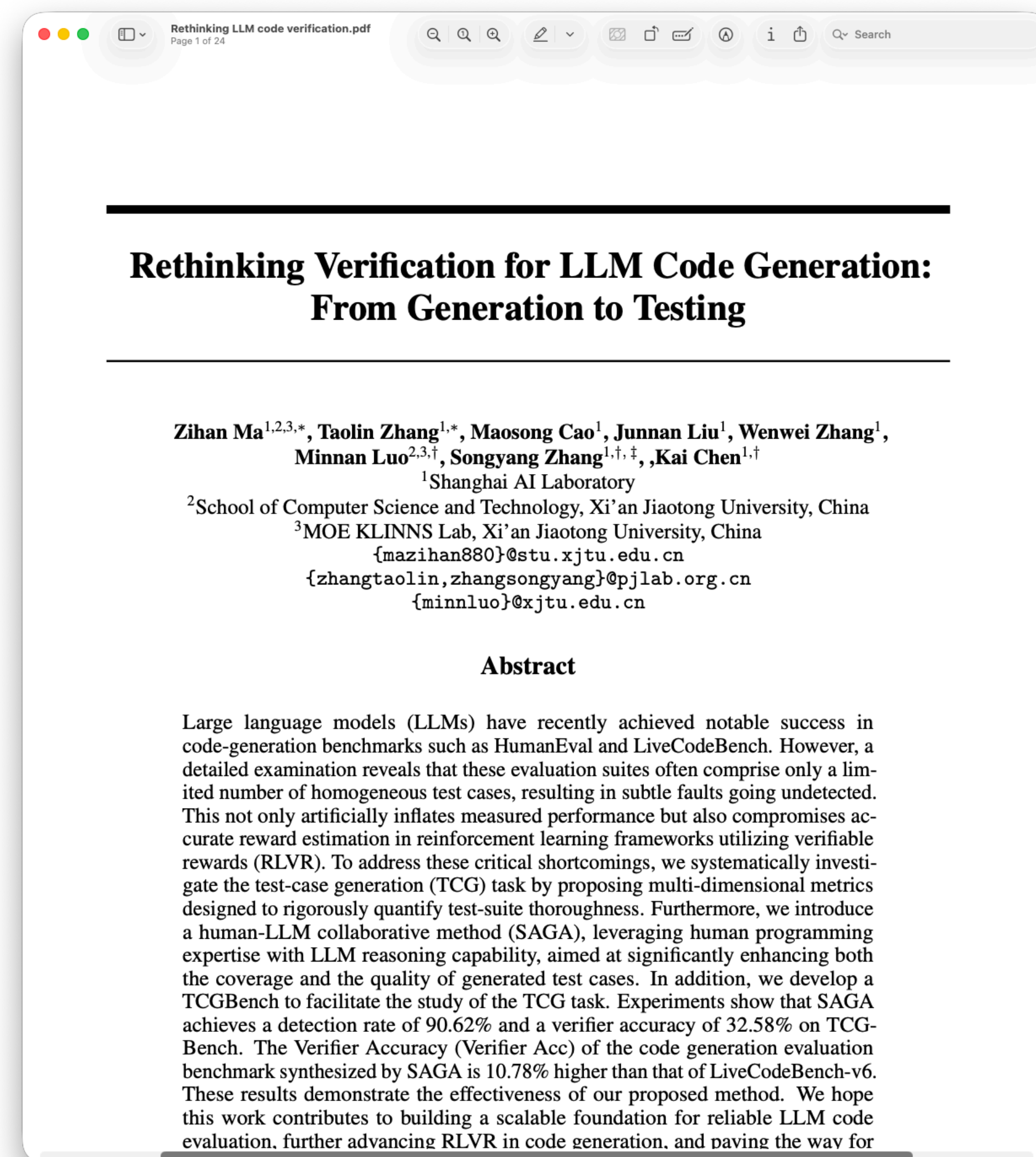
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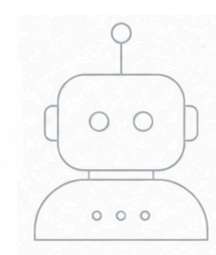


Are LLMs Good At Testing Software?

- 1840 programming problems
 - From competitive coding platforms
- 36.6 incorrect solutions per problem
 - ~67K solutions analyzed
- Evaluate how well LLMs write tests
 - For these known problem-solution combos

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Low Quality of Generated Tests



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Low Quality of Generated Tests

- **Detection rate:** Given a wrong solution, does the test suite fail?

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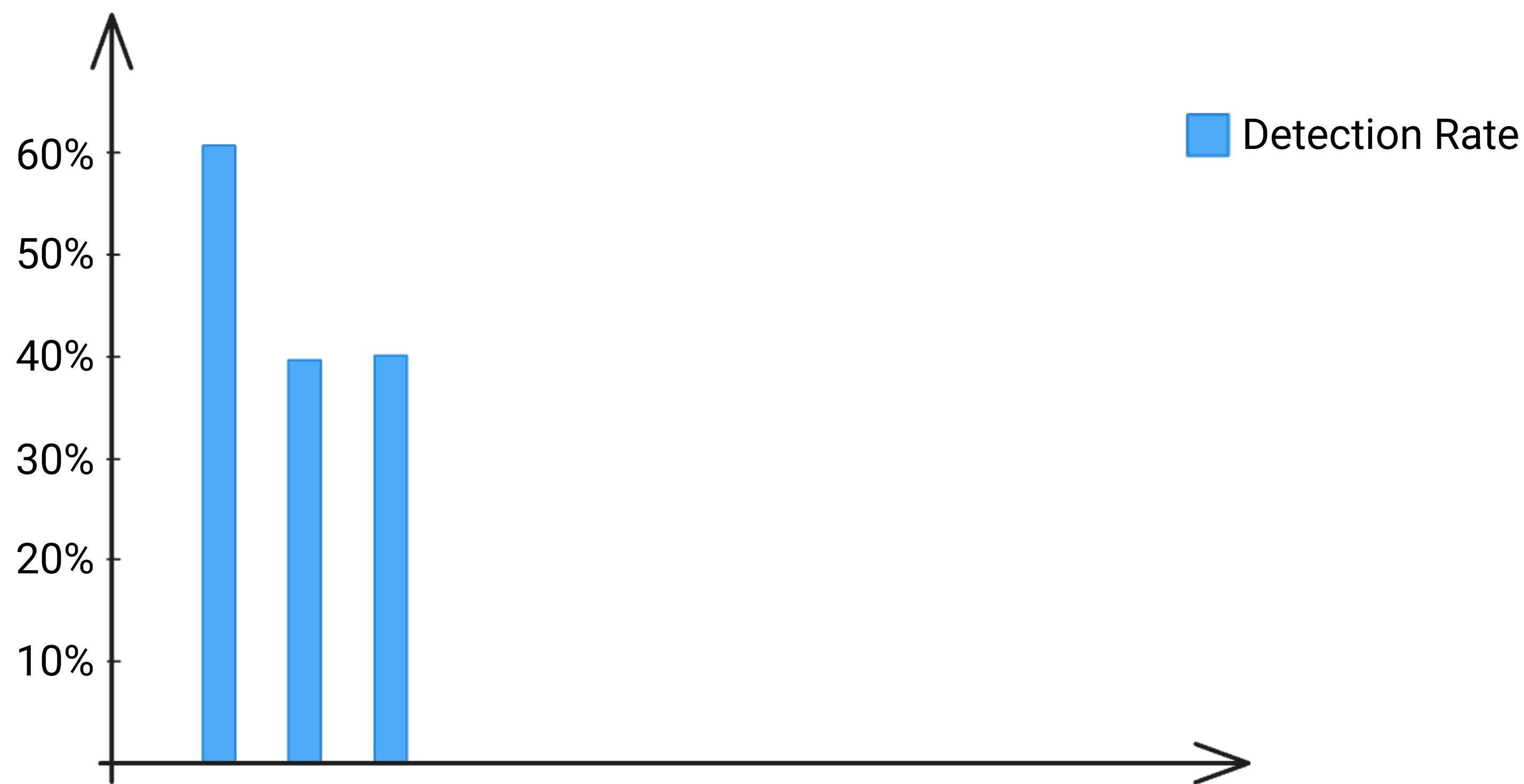
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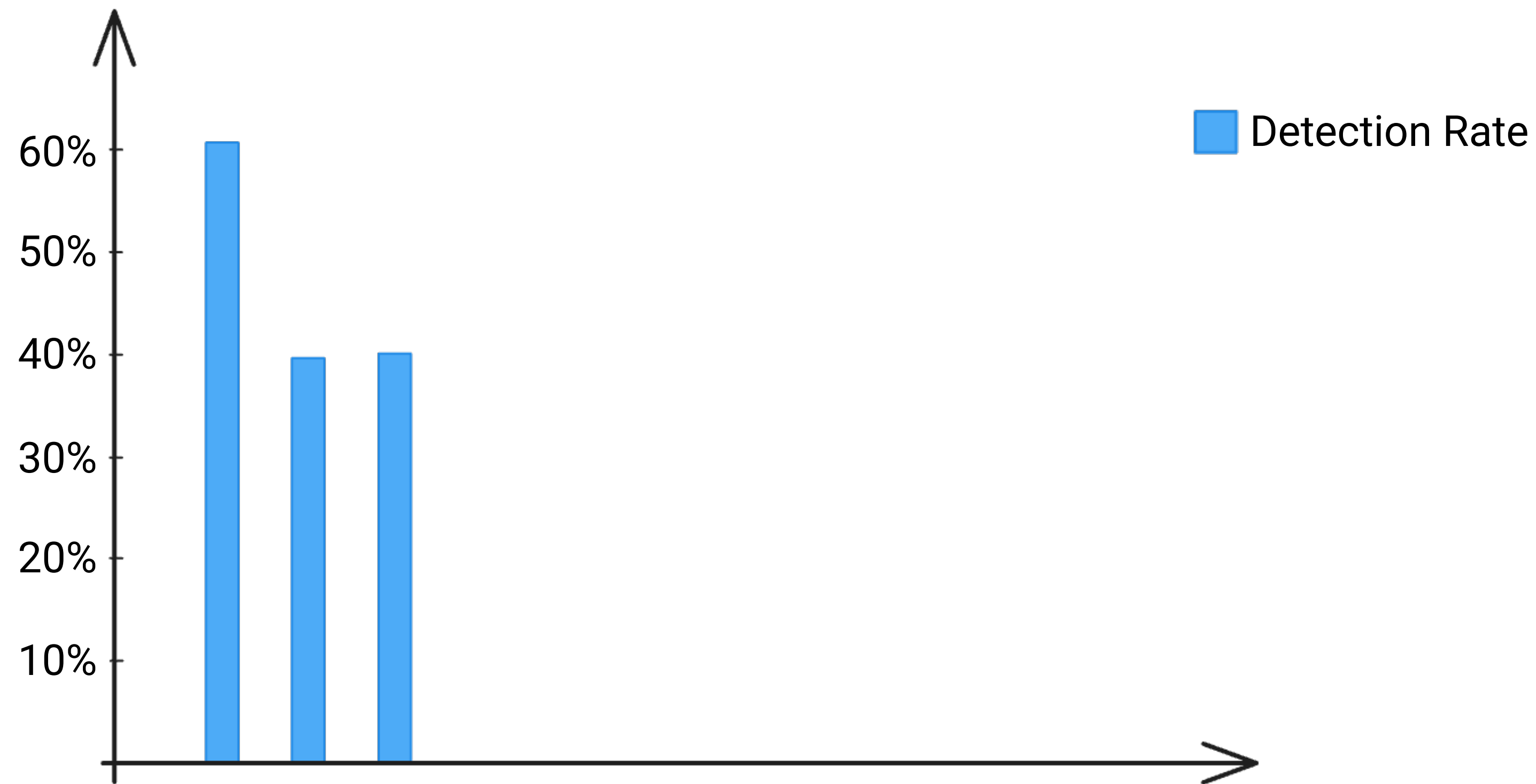
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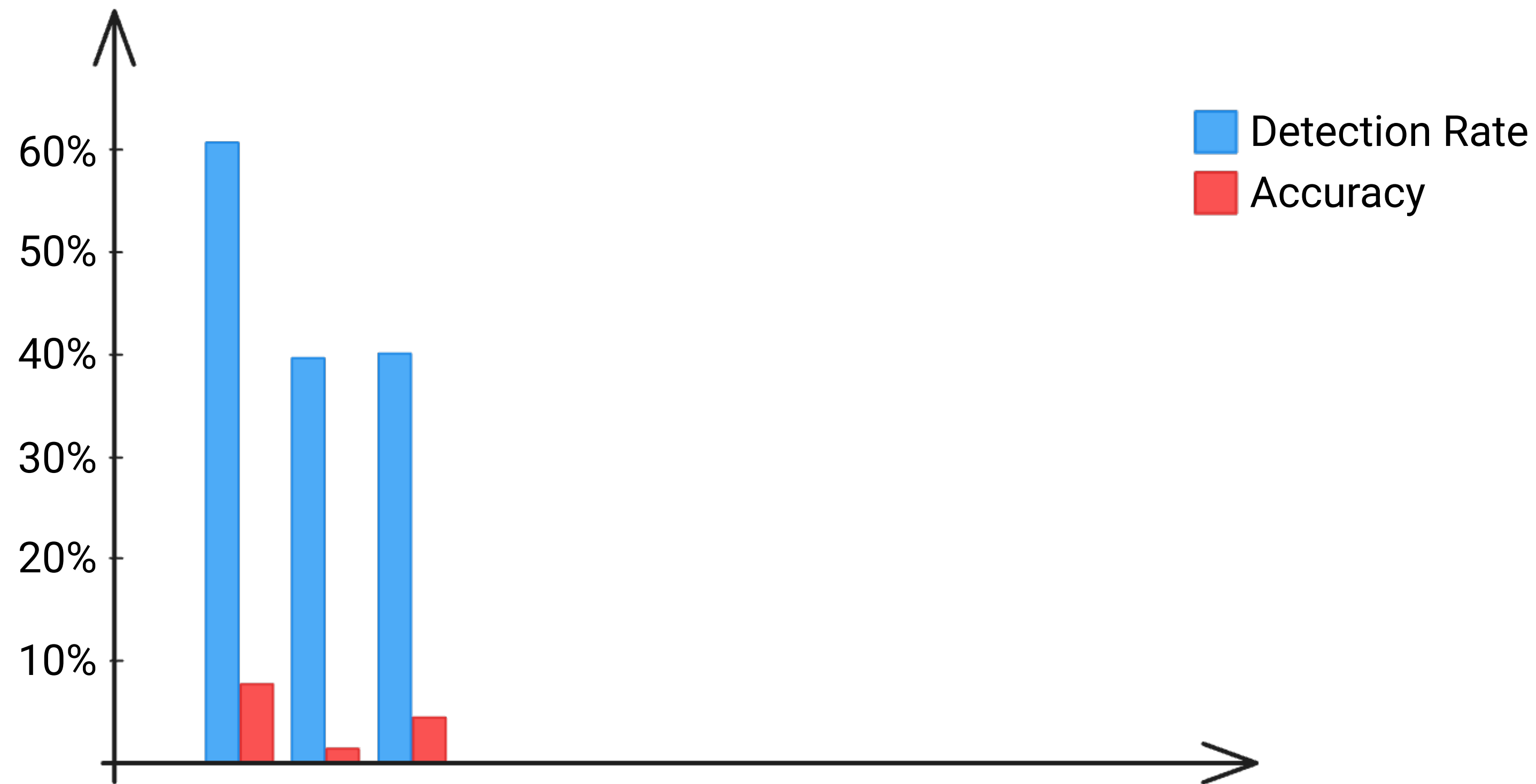
- **Detection rate:** Given a wrong solution, does the test suite fail?
- **Accuracy:** Given a problem, does it detect all wrong solutions?



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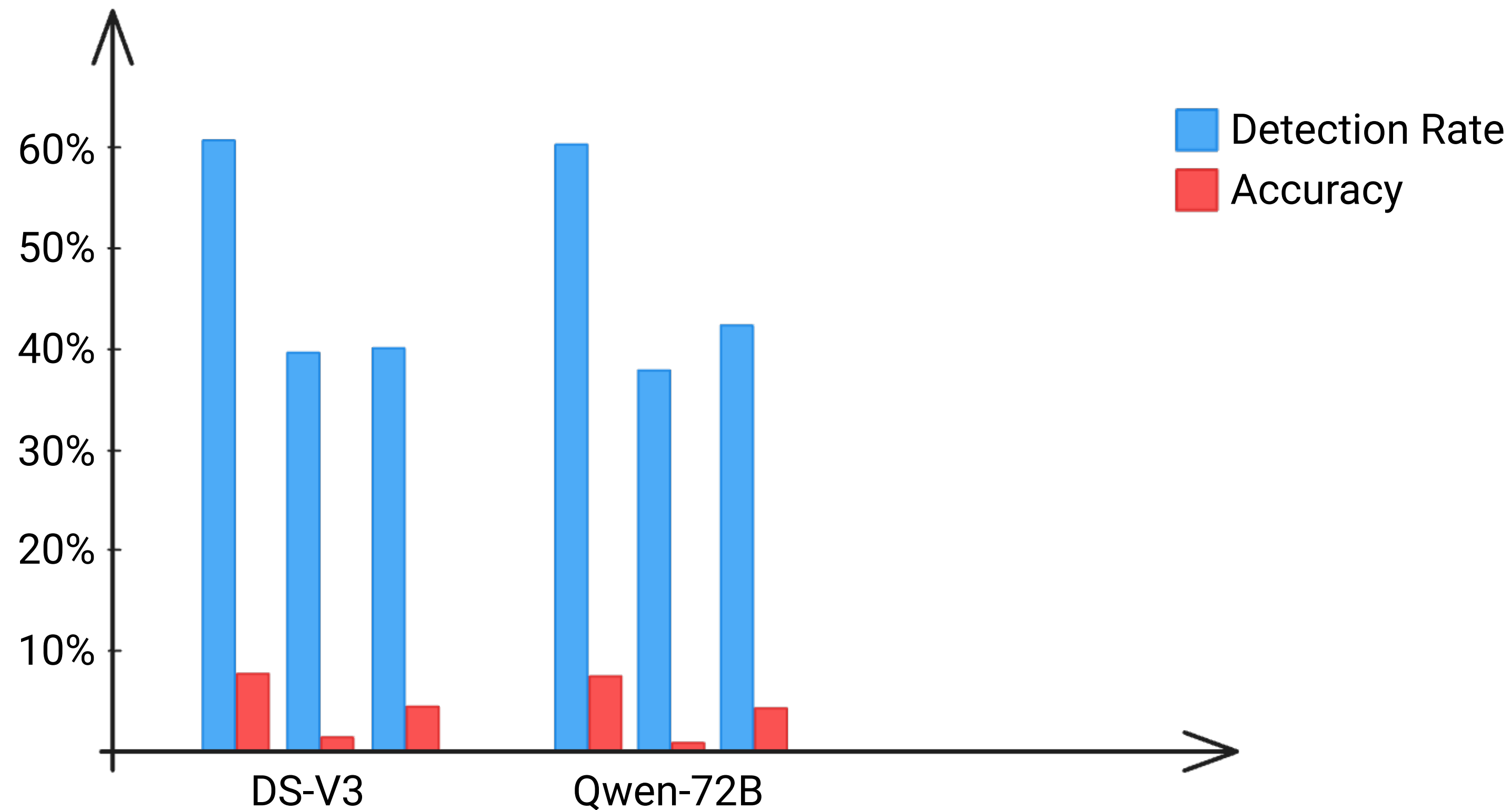
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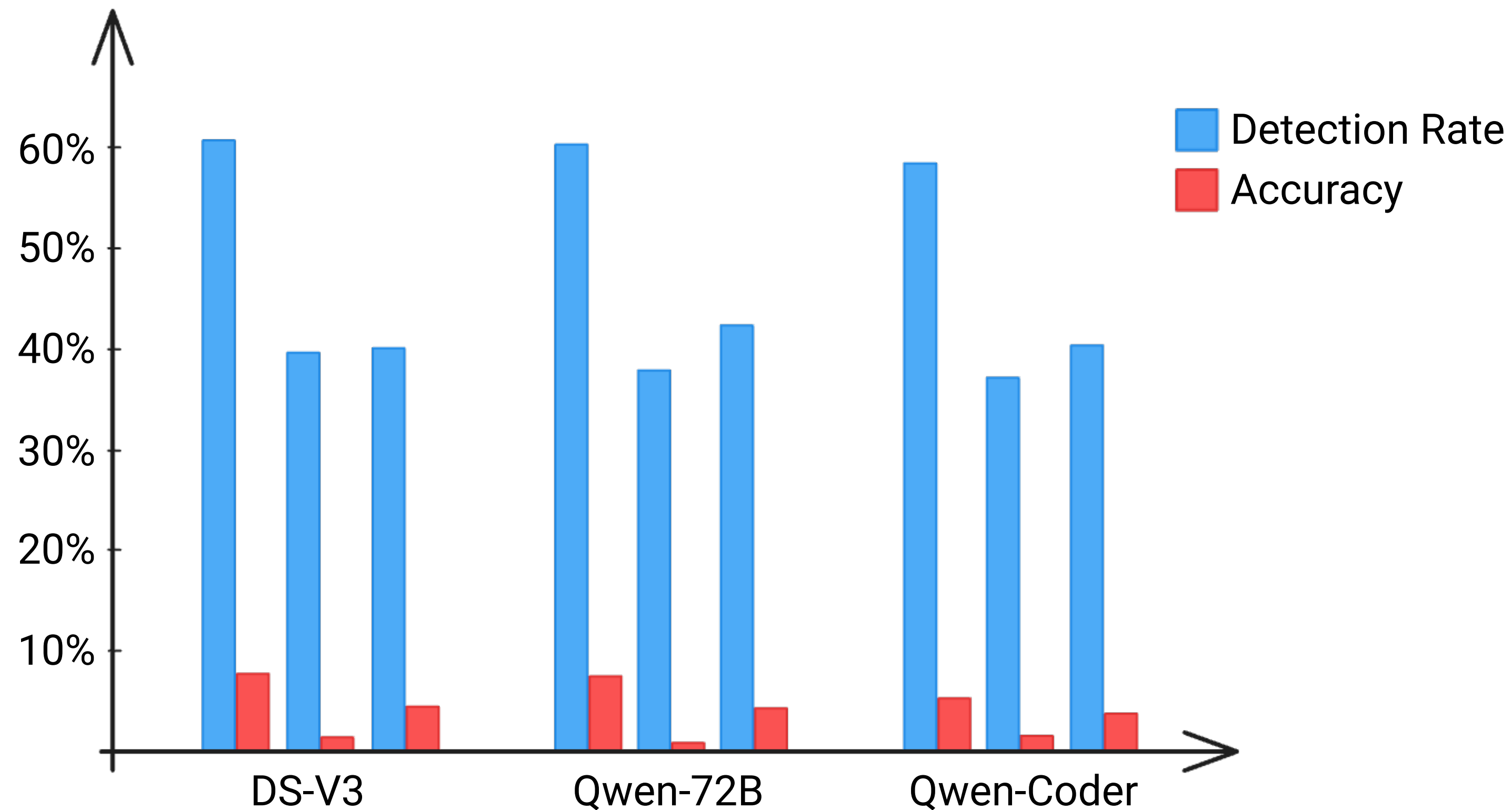
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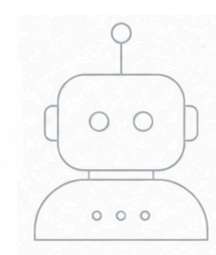
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Self-Evaluation Blind Spot



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Self-Evaluation Blind Spot

DS-V3			
Qwen-72B			
Qwen-Coder			
	DS-V3	Qwen-72B	Qwen-Coder

Source: <https://arxiv.org/abs/2507.06920>, presented at NeurIPS 2025

Self-Evaluation Blind Spot

Solution Model			
	Test Generation Model		
	DS-V3	Qwen-72B	Qwen-Coder
	DS-V3	Qwen-72B	Qwen-Coder
DS-V3			
Qwen-72B			
Qwen-Coder			

Source: <https://arxiv.org/abs/2507.06920>, presented at NeurIPS 2025

Self-Evaluation Blind Spot

Solution Model	DS-V3	85.3		
	Qwen-72B		70.8	
	Qwen-Coder			68.6
		DS-V3	Qwen-72B	Qwen-Coder
		Test Generation Model		

Source: <https://arxiv.org/abs/2507.06920>, presented at NeurIPS 2025

Self-Evaluation Blind Spot

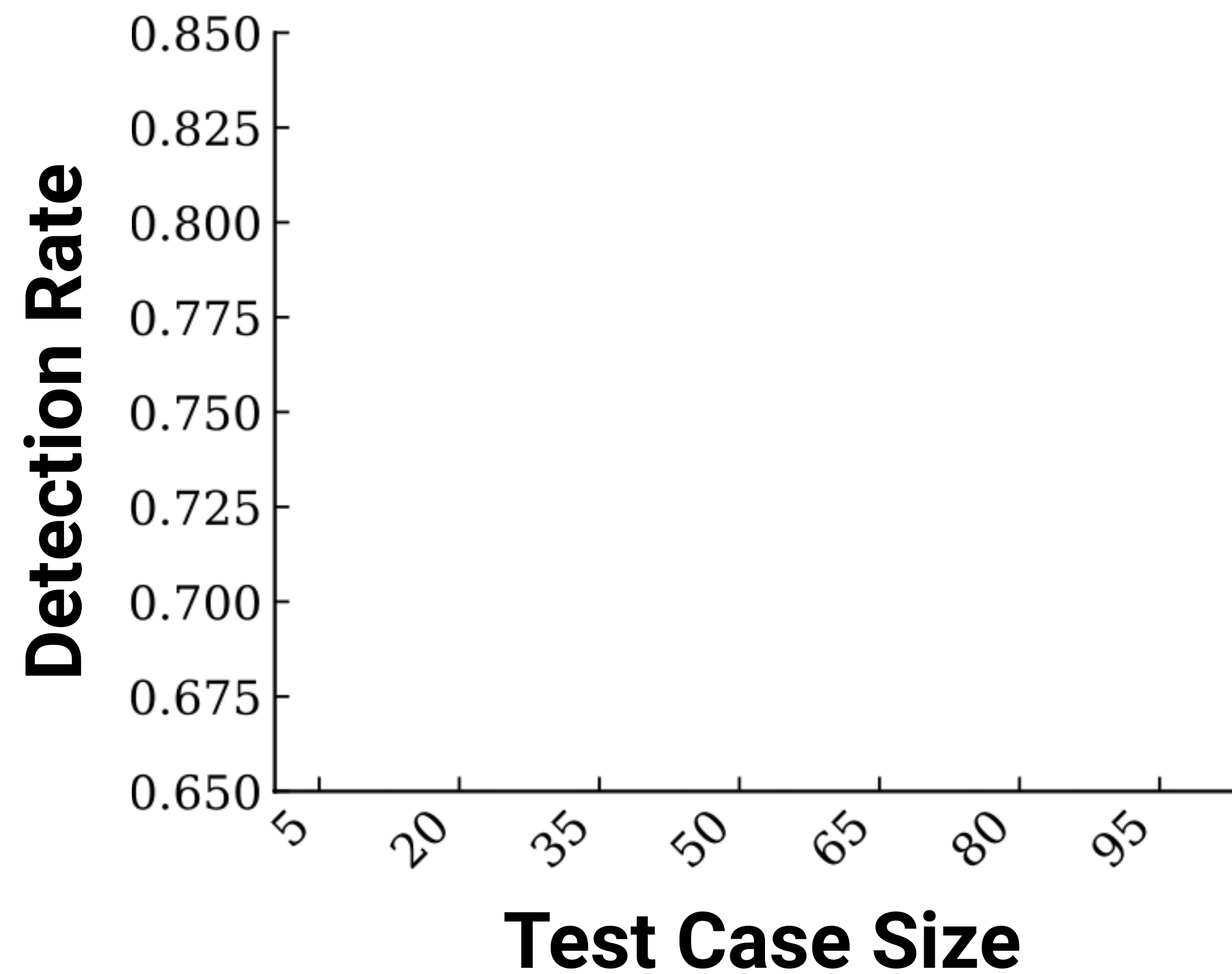
Solution Model				
	DS-V3	Qwen-72B	Qwen-Coder	Official
	DS-V3	Qwen-72B	Qwen-Coder	Official
	Official	Official	Official	Official
DS-V3	85.3			76.7
Qwen-72B		70.8		49.9
Qwen-Coder			68.6	47.5

Source: <https://arxiv.org/abs/2507.06920>, presented at NeurIPS 2025

Not a Factor of the Number of Tests

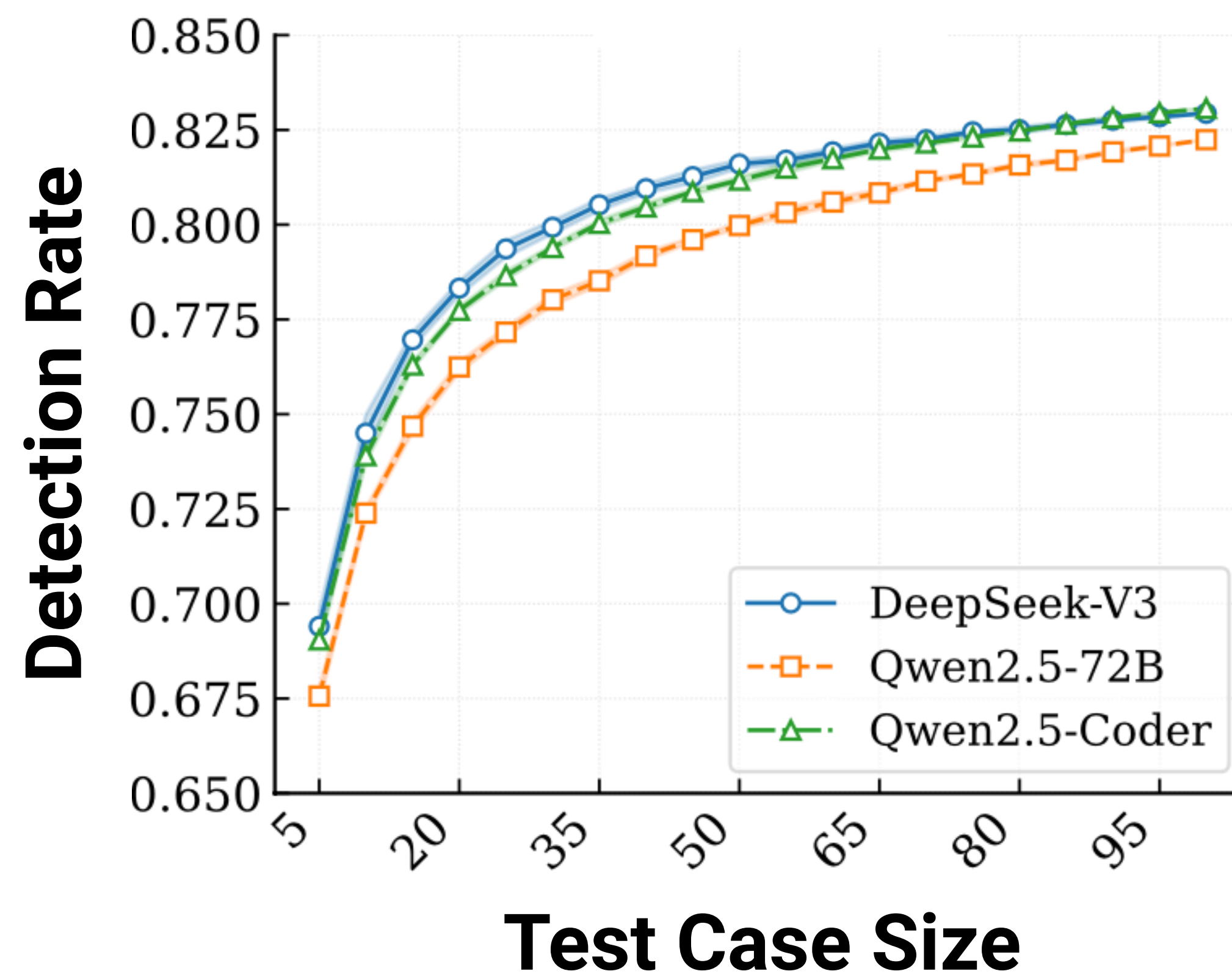
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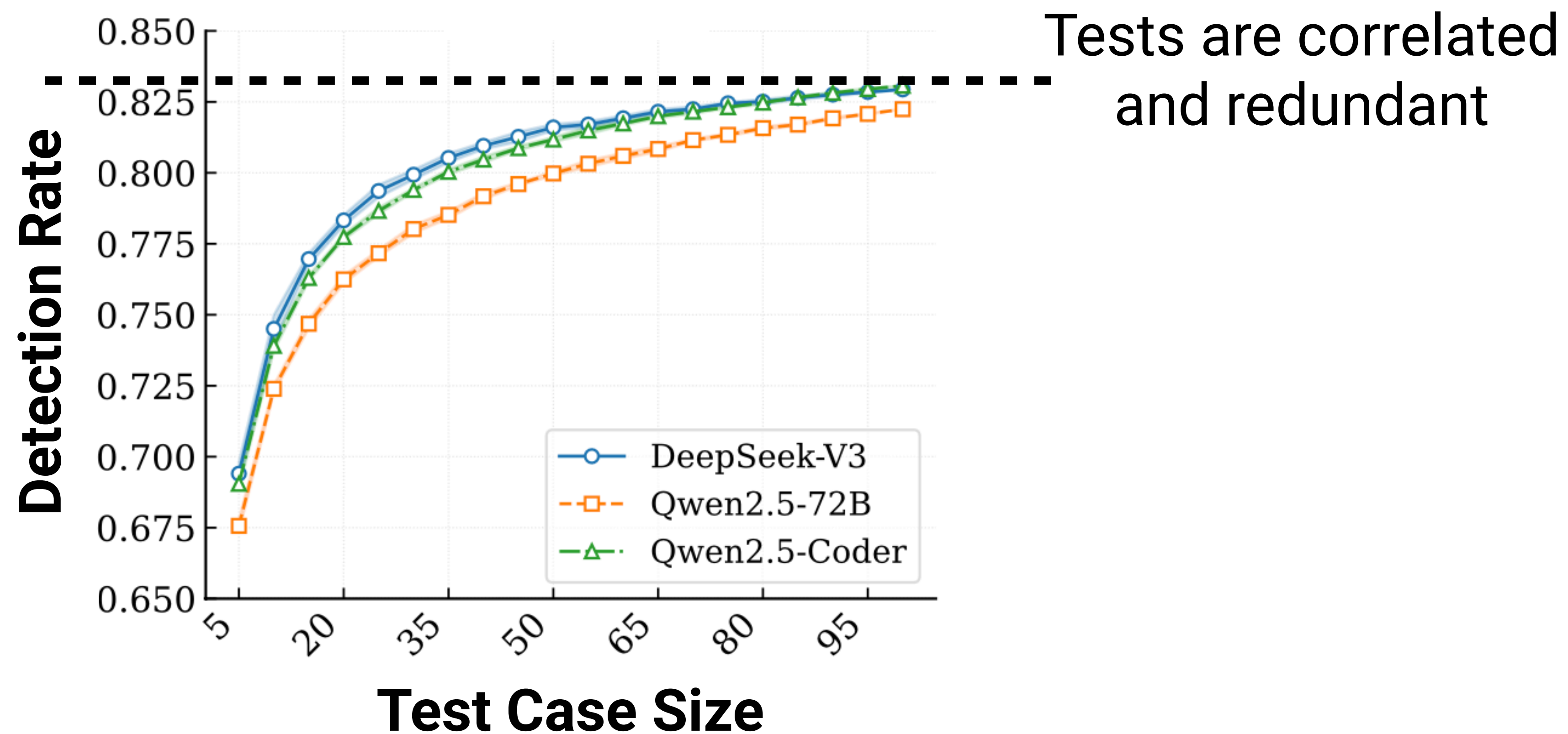
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Are LLMs Good At Testing Software?



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Are LLMs Good At Testing Software?

- LLMs are not great at catching bugs
 - ~60% chance of identifying a buggy solution



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 - Cases missed in source are also missed in tests



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Are LLMs Good At Testing Software?

- LLMs are not great at catching bugs
 - ~60% chance of identifying a buggy solution
- Inflated success rate when LLMs test their own solutions
 - Cases missed in source are also missed in tests
- Generating more tests doesn't work
 - Detection rate plateaus, indicating persistent blind spots

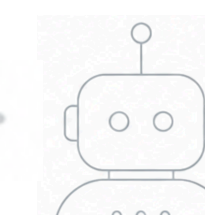


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Speed At Odds With Quality





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- Code generation is fast
 - AI writes code faster than we can read it





Speed At Odds With Quality

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 - AI writes code faster than we can read it
 - But it is also probabilistic
 - If we can't read it, we can't verify its correctness
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- 



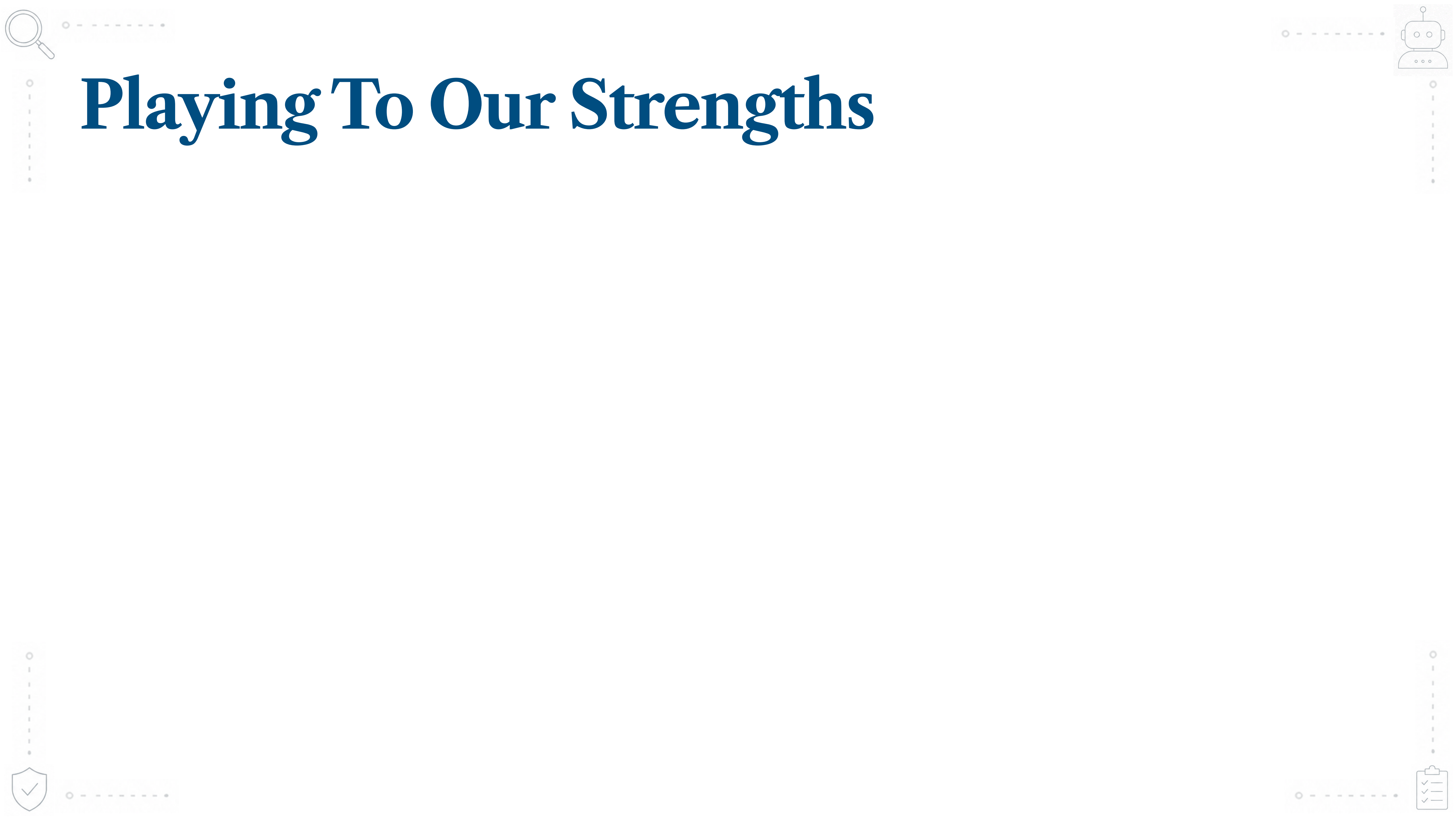
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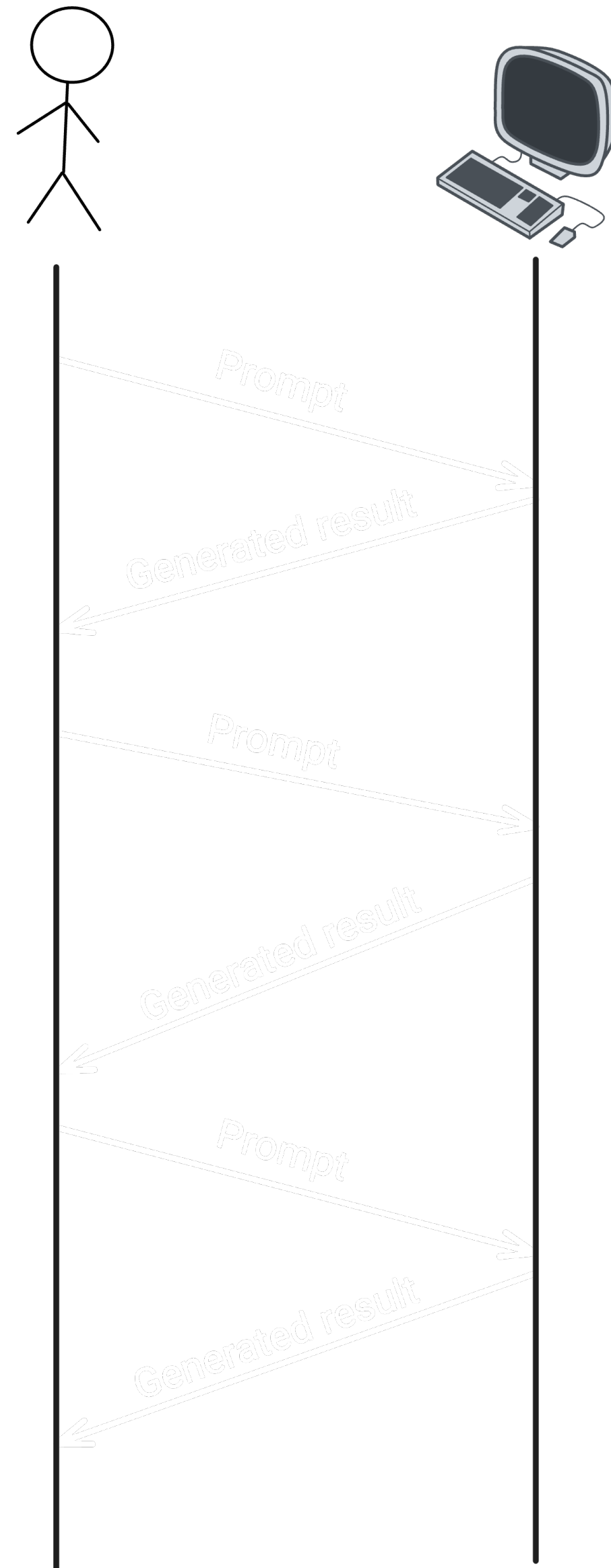
Speed At Odds With Quality

- Code generation is fast
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 - If we can't read it, we can't verify its correctness
 - LLMs are not good at testing software
 - Human comprehension
 - Our ability to understand and make meaning from information
- 
- 

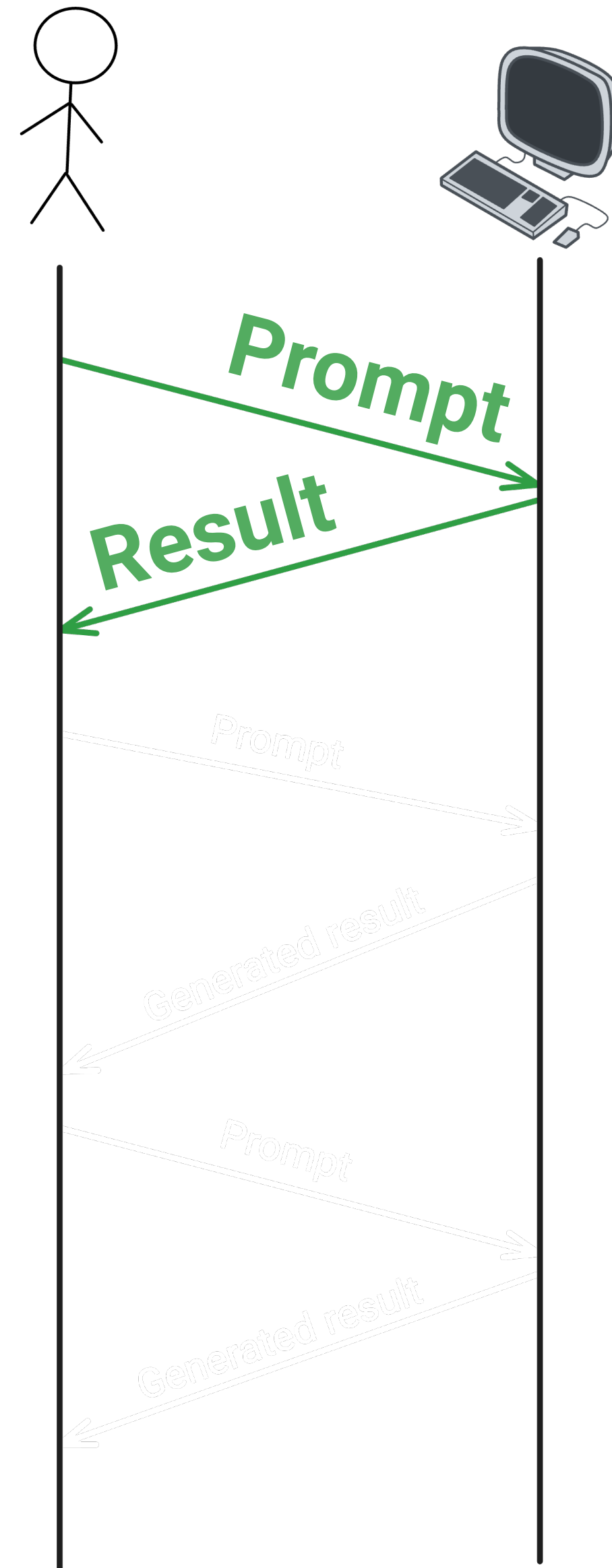


Playing To Our Strengths

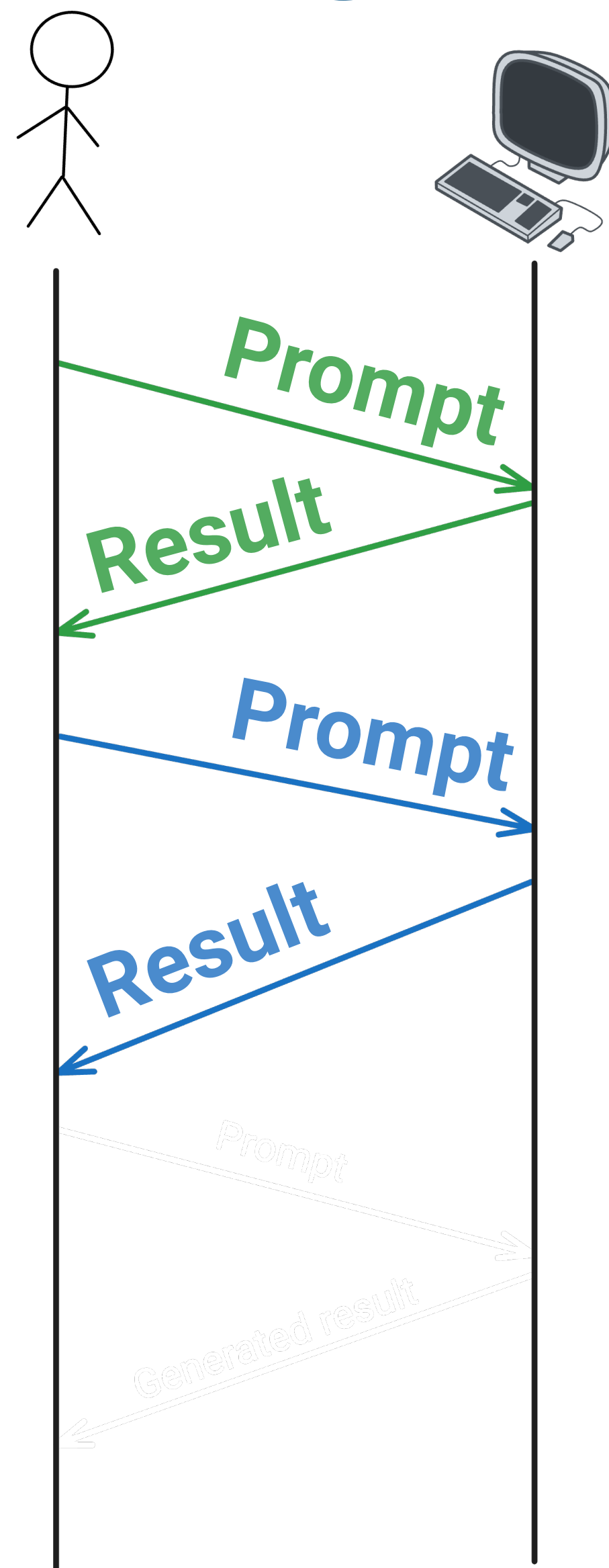
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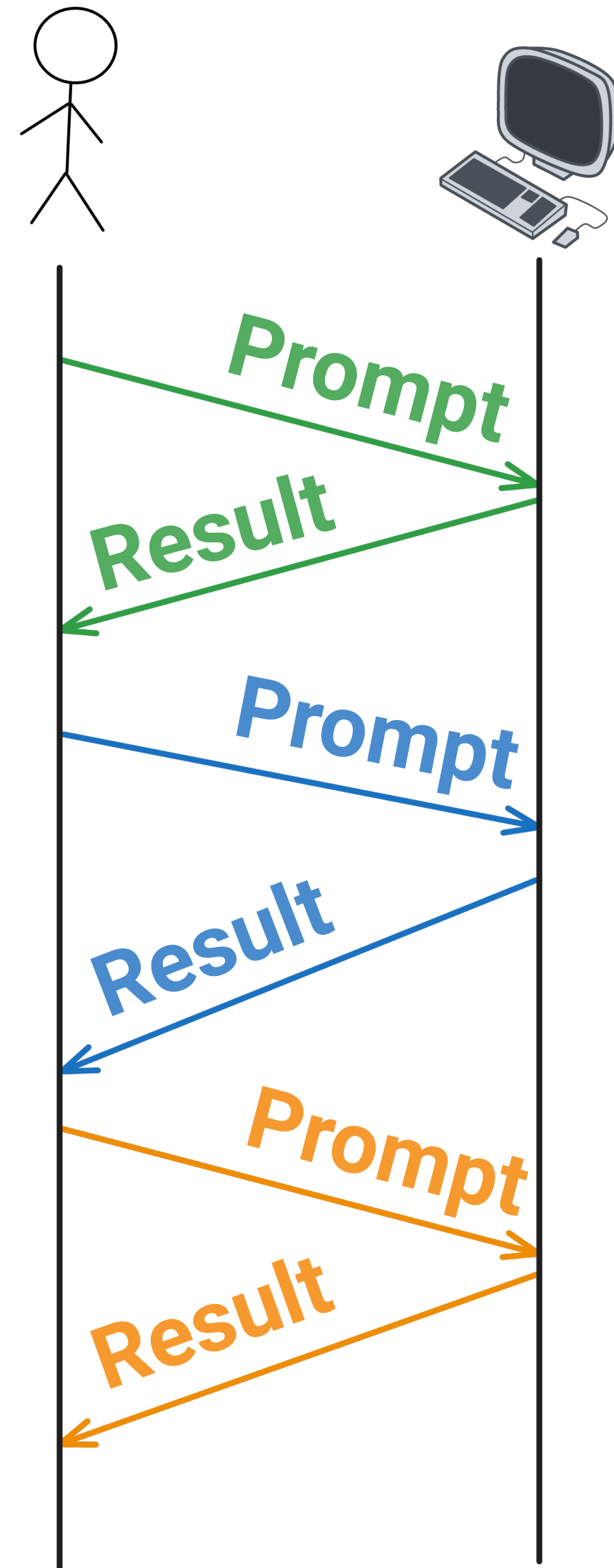
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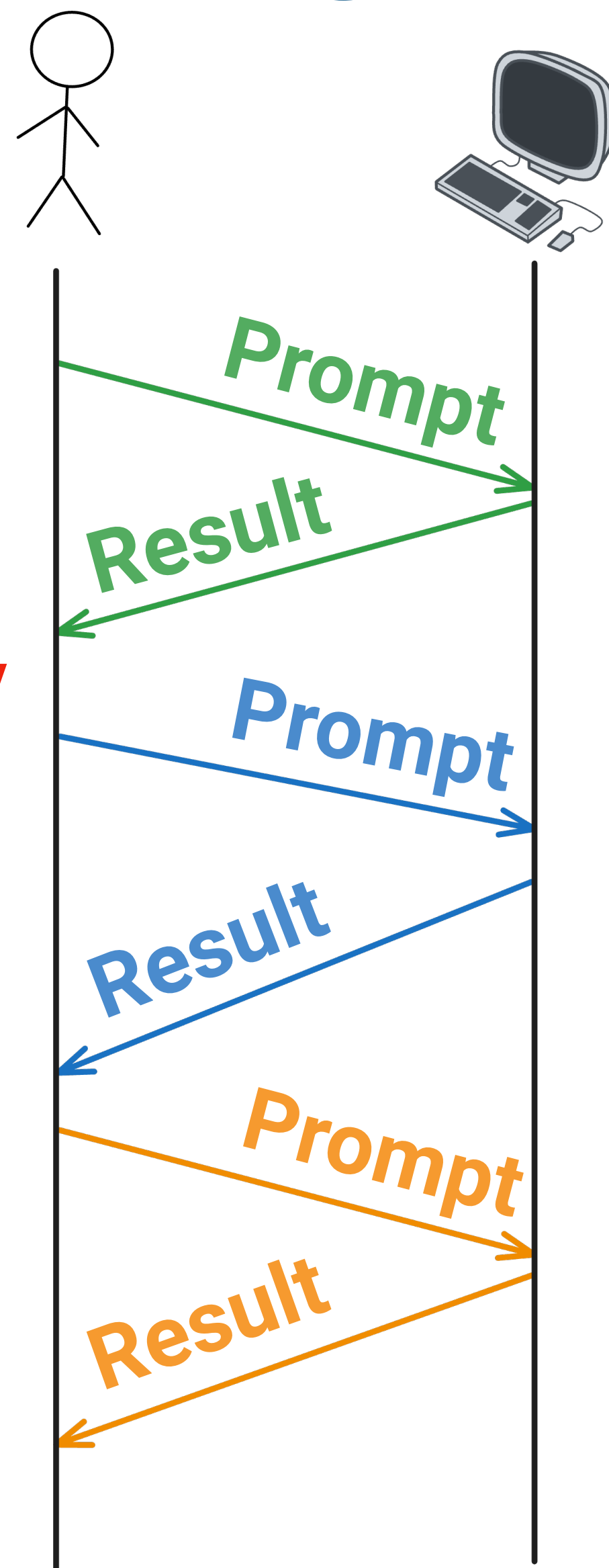


Playing To Our Strengths



Playing To Our Strengths

Happens too quickly



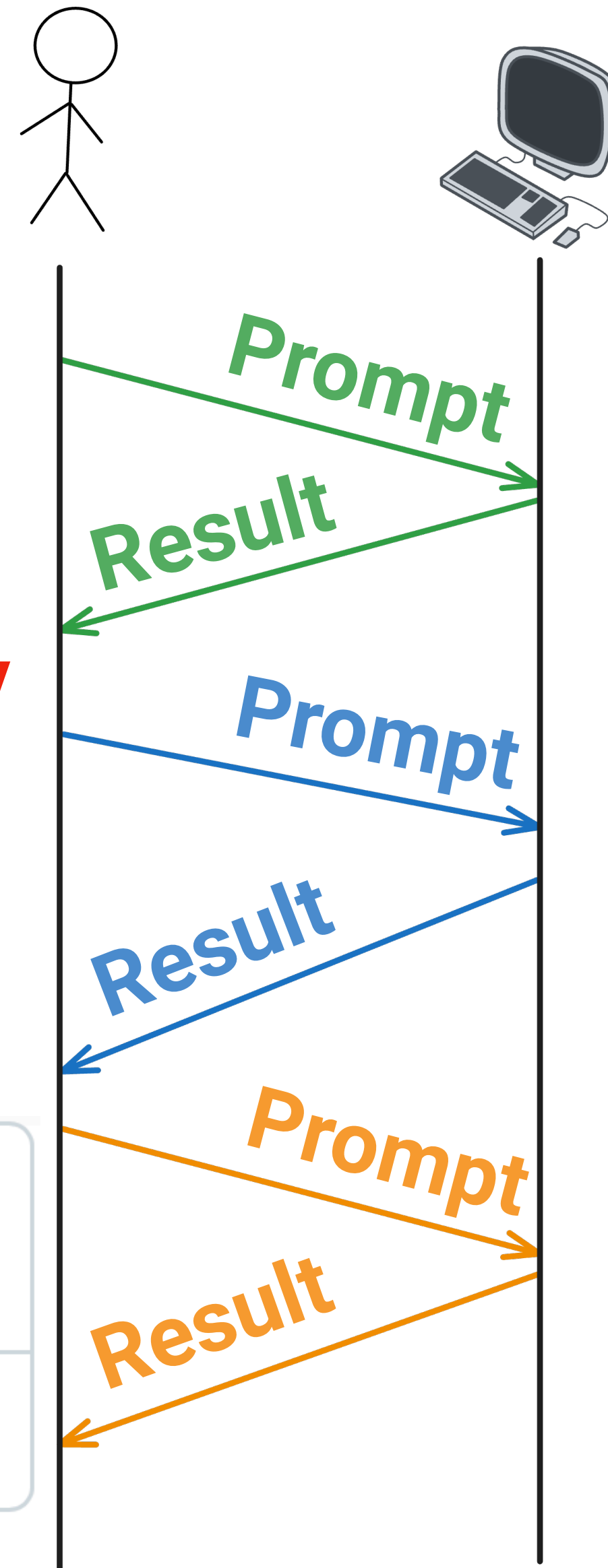
Playing To Our Strengths

Happens too quickly

feat: implement user, meeting,
invitation endpoints

25 files changed

+1,030 -32 ■ ■ ■ ■ ■



Playing To Our Strengths

Active Participant

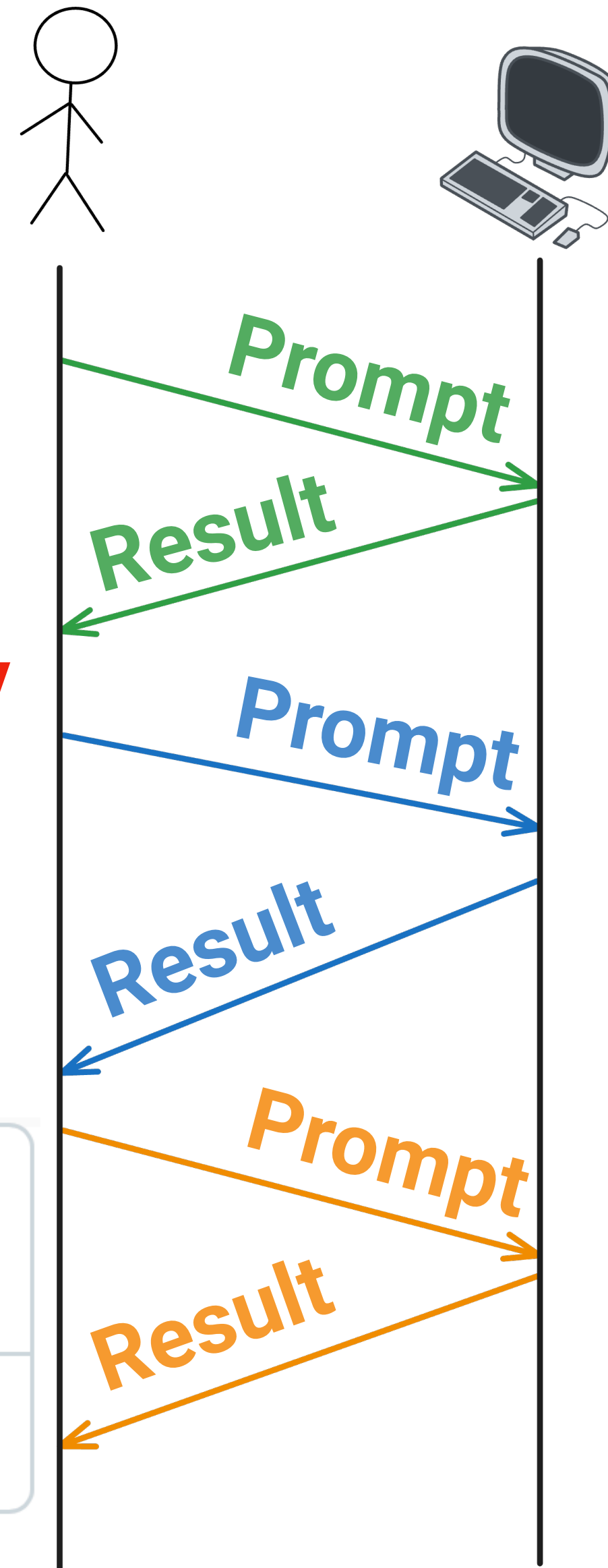
***Superficial* Overseer**

Happens too quickly

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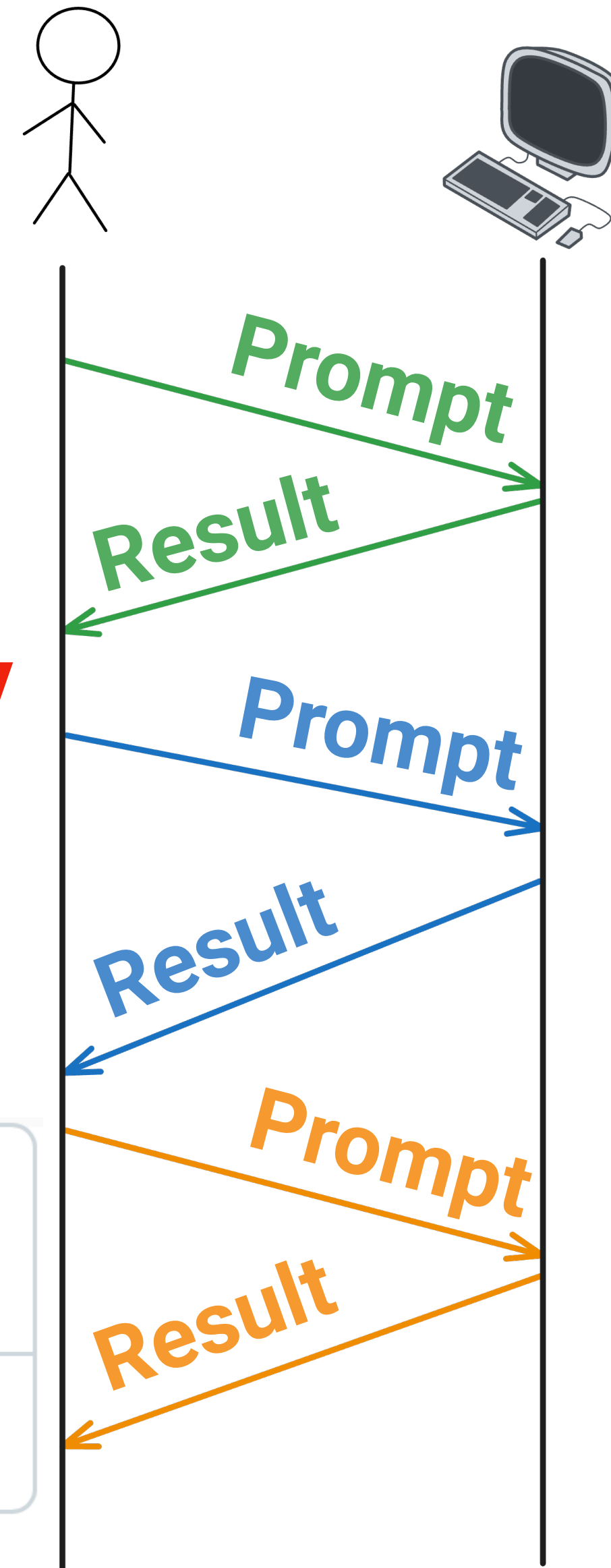
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Active Participant



Superficial Overseer

Overseeing does not **engage** our
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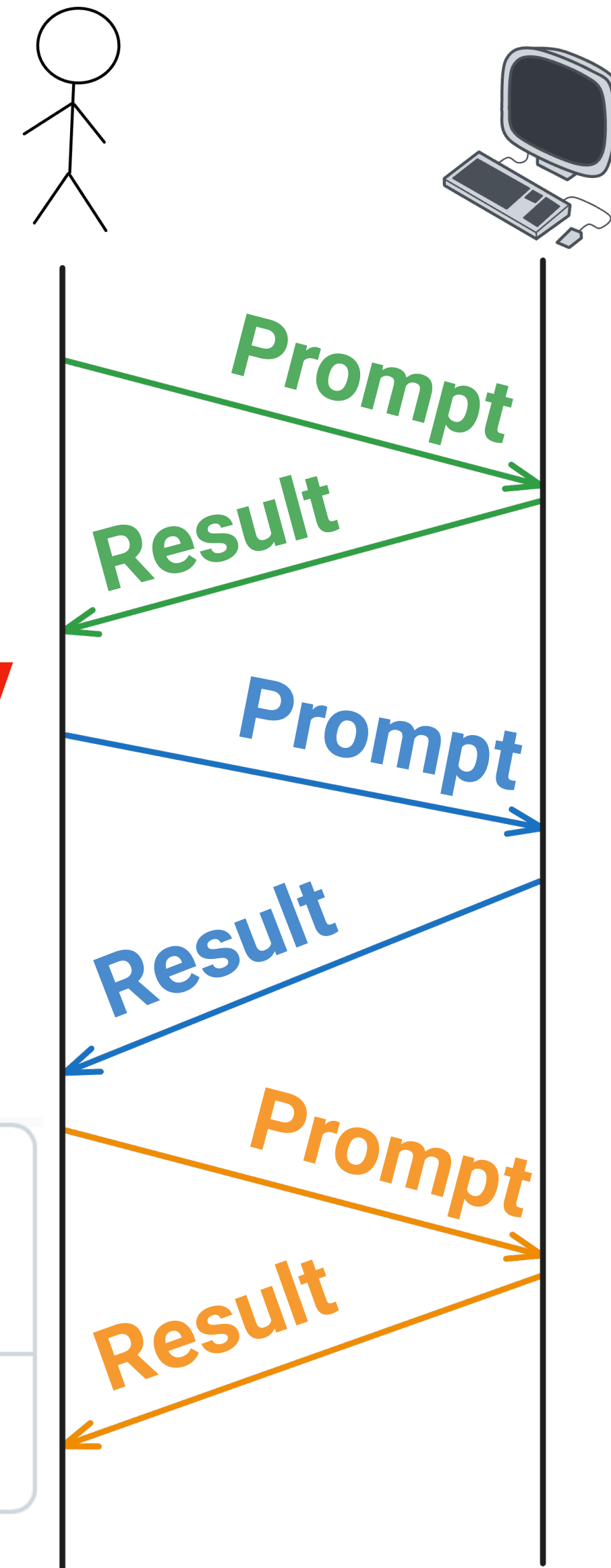
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Active Participant

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How did I miss that?

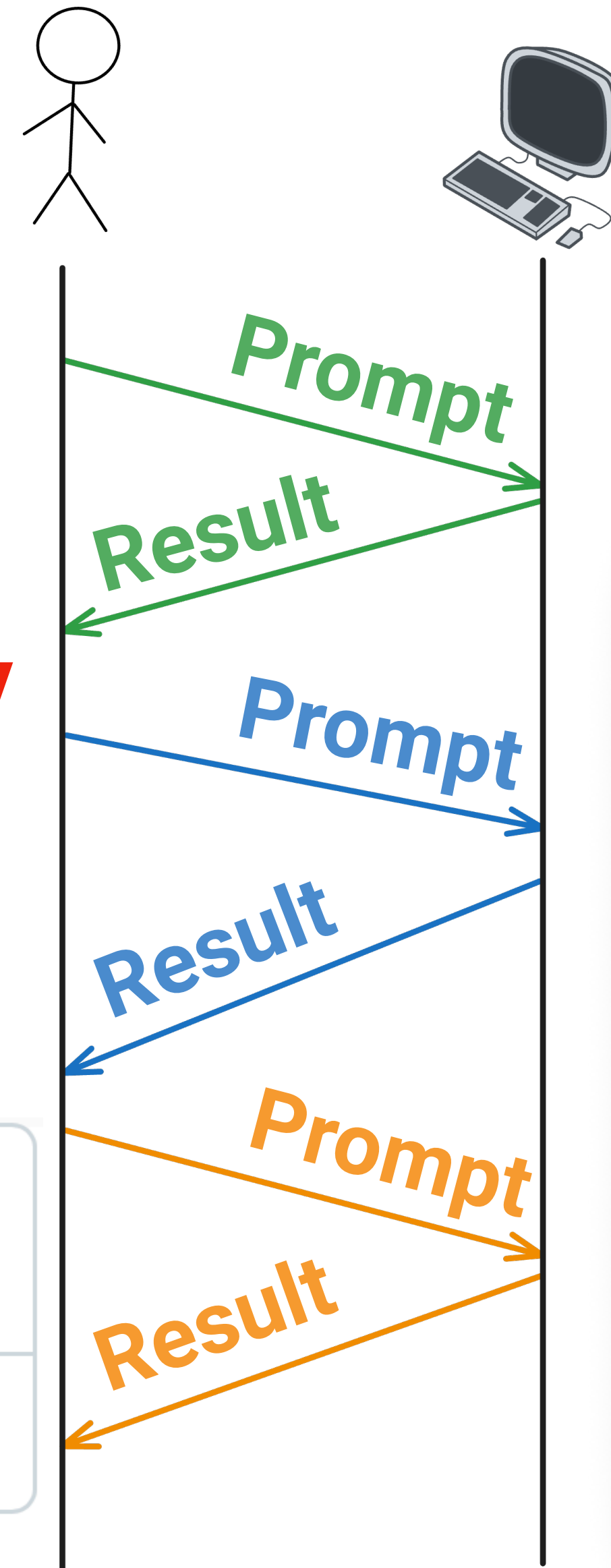
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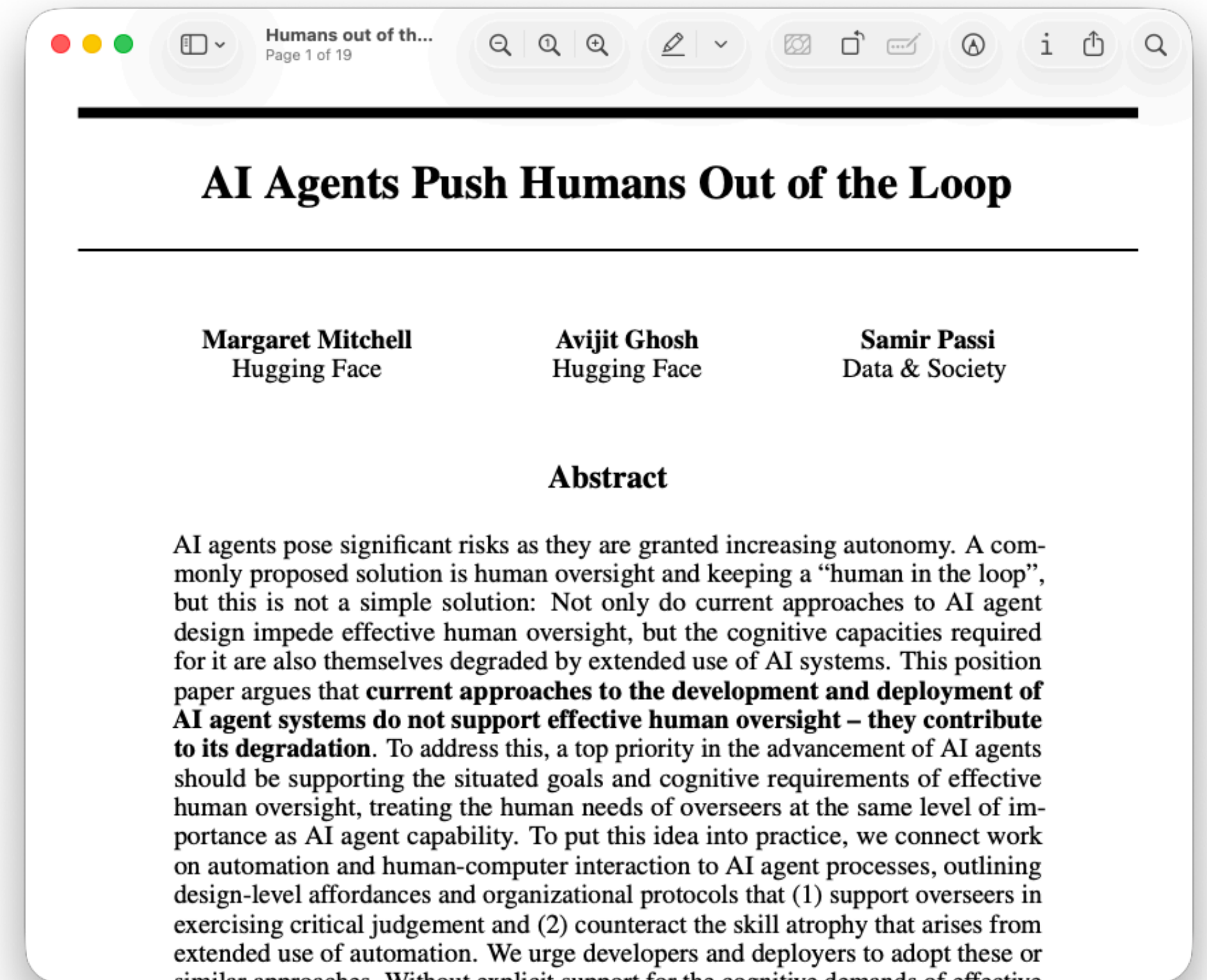
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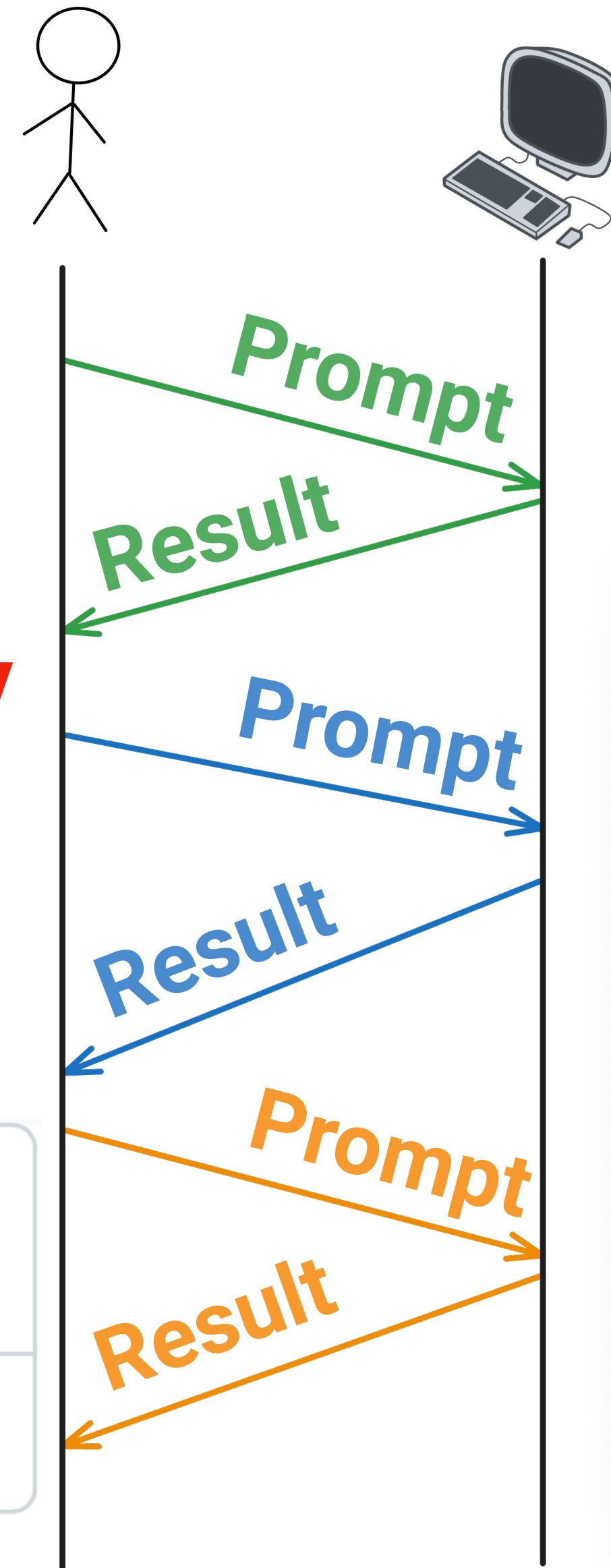
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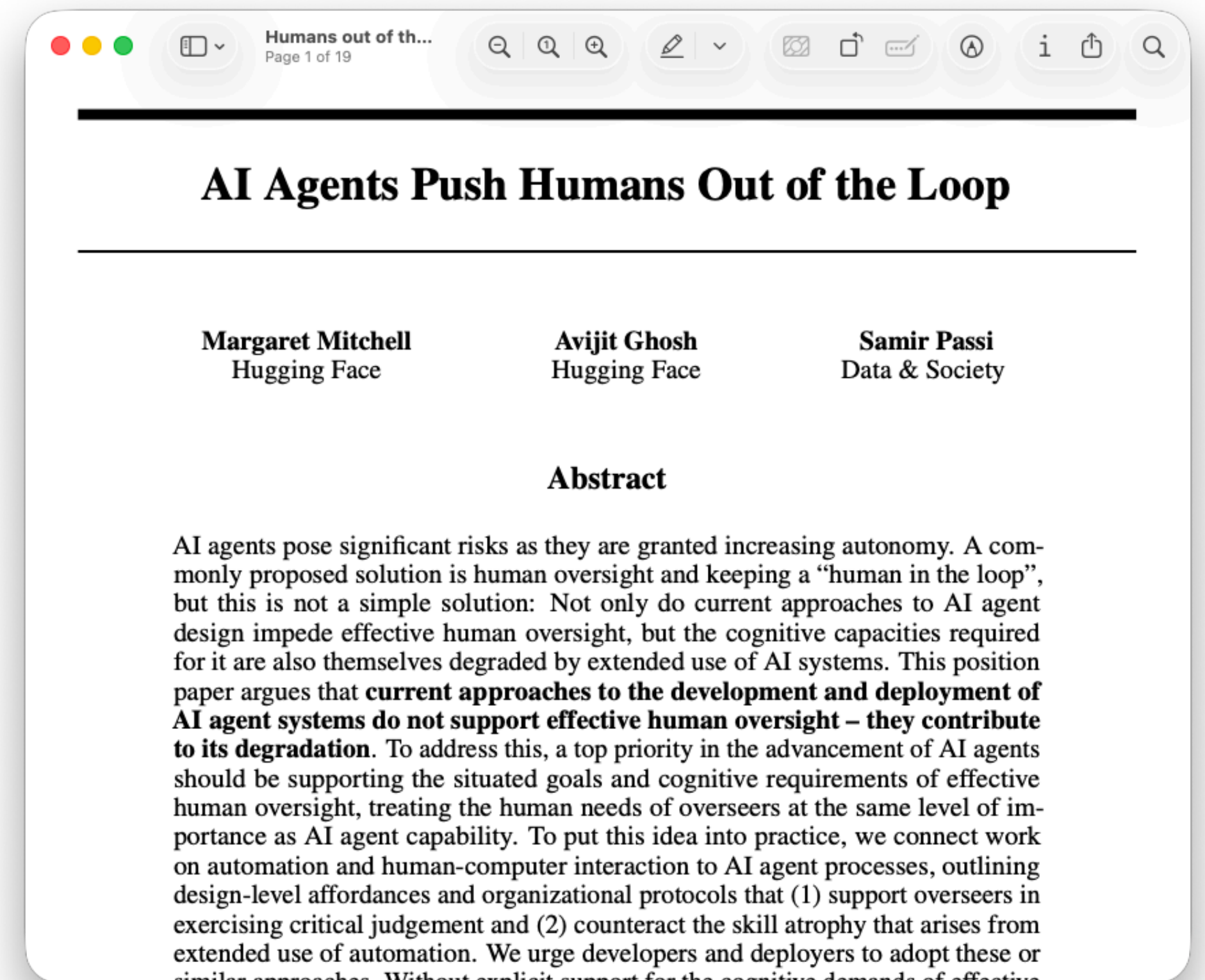
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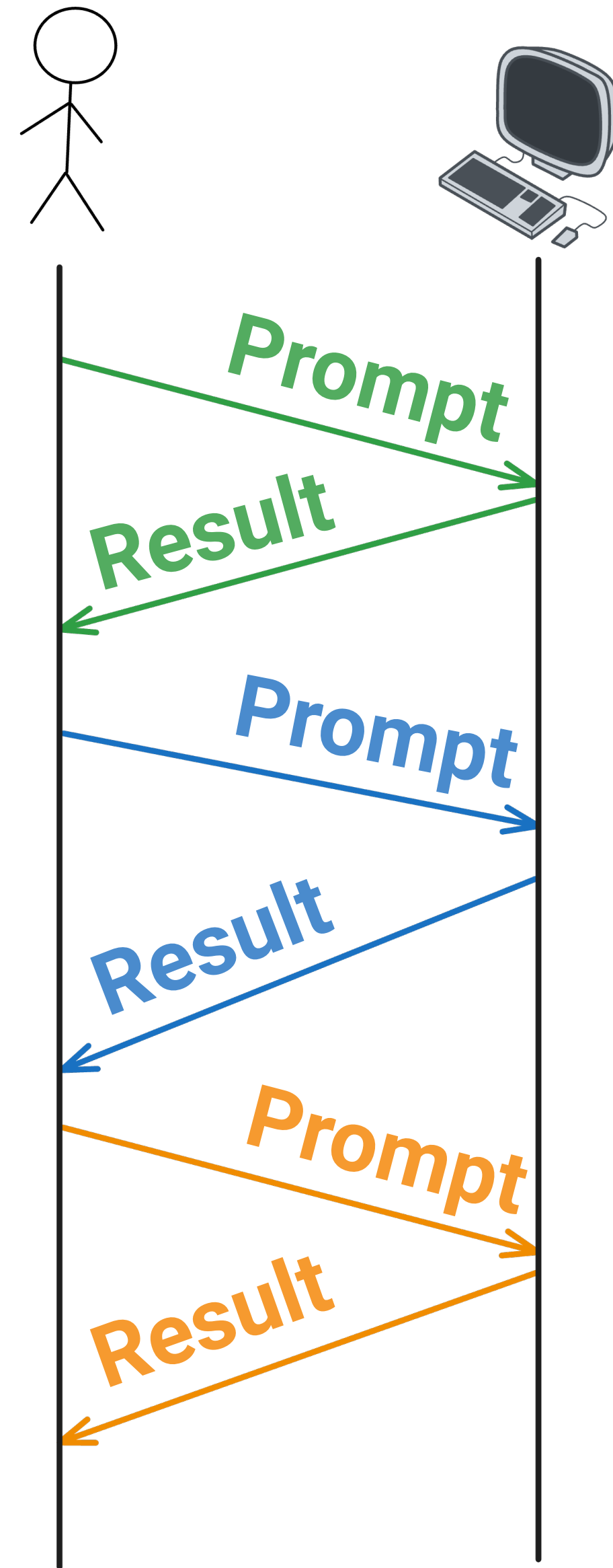


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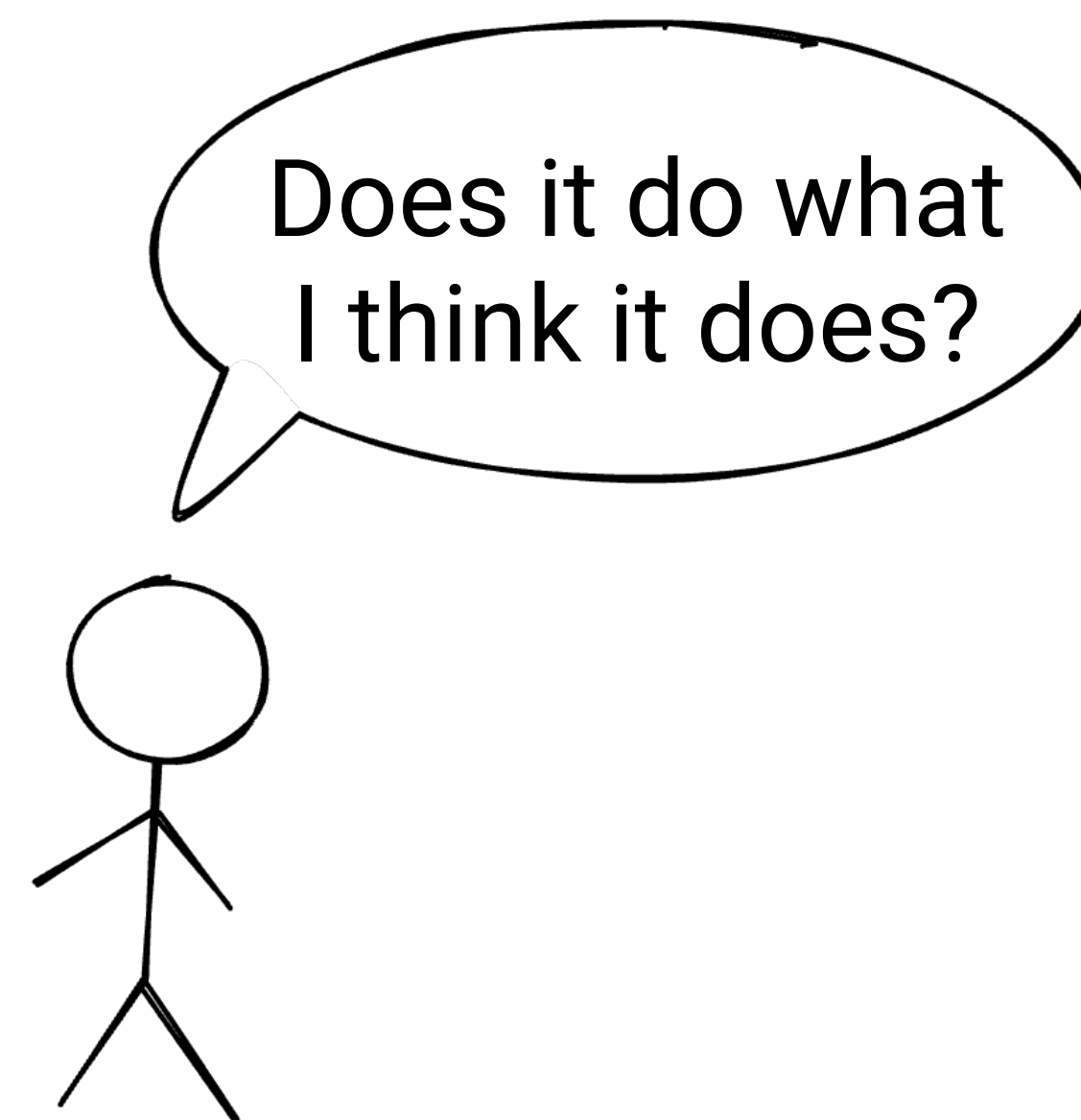
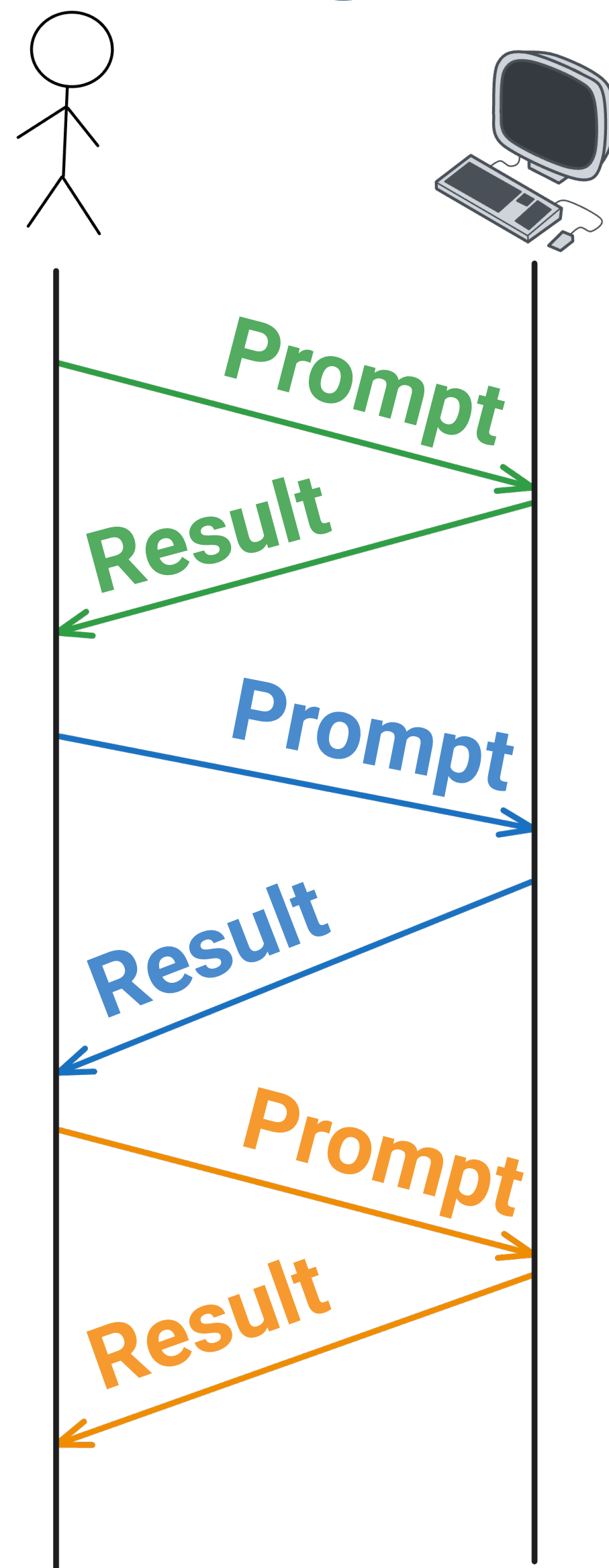
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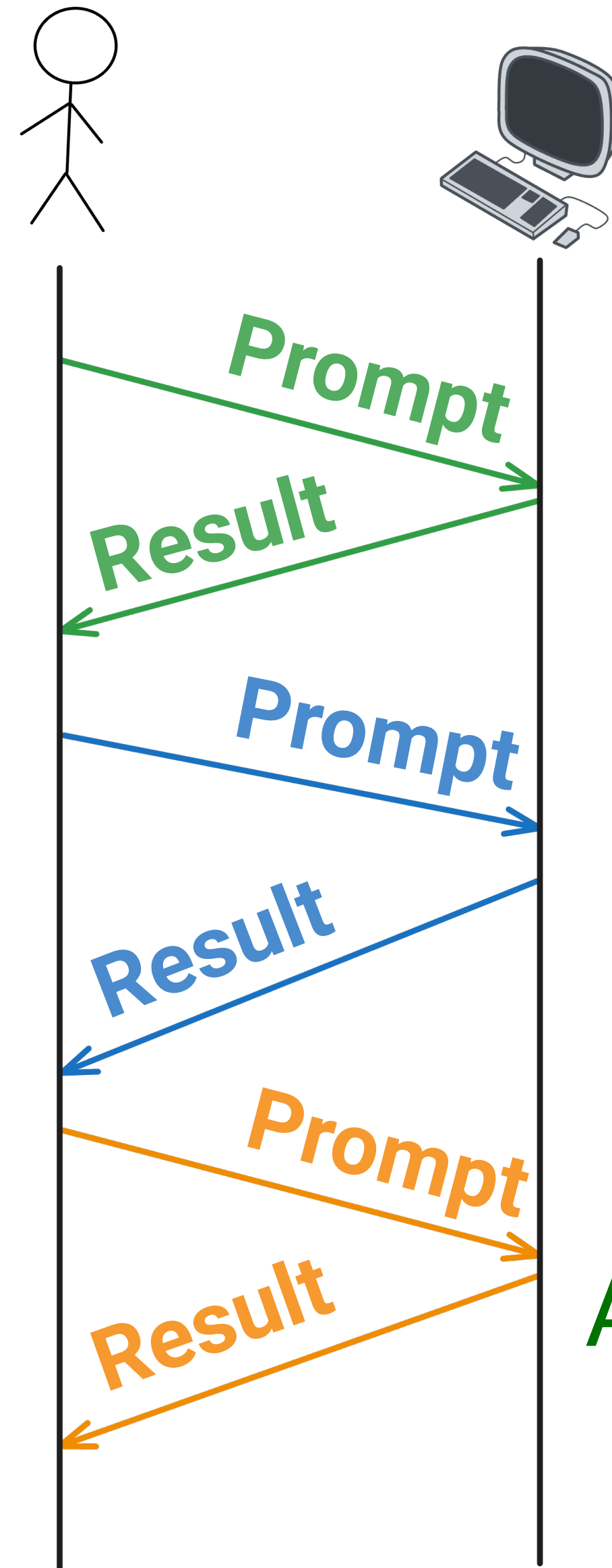
Playing To Our Strengths



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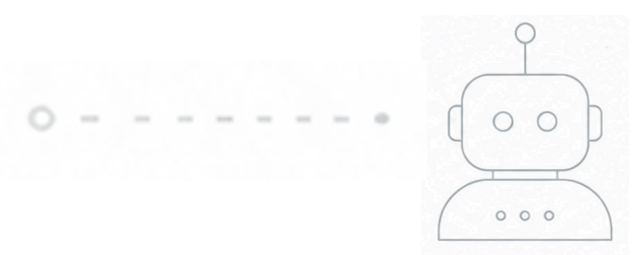


Playing To Our Strengths



Does it do what
I think it does?

Comprehension:
Assessing the fundamentals



Embedding Comprehension

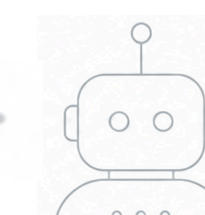
into Software Engineering





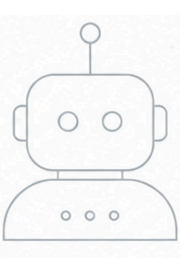
Fundamental Properties





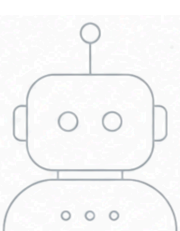
Fundamental Properties

- A user never receives stack traces
- 
- 



Fundamental Properties

- A user never receives stack traces
 - Every element in a sorted list $\leq$ the next element
- 
- 



Fundamental Properties

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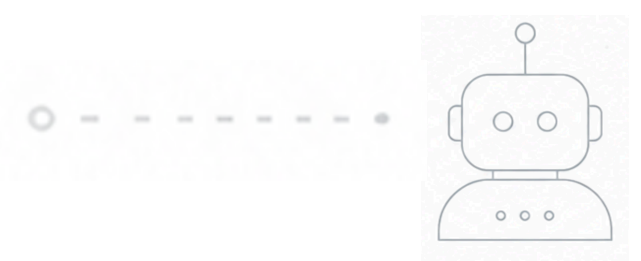
Fundamental Properties

- A user never receives stack traces
 - Every element in a sorted list $\leq$ the next element
 - One person cannot be in two meetings at the same time
 - The amount of money remains the same after account transfers
- 
- 



Fundamental Properties of Addition





Fundamental Properties of Addition

- Zero is the additive identity
 - $a + 0 = a$





Fundamental Properties of Addition

- Zero is the additive identity
 - $a + 0 = a$
 - Adding something to its inverse gives back the identity
 - $a + (-a) = 0$
- 
- 



Fundamental Properties of Addition

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 - $a + b = b + a$
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Fundamental Properties of Addition

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 - $a + 0 = a$
 - Adding something to its inverse gives back the identity
 - $a + (-a) = 0$
 - Addition is commutative
 - $a + b = b + a$
 - Addition is associative
 - $(a + b) + c = a + (b + c)$
- 
- 



Testing the Properties of Addition

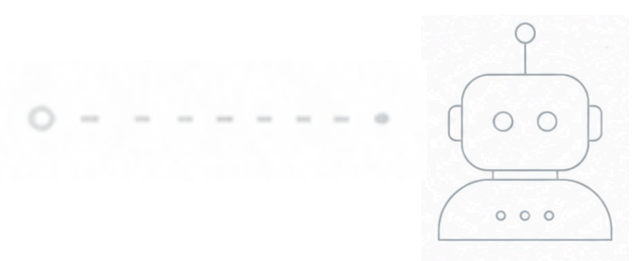




Testing the Properties of Addition

```
assertEquals(a, add(a, 0), "Additive Identity");
```





Testing the Properties of Addition

```
assertEquals(a, add(a, 0), "Additive Identity");  
assertEquals(0, add(a, -a), "Additive Inverse");
```





Testing the Properties of Addition

```
assertEquals(a, add(a, 0), "Additive Identity");  
assertEquals(0, add(a, -a), "Additive Inverse");  
assertEquals(add(a, b), add(b, a), "Commutativity");
```

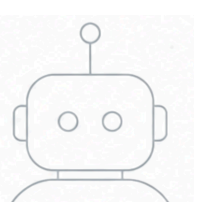


Testing the Properties of Addition

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assertEquals(a, add(a, 0), "Additive Identity");
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assertEquals(add(a, b), add(b, a), "Commutativity");
assertEquals(add(1, add(a, b)), add(add(1, a), b), "Associativity");
```

Testing the Properties of Addition

```
        @ForAll double a, @ForAll double b
assertEquals(a, add(a, 0), "Additive Identity");
assertEquals(0, add(a, -a), "Additive Inverse");
assertEquals(add(a, b), add(b, a), "Commutativity");
assertEquals(add(1, add(a, b)), add(add(1, a), b), "Associativity");
```

Testing the Properties of Addition

```
void propertyBasedAdditionTest(@ForAll double a, @ForAll double b) {  
    assertEquals(a, add(a, 0), "Additive Identity");  
    assertEquals(0, add(a, -a), "Additive Inverse");  
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Testing the Properties of Addition

@Property

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}
```

@Test

```
void exampleBasedAdditionTest() {  
    assertEquals(1.5, add(1.5, 0));  
    assertEquals(-101.8, add(-100, -1.8));  
    assertEquals(99, add(-1, 100));  
    assertEquals(230.5, add(130.5, 100));  
}
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Sampled values are tested

Testing the Properties of Addition

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}
```

The problem space is explored

@Test

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void exampleBasedAdditionTest() {  
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}
```

Verifies for multiple values
of **a** and **b**

@Test

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Testing the Properties of Addition

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Verifies for multiple values
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@Test

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void exampleBasedAdditionTest() {  
    assertEquals(1.5, add(1.5, 0));  
    assertEquals(-101.8, add(-100, -1.8));  
    assertEquals(99, add(-1, 100));  
    assertEquals(230.5, add(130.5, 100));  
}
```

Only verifies for 1.5 and 0

Testing the Properties of Addition

@Property

```
void propertyBasedAdditionTest(@ForAll double a, @ForAll double b) {  
    assertEquals(a, add(a, 0), "Additive Identity");  
    assertEquals(0, add(a, -a), "Additive Inverse");  
    assertEquals(add(a, b), add(b, a), "Commutativity");  
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}
```



@Test

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}
```



Testing the Properties of Addition

@Property

void proper

assertEq

assertEq

assertEq

assertEq

}

@Test

void exampl

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assertEq

assertEq

}

tries = 285

checks = 285

generation = RANDOMIZED

...

Original Sample

a: 15.6

b: -1.49

Original Error

org.opentest4j.AssertionFailedError:

Associativity $\implies$ expected: <15.11> but was: <15.1100000000000001>

|-----jqwik-----

| # of calls to property

| # of not rejected calls

| parameters are randomly generated

Testing the Properties of Addition

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|-----jqwik-----

| # of calls to property

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jsell>

Testing the Properties of Addition

@Property

```
void proper
    assertEquals
    assertEquals
    assertEquals
    assertEquals
}
```

```
tries = 285
checks = 285
generation = RANDOMIZED
...
```

Original Sample

a: 15.6
b: -1.49

Original Error

org.opentest4j.AssertionFailedError:

Associativity $\Rightarrow$ expected: <15.11> but was: <15.1100000000000001>

-----jqwik-----
| # of calls to property
| # of not rejected calls
| parameters are randomly generated

jshe11> 1 + (15.6 + -1.49)
\$1 $\Rightarrow$ 15.11

Testing the Properties of Addition

@Property

```
void proper
    assertEquals
    assertEquals
    assertEquals
    assertEquals
}
```

```
tries = 285
checks = 285
generation = RANDOMIZED
...
```

Original Sample

```
-----
a: 15.6
b: -1.49
```

@Test

```
void example
    assertEquals
    assertEquals
    assertEquals
    assertEquals
}
```

Original Error

```
-----
org.opentest4j.AssertionFailedError:
```

Associativity $\Rightarrow$ expected: <15.11> but was: <15.1100000000000001>

```
|-----jqwik-----
| # of calls to property
| # of not rejected calls
| parameters are randomly generated
```

```
jshe11> 1 + (15.6 + -1.49)
$1  $\Rightarrow$  15.11
```

```
jshe11> (1 + 15.6) + -1.49
$2  $\Rightarrow$  15.1100000000000001
```


Testing the Properties of Addition

@Property

```
void proper
    assertEquals
    assertEquals
    assertEquals
    assertEquals
}
```

```
tries = 285
checks = 285
generation = RANDOMIZED
...
```

Original Sample

```
a: 15.6
b: -1.49
```

Original Error

org.opentest4j.AssertionFailedError:

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|-----jqwik-----
| # of calls to property
| # of not rejected calls
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```
jsHELL> 1 + (15.6 + -1.49)
$1  $\Rightarrow$  15.11
```

```
jsHELL> (1 + 15.6) + -1.49
$2  $\Rightarrow$  15.1100000000000001
```

```
jsHELL> 15.6d == 15.6f
$3  $\Rightarrow$  false
```

Testing the Properties of Addition

@Property

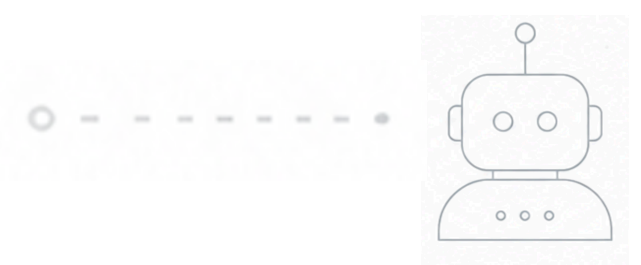
```
void propertyBasedAdditionTest(@ForAll double a, @ForAll double b) {  
    assertEquals(a, add(a, 0), "Additive Identity");  
    assertEquals(0, add(a, -a), "Additive Inverse");  
    assertEquals(add(a, b), add(b, a), "Commutativity");  
    assertEquals(add(1, add(a, b)), add(add(1, a), b), "Associativity");  
}
```



@Test

```
void exampleBasedAdditionTest() {  
    assertEquals(1.5, add(1.5, 0));  
    assertEquals(-101.8, add(-100, -1.8));  
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    assertEquals(230.5, add(130.5, 100));  
}
```

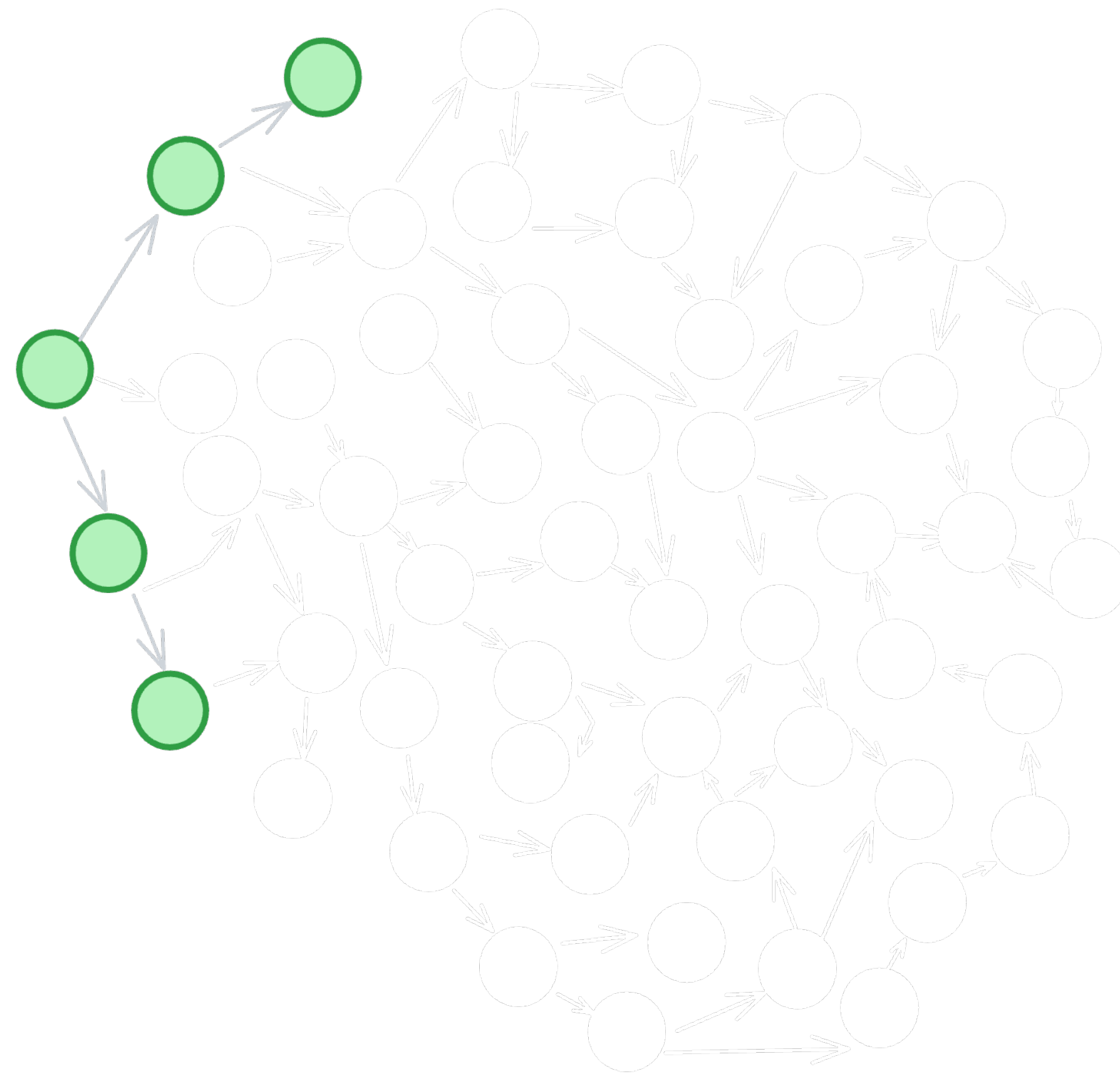




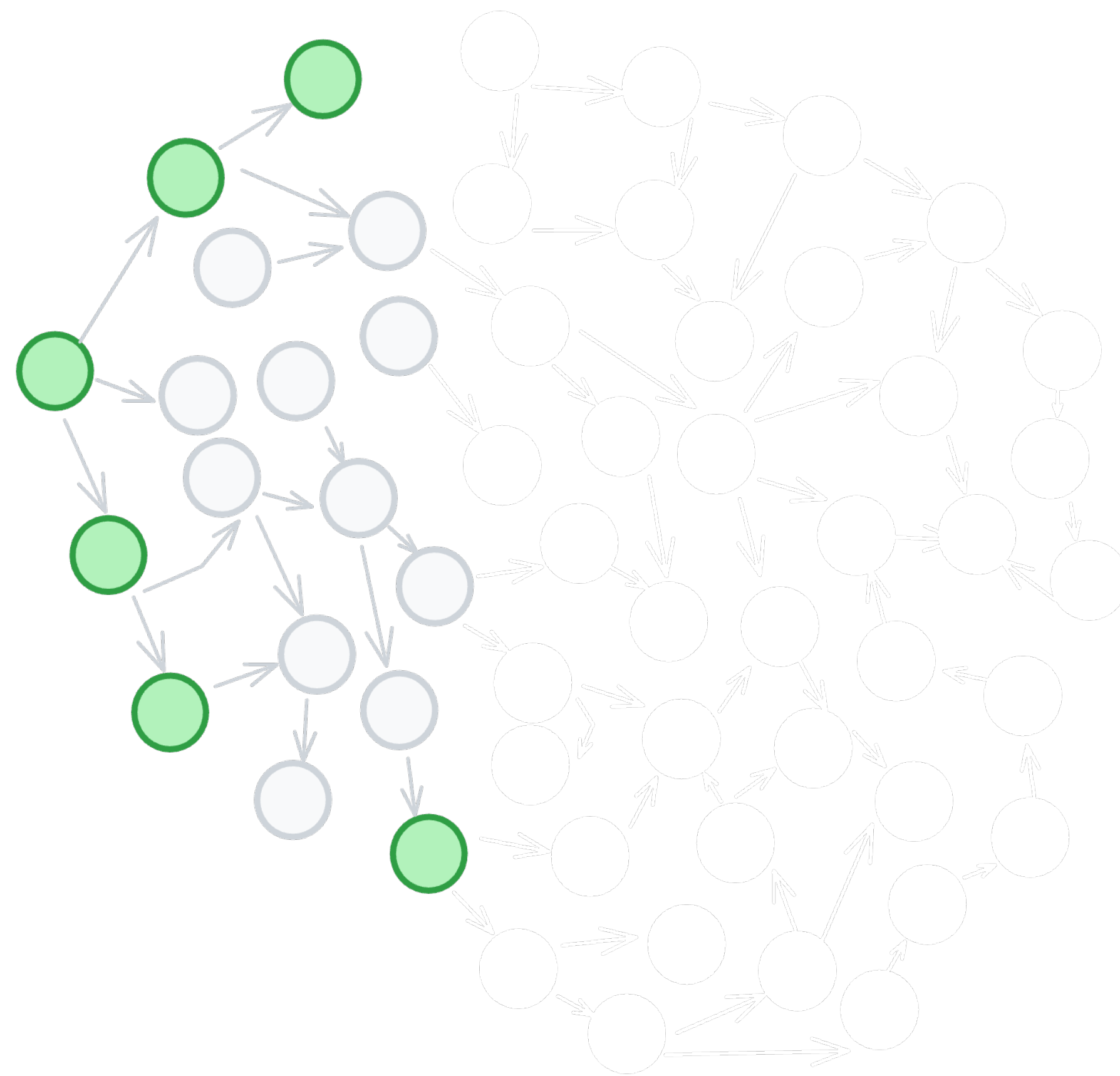
Examples → Exploration



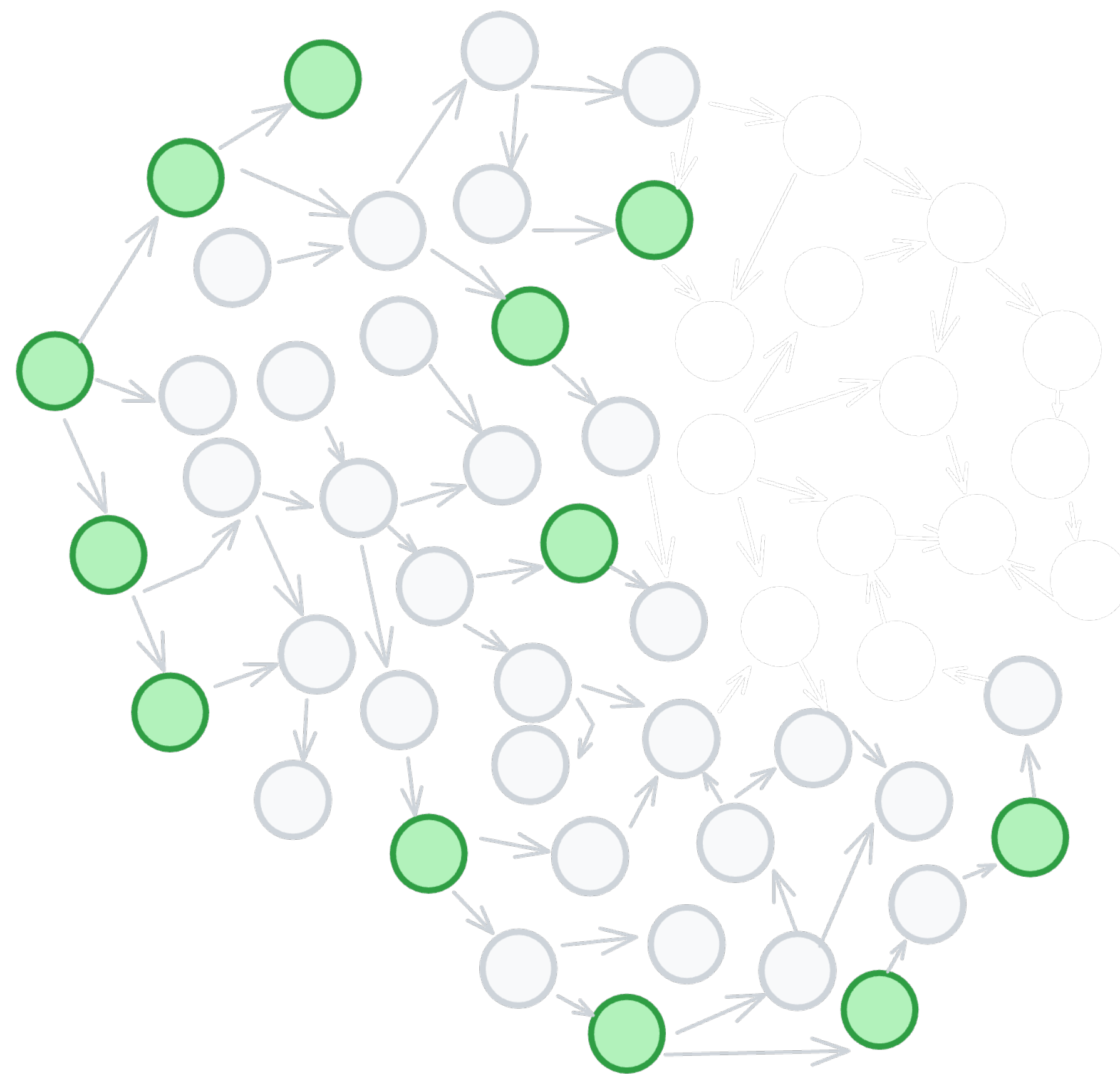
Examples → Exploration



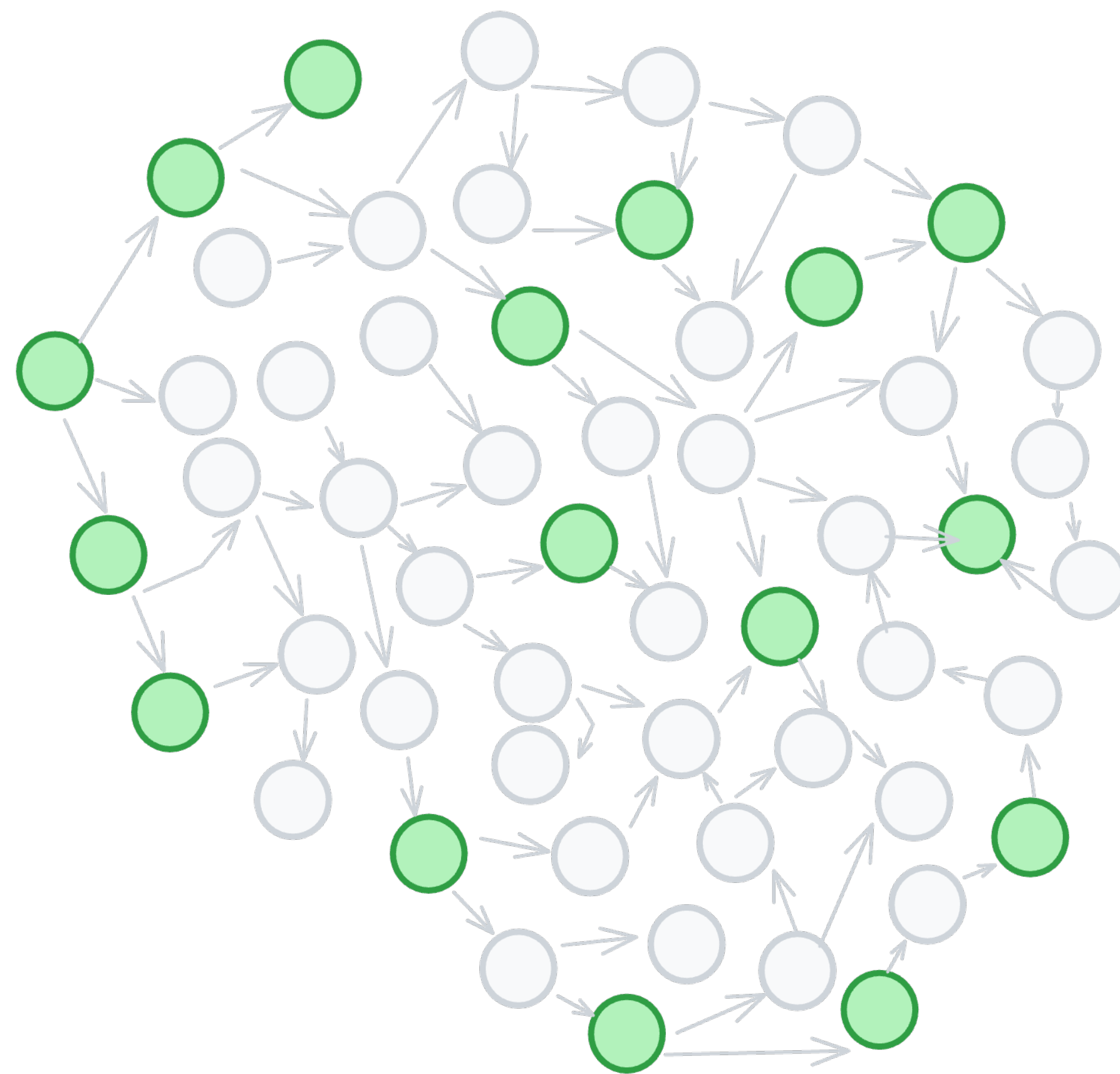
Examples → Exploration



Examples → Exploration

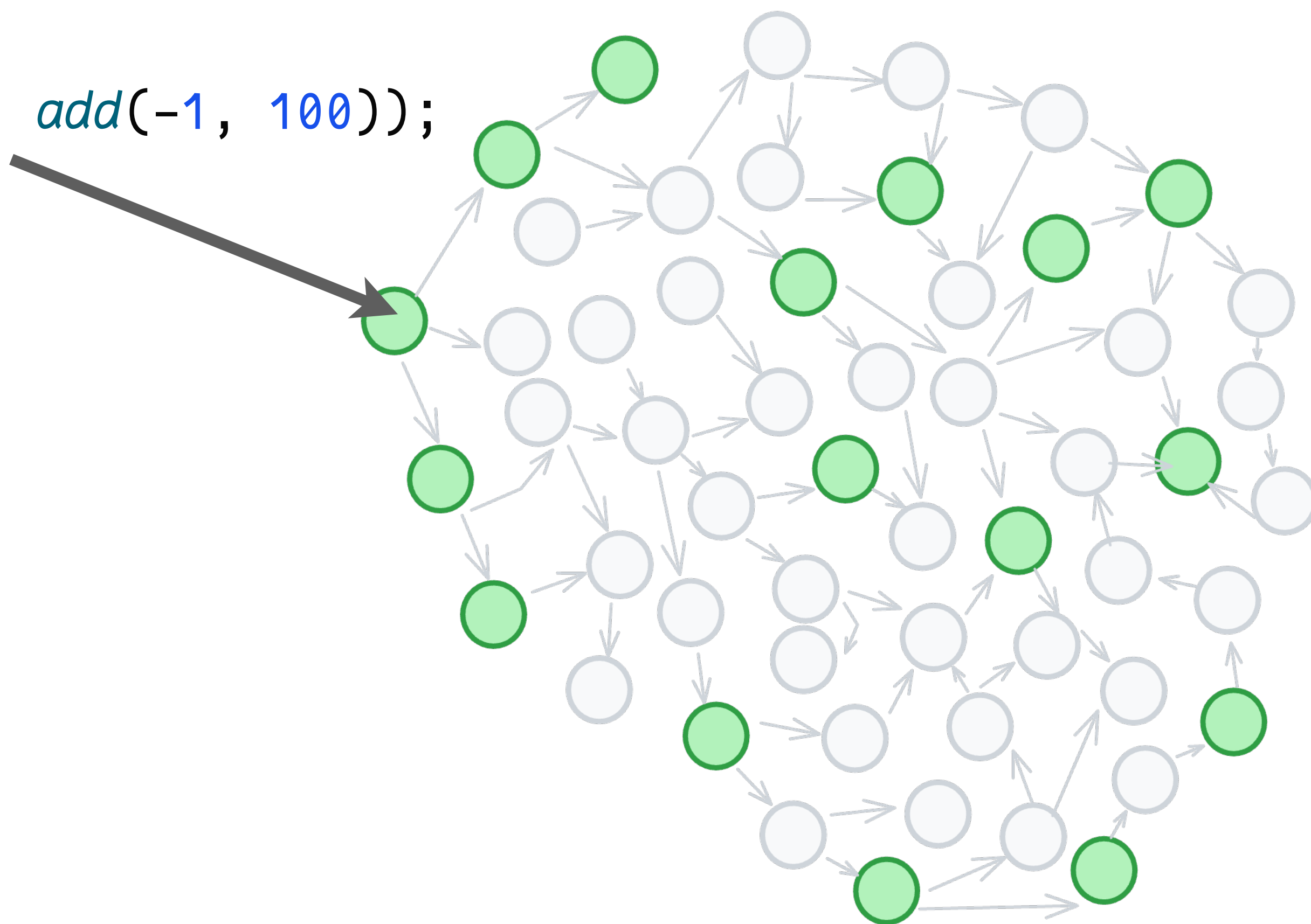


Examples → Exploration



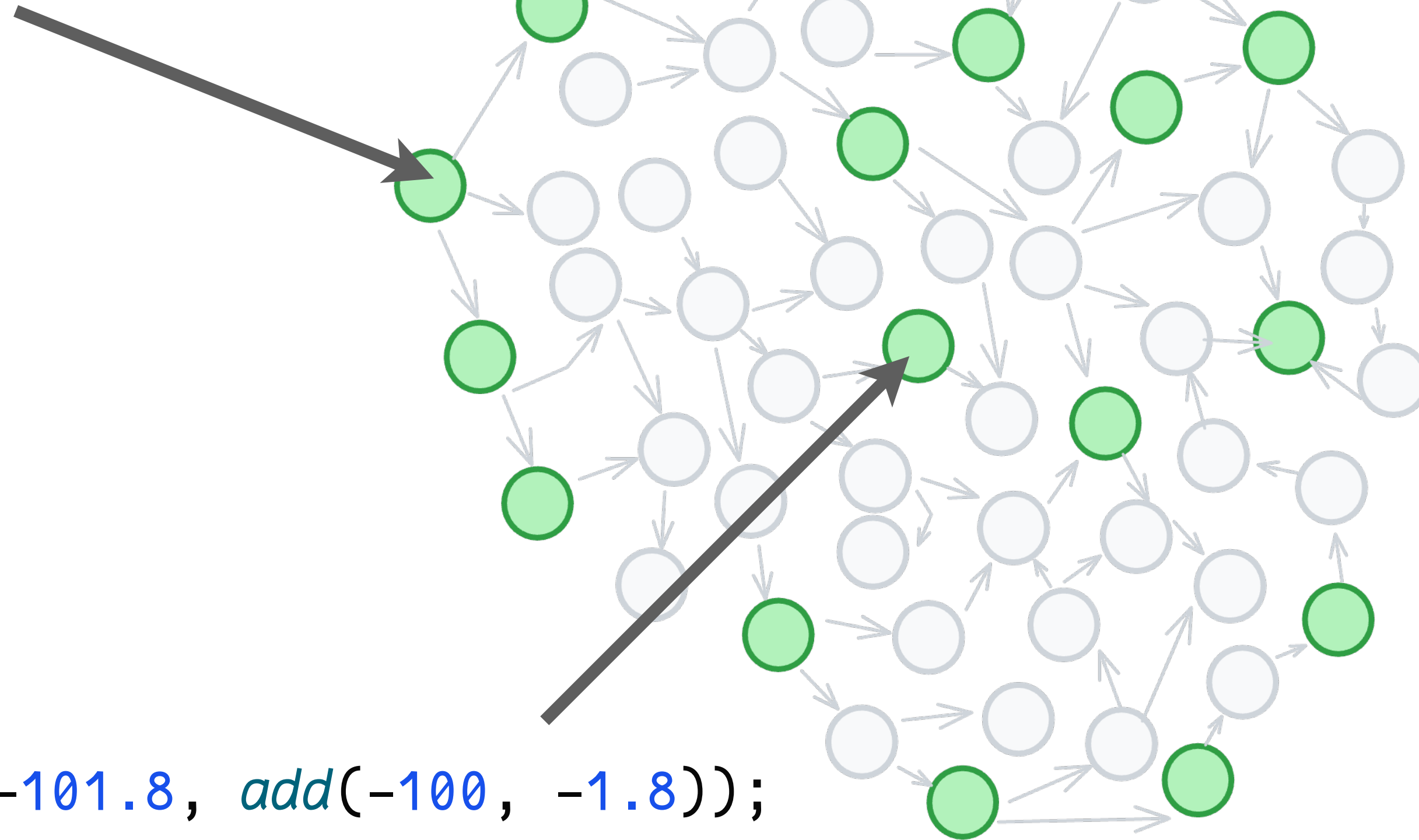
Examples → Exploration

`assertEquals(99, add(-1, 100));`



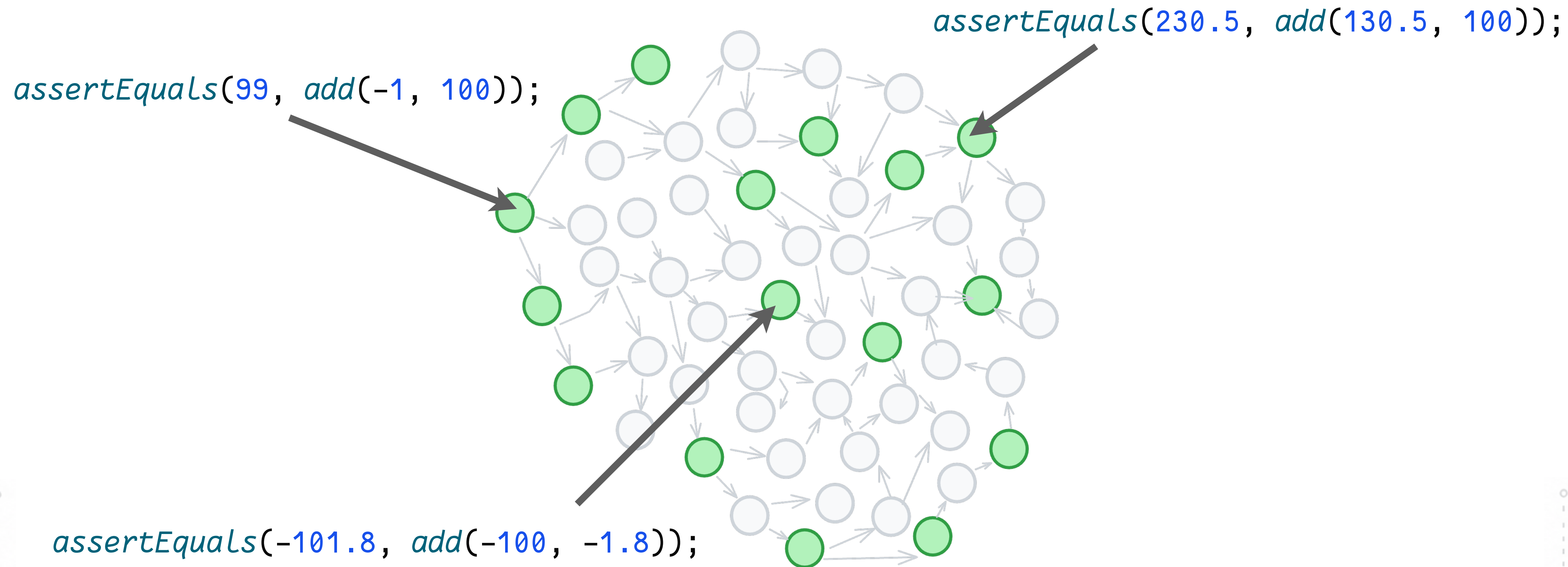
Examples → Exploration

`assertEquals(99, add(-1, 100));`

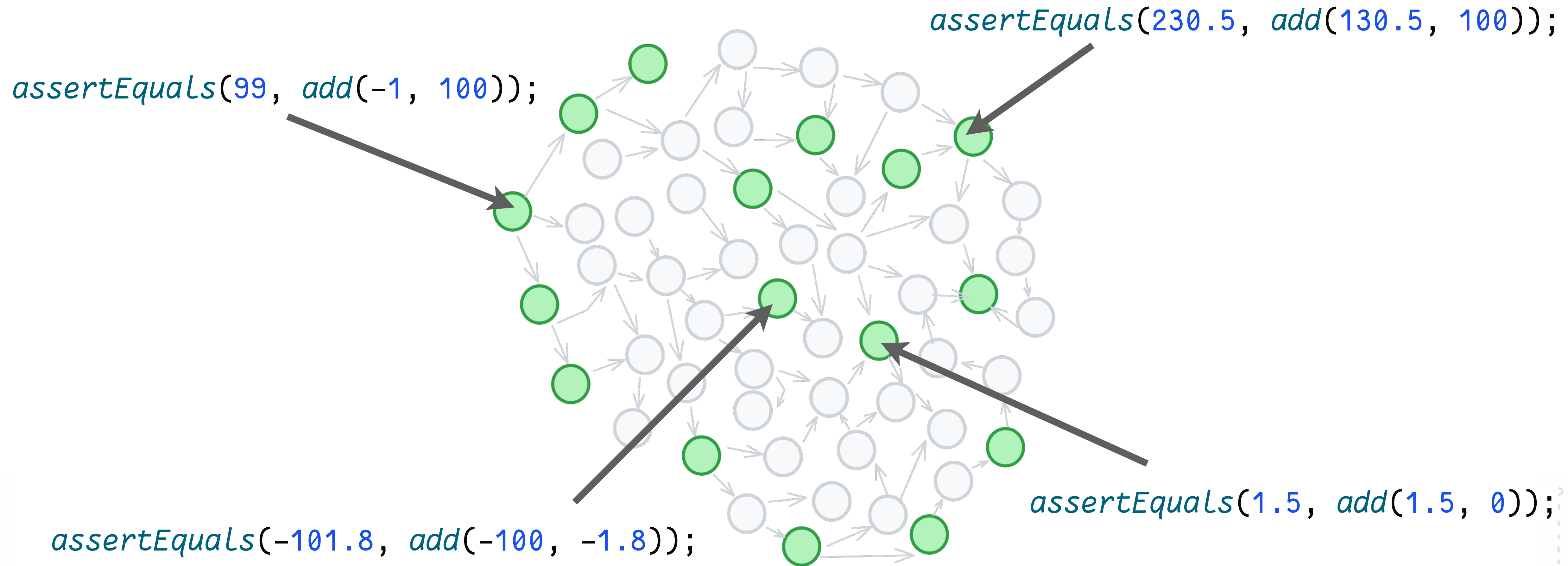


`assertEquals(-101.8, add(-100, -1.8));`

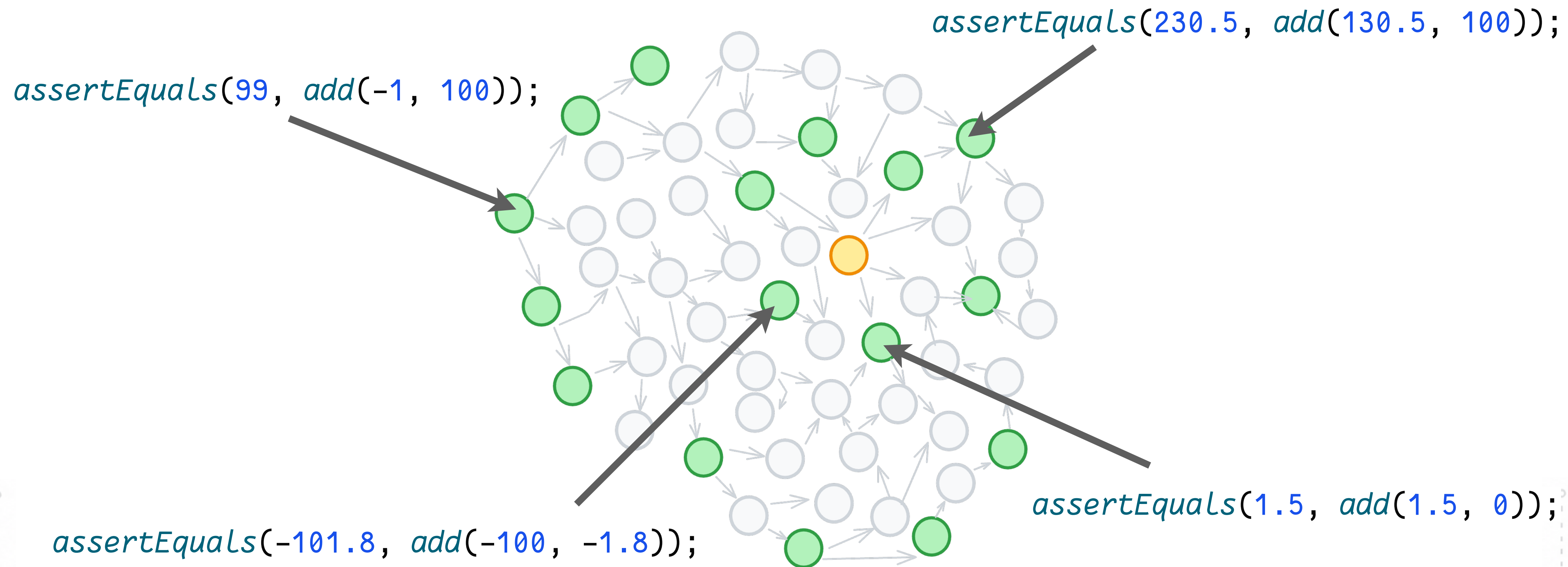
Examples → Exploration



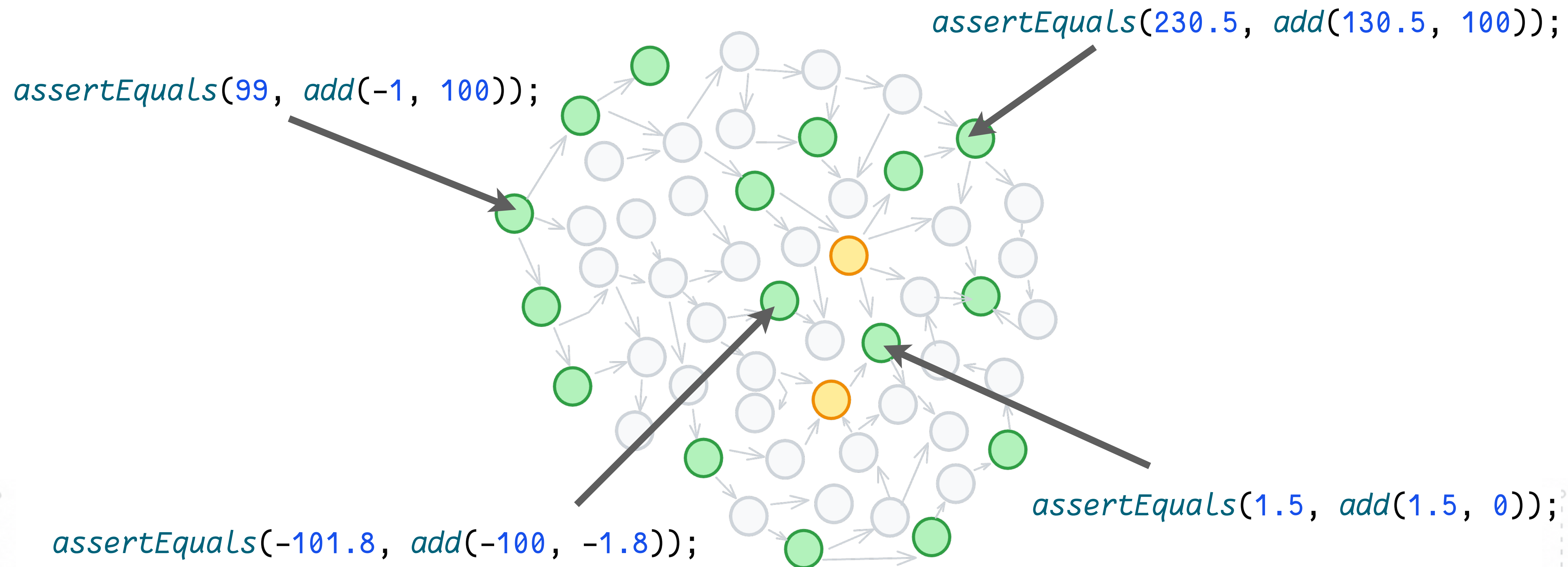
Examples → Exploration



Examples → Exploration

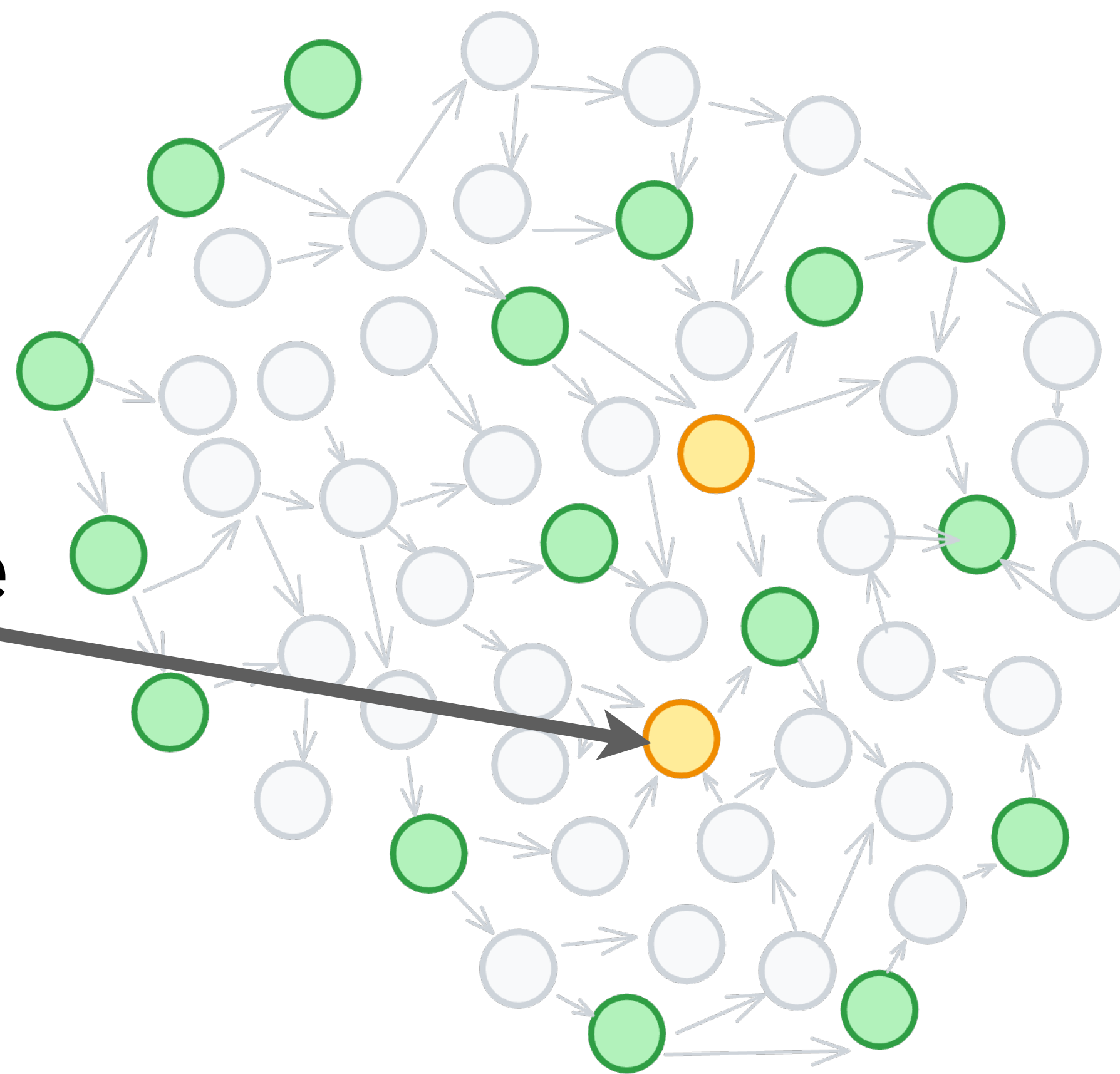


Examples → Exploration



Examples → Exploration

What if the LLM didn't write
a test case for this?





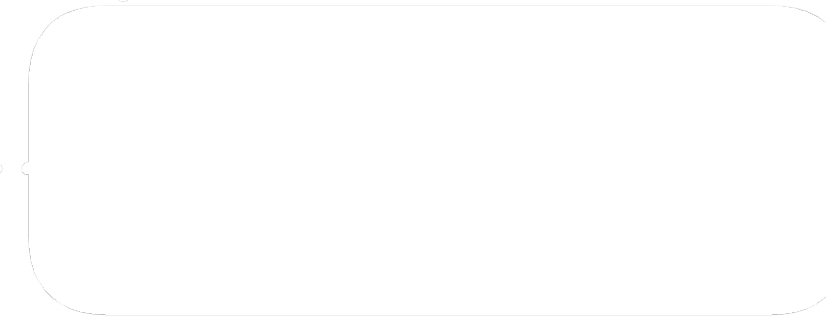
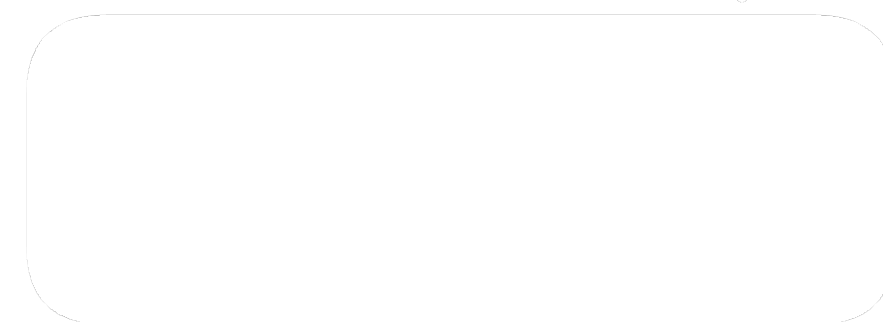
Examples → Exploration



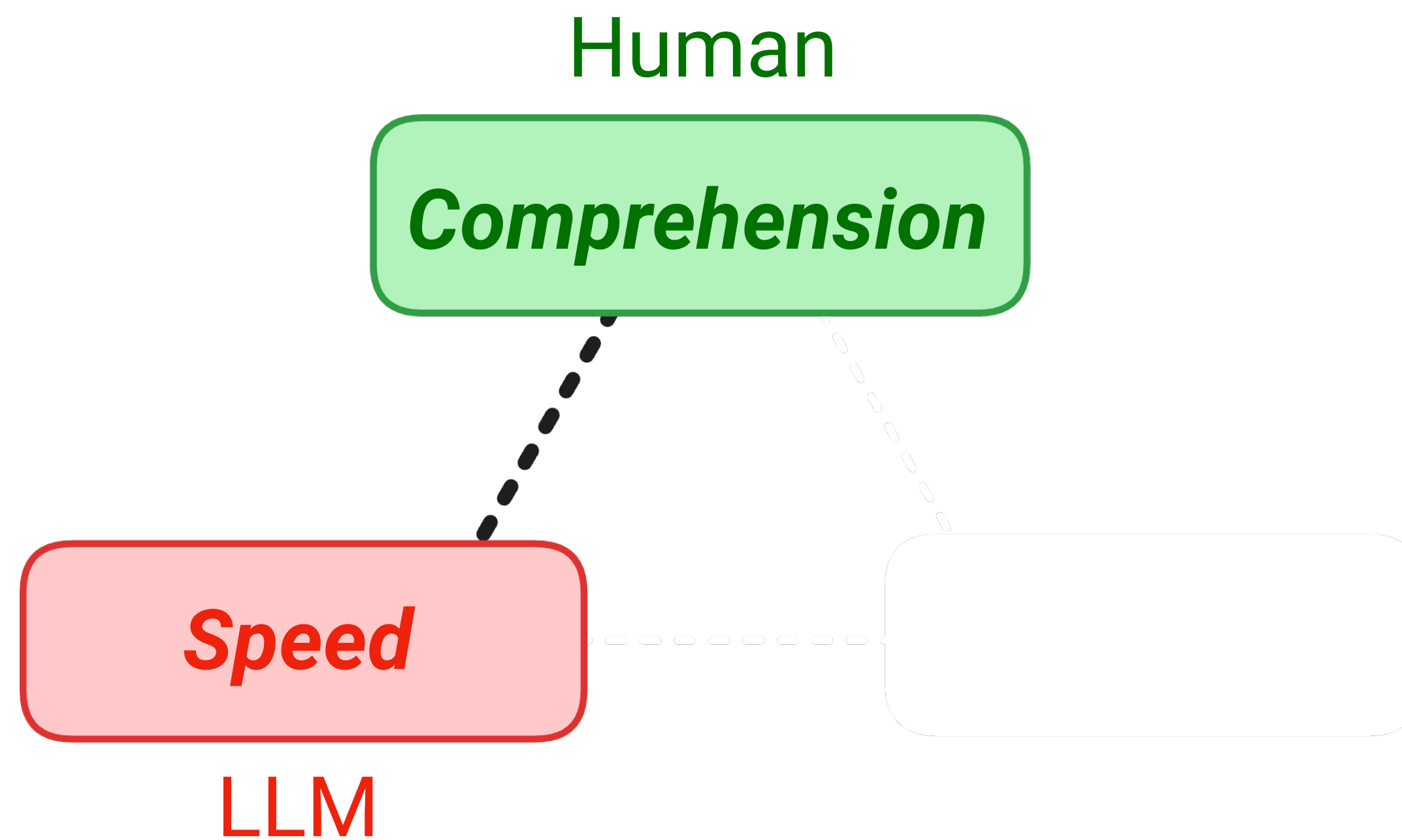
Examples → Exploration

Human

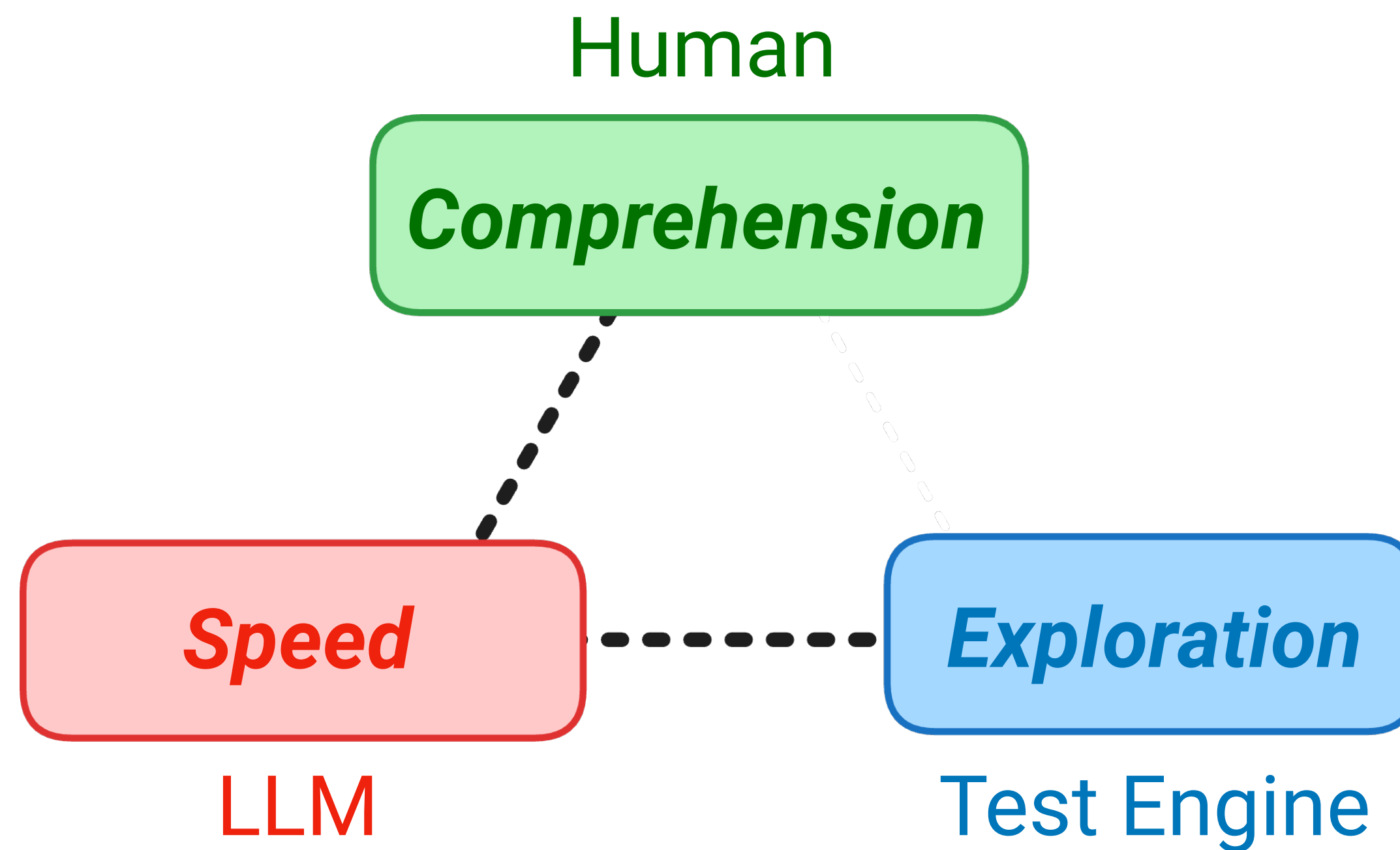
Comprehension



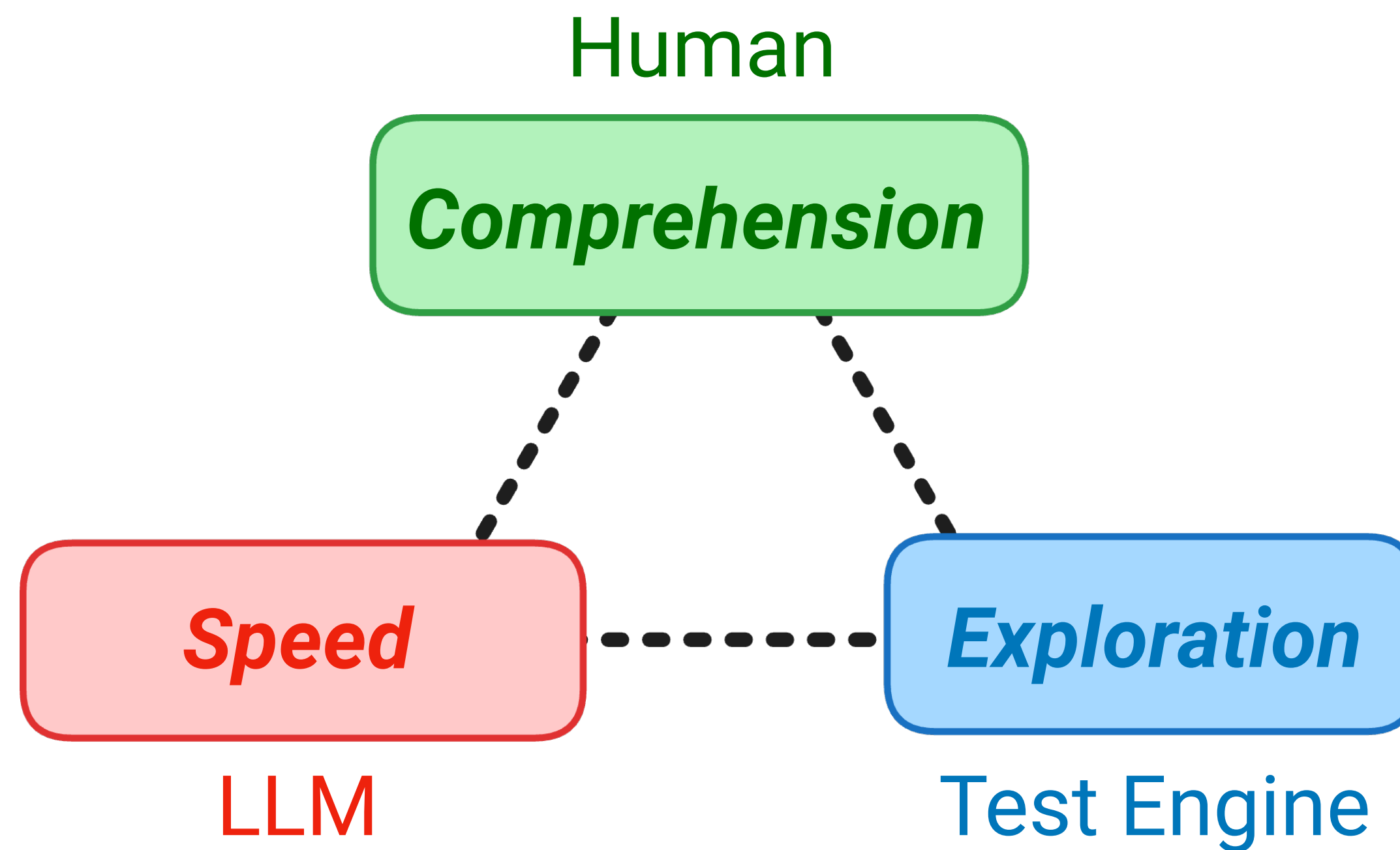
Examples → Exploration



Examples → Exploration



Examples → Exploration





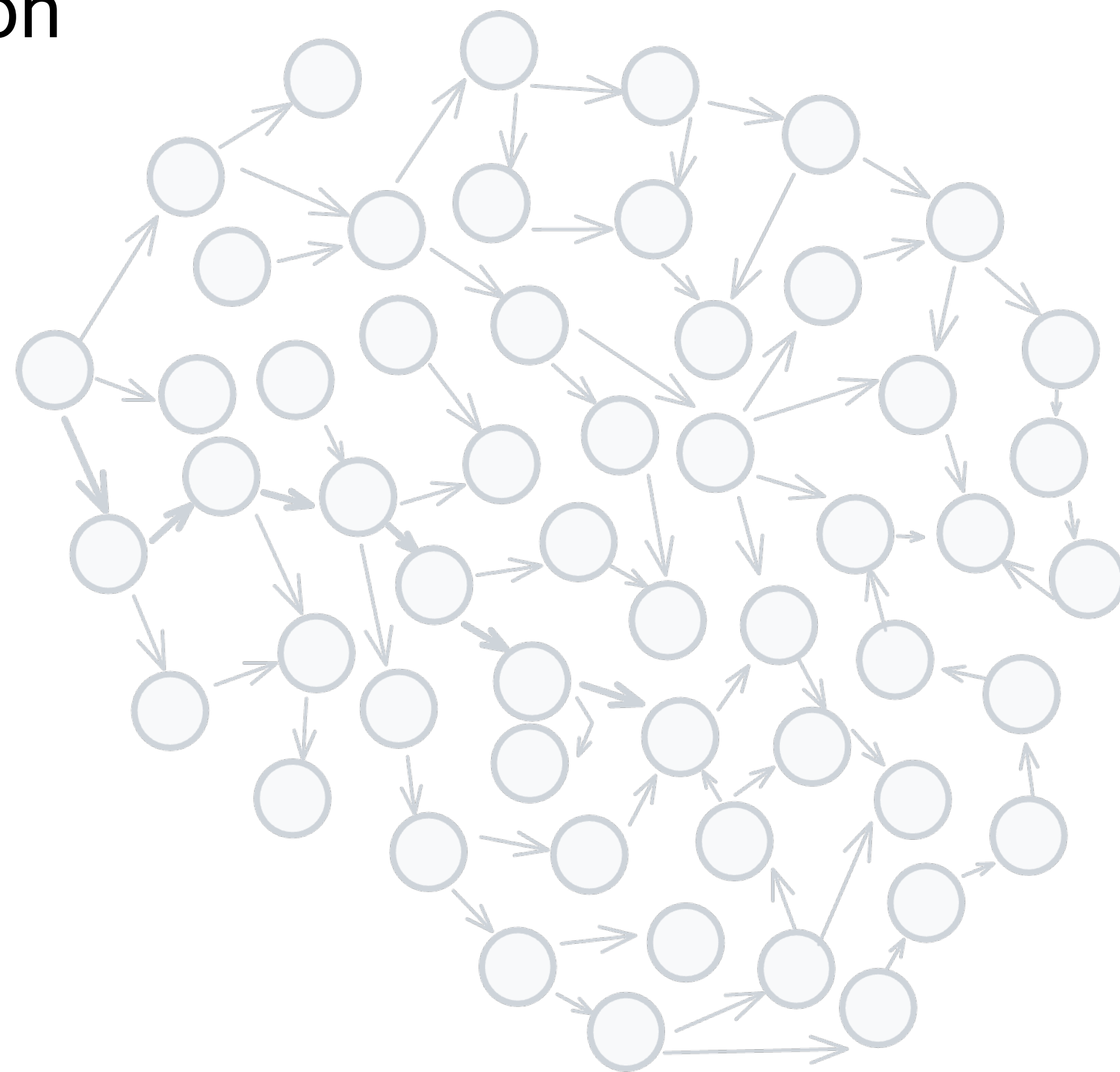
Examples → Exploration

LLMs generate a solution



Examples → Exploration

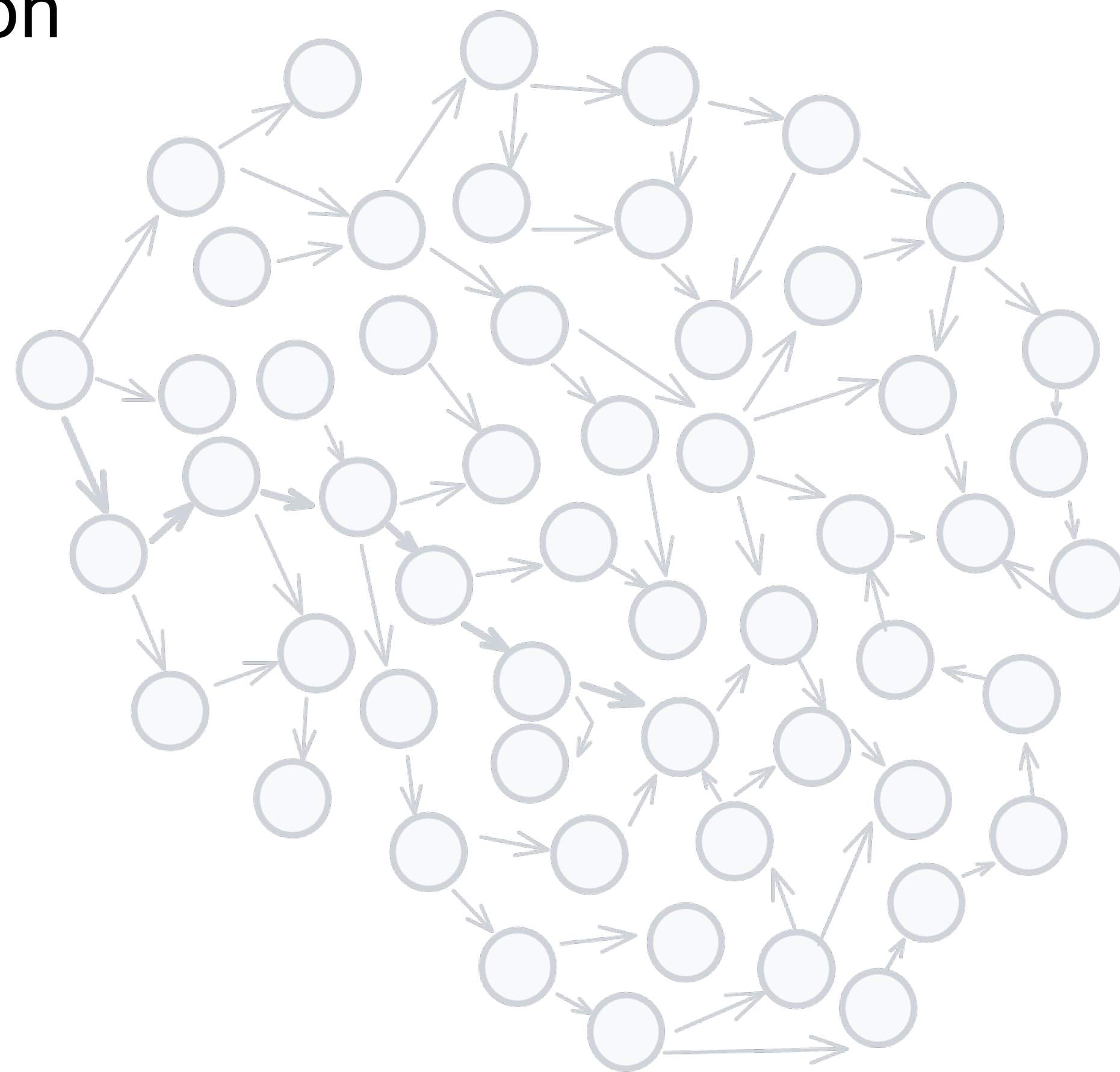
LLMs generate a solution



Examples → Exploration

LLMs generate a solution

Humans define what
must always hold true

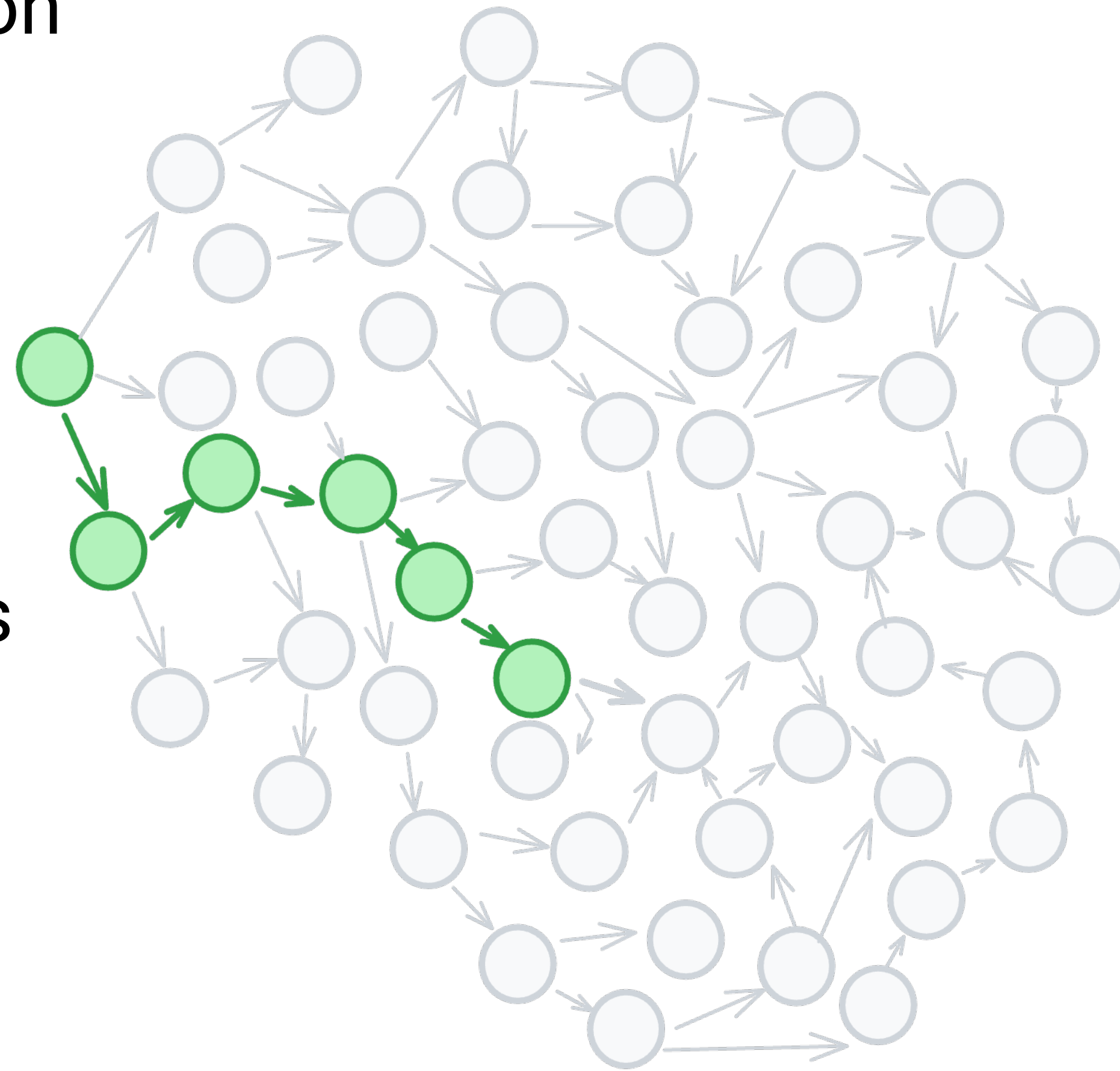


Examples → Exploration

LLMs generate a solution

Humans define what
must always hold true

The test engine explores
the problem space

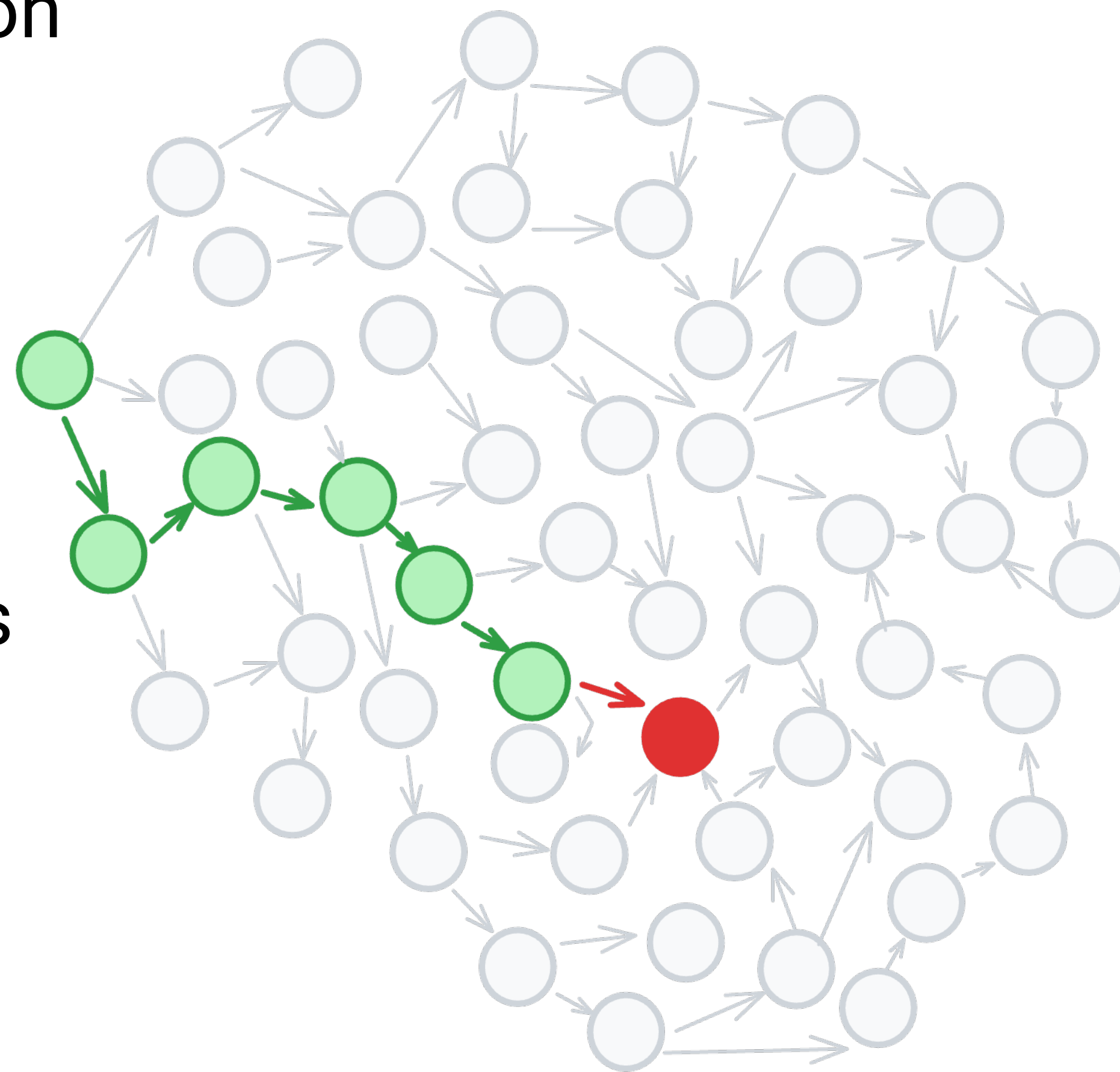


Examples → Exploration

LLMs generate a solution

Humans define what
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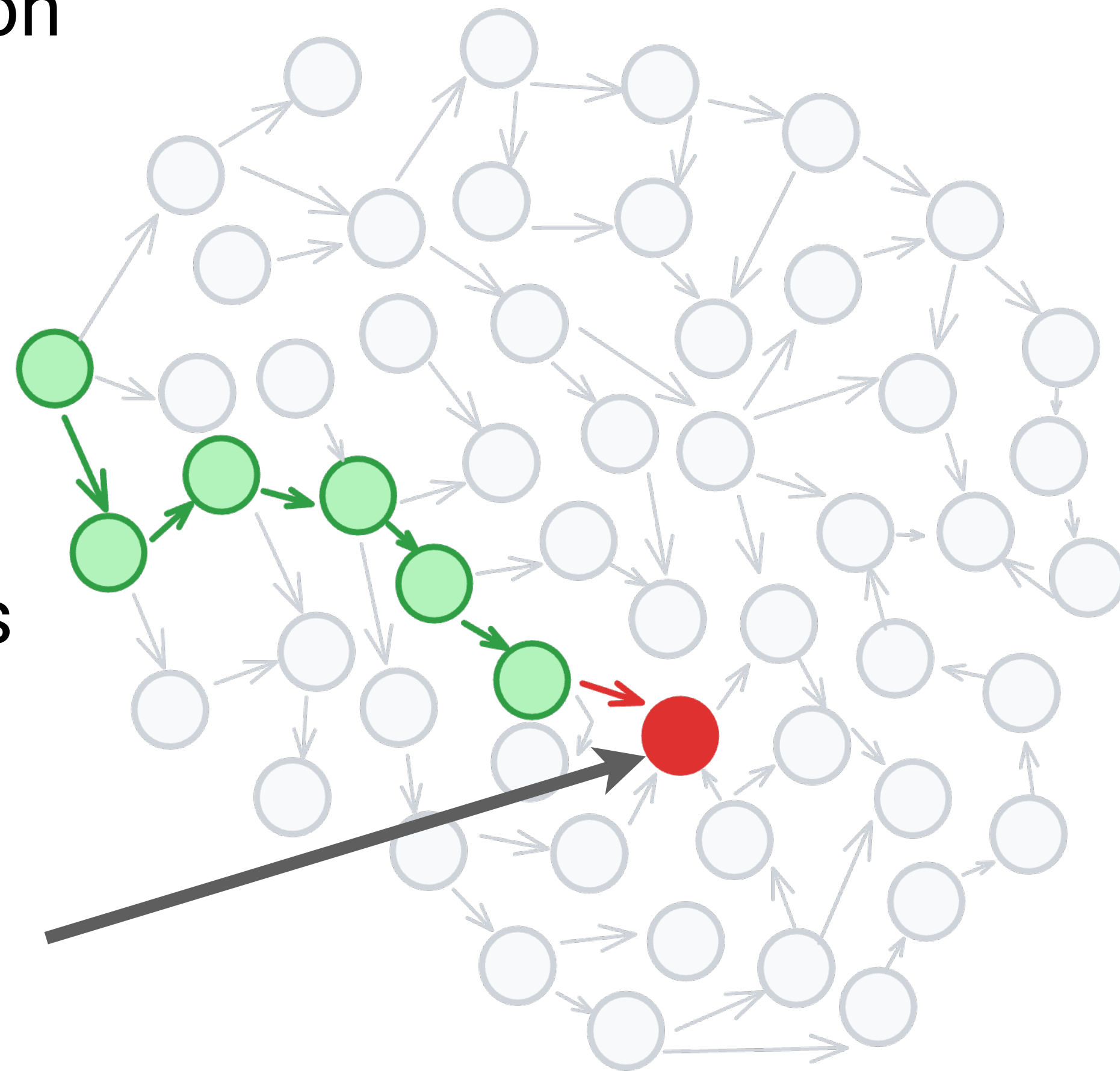
Examples → Exploration

LLMs generate a solution

Humans define what
must always hold true

The test engine explores
the problem space

The test engine finds
a counter example



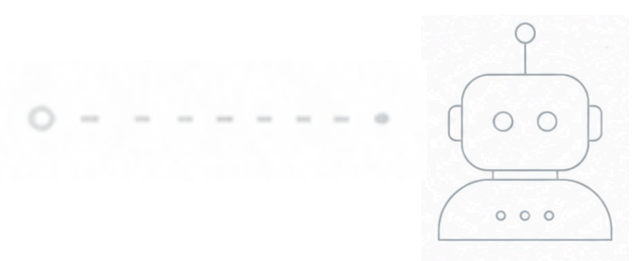


Failure



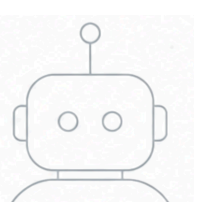
Failure → Shrinking





Failure → Shrinking → Understanding





Failure → Shrinking → Understanding

Original Sample

```
integers: [-778383, 138, -958125, -9, -2125, 1430, -2, 16, 1, -50110, -91381,
           249644599, -5995, 2396, -1997, 24, 23, -121416, -868149, -38934,
           8045, -373, 147465, -3, -58, 514, -177, 97, 1386116950, 2953, 3804]
```

Expecting actual:

1386116950

to be less than or equal to:

3804



Failure → Shrinking → Understanding

Original Sample

```
integers: [-778383, 138, -958125, -9, -2125, 1430, -2, 16, 1, -50110, -91381,  
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```

Expecting actual:

1386116950

to be less than or equal to:

3804

Failure → Shrinking → Understanding

Shrunk Sample (11 steps)

integers: [0, -1]

Expecting actual:

0

to be less than or equal to:

-1

Original Sample

integers: [-778383, 138, -958125, -9, -2125, 1430, -2, 16, 1, -50110, -91381,
249644599, -5995, 2396, -1997, 24, 23, -121416, -868149, -38934,
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Failure → Shrinking → Understanding

Shrunk Sample (11 steps)

integers: [0, -1]

Expecting actual:

0

to be less than or equal to:

-1

```
for (int i = 0; i < n - 1; i++) {  
    for (int j = 0; j < n - i - 2; j++) {  
        if (arr.get(j) > arr.get(j + 1)) {  
            int temp = arr.get(j);  
            arr.set(j, arr.get(j + 1));  
            arr.set(j + 1, temp);  
        }  
    }  
}
```

Failure → Shrinking → Understanding

Shrunk Sample (11 steps)

integers: [0, -1]

Expecting actual:

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for (int i = 0; i < n - 1; i++) {  
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            arr.set(j + 1, temp);  
        }  
    }  
}
```

Last item never gets sorted

Failure → Shrinking → Understanding

Shrunk Sample (11 steps)

integers: [0, -1]

Expecting actual:

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            arr.set(j, arr.get(j + 1));  
            arr.set(j + 1, temp);  
        }  
    }  
}
```

Last item never gets sorted
Shrunk samples are easier to **understand**



Could LLMs Do It All?



Could LLMs Do It All?

From Prompts to Properties: Rethinking LLM Code Generation with Property-Based Testing

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Virginia Tech
Blacksburg, Virginia, USA

Abstract

Large Language Models (LLMs) have shown promise in automated code generation, but ensuring correctness remains a significant challenge. Traditional unit testing evaluates functional correctness but often fails to capture deeper logical constraints. We apply Property-Based Testing (PBT) as an alternative evaluation strategy to StarCoder and CodeLlama on MBPP and HumanEval. Our results reveal that while pass@k evaluation shows moderate success, PBT exposes additional correctness gaps. A significant portion of generated solutions only partially adhere to correctness properties (30–32%), while 18–23% fail outright. Property extraction is also imperfect, with 9–13% of constraints missing. These findings highlight that unit test-based evaluations may overestimate solution correctness by not capturing fundamental logical errors. Our study demonstrates that combining unit testing with PBT can offer a more comprehensive assessment of generated code correctness, revealing limitations that traditional verification approaches miss.

CCS Concepts

• **Software and its engineering** → **Functionality**.

Keywords

Code generation, property-based-testing, Large language model(LLM)

ACM Reference Format:

Dibyendu Brinto Bose. 2025. From Prompts to Properties: Rethinking LLM Code Generation with Property-Based Testing. In *33rd ACM International Conference on the Foundations of Software Engineering (FSE Companion '25)*, June 23–28, 2025, Trondheim, Norway. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3696630.3728702>

1 Introduction

Recent advances in Large Language Models (LLMs) have transformed software engineering. For instance, LLMs are increasingly used to support software development tasks [20, 24], such as code

by GitHub Copilot.¹ However, LLM-generated code can face numerous issues, such as hallucination [12], security [23] and miss important functionalities [2]. Thus, techniques are needed to assess the quality of LLM-generated code.

The predominant method for evaluating and refining code generated by LLMs relies on unit testing [17, 18], which primarily verifies specific input-output pairs. While unit tests serve as a useful validation tool, they capture only a limited aspect of computational thinking. This approach overlooks the deeper structural and behavioral correctness of the code, which is essential for robust software development [3]. As a result, LLM-generated code may pass unit tests while still containing fundamental logical errors, failing in edge cases or unforeseen scenarios [15].

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Source: <https://dl.acm.org/doi/10.1145/3696630.3728702>, presented at FSE 2025

Could LLMs Do It All?

- LLMs for property extraction

From Prompts to Properties: Rethinking LLM Code Generation with Property-Based Testing

Dibyendu Brinto Bose
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Virginia Tech
Blacksburg, Virginia, USA

Abstract

Large Language Models (LLMs) have shown promise in automated code generation, but ensuring correctness remains a significant challenge. Traditional unit testing evaluates functional correctness but often fails to capture deeper logical constraints. We apply Property-Based Testing (PBT) as an alternative evaluation strategy to StarCoder and CodeLlama on MBPP and HumanEval. Our results reveal that while pass@k evaluation shows moderate success, PBT exposes additional correctness gaps. A significant portion of generated solutions only partially adhere to correctness properties (30–32%), while 18–23% fail outright. Property extraction is also imperfect, with 9–13% of constraints missing. These findings highlight that unit test-based evaluations may overestimate solution correctness by not capturing fundamental logical errors. Our study demonstrates that combining unit testing with PBT can offer a more comprehensive assessment of generated code correctness, revealing limitations that traditional verification approaches miss.

CCS Concepts

• Software and its engineering → Functionality.

Keywords

Code generation, property-based-testing, Large language model(LLM)

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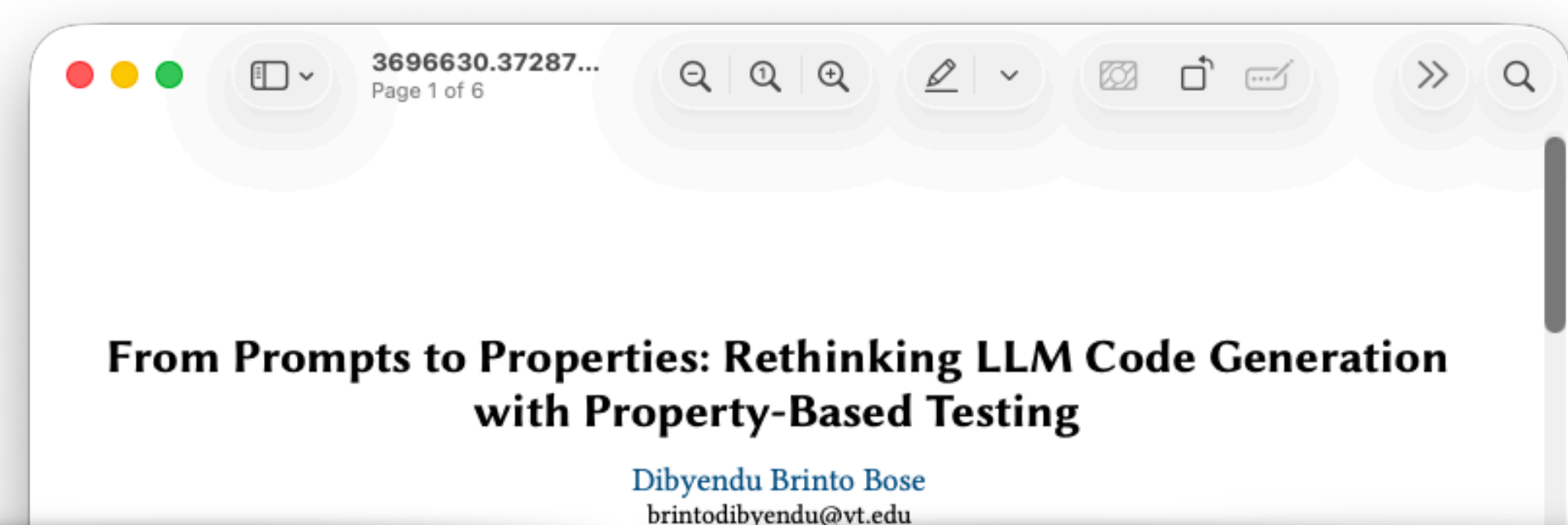
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- LLMs for property extraction
- LLMs for property compliance



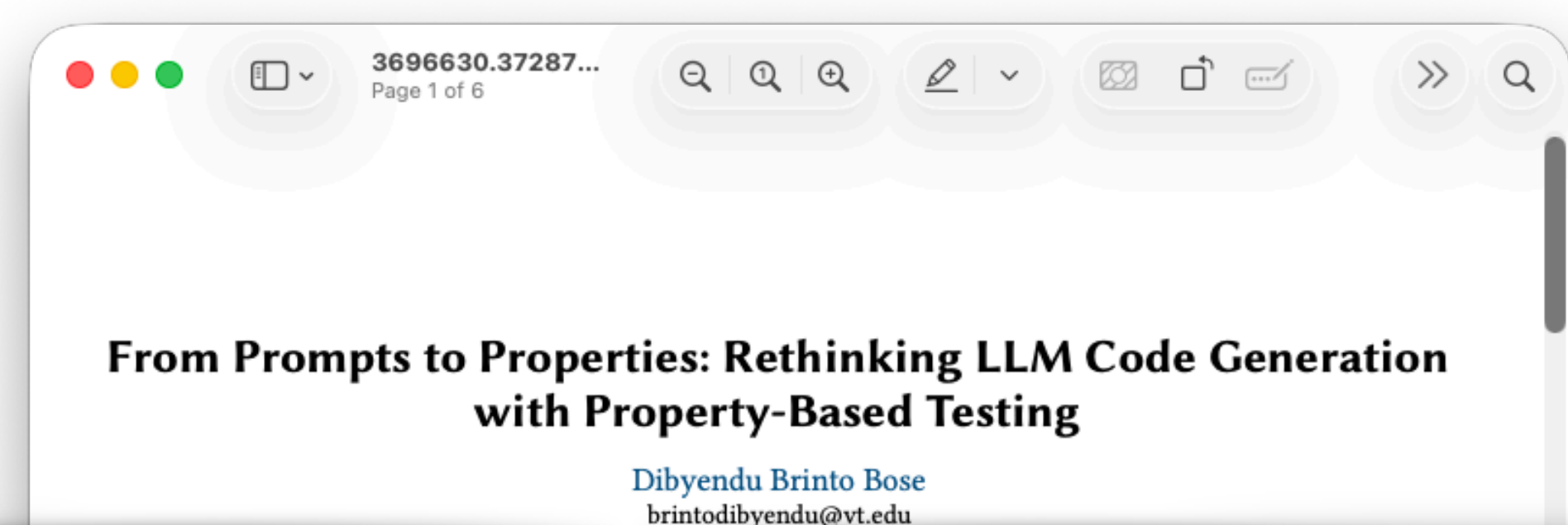
From Prompts to Properties: Rethinking LLM Code Generation with Property-Based Testing

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Implications Developers increasingly use LLMs to generate code, yet report lacking trust in the accuracy of generated output [1]. Our results have important implications for validating LLM-generated code. First, unit test-based evaluation alone provides an incomplete correctness measure, failing to capture deeper structural flaws. PBT offers a more nuanced perspective by enforcing logical constraints, ensuring that generated code adheres to expected behaviors beyond input-output mappings. However, the inconsistency in property extraction suggests the need for improved property refinement mechanisms, such as ranking, filtering, or human-in-the-loop validation, to enhance the quality of extracted constraints. In addition, LLMs struggle with strict property adherence, indicating that more effective mechanisms are needed to guide models toward better constraint integration.

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Could LLMs Do It All?

- LLMs for property extraction
 - 76% valid property extraction
- LLMs for property compliance

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Table 2: Property Extraction Performance: Valid, Redundant, and Missing Properties Across MBPP and HumanEval datasets (100 samples).

Model	Dataset	Valid (%)	Redundant (%)	Missing (%)
StarCoder	MBPP	76	15	9
CodeLlama	MBPP	72	18	10
StarCoder	HumanEval	74	14	12
CodeLlama	HumanEval	70	17	13

Code generation, property-based-testing, Large language model(LLM)

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Could LLMs Do It All?

- LLMs for property extraction
 - 76% valid property extraction
- LLMs for property compliance
 - 52% property compliance

3696630.37287...
Page 1 of 6

From Prompts to Properties: Rethinking LLM Code Generation with Property-Based Testing

Dibyendu Brinto Bose
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Table 3: Property-Based Testing Compliance: Fully Passed, Partially Passed, and Failed Cases Across MBPP and HumanEval datasets (100 samples).

Model	Dataset	Property-Based Testing Compliance (%)		
		Fully Passed	Partially Passed	Failed
StarCoder	MBPP	52	30	18
CodeLlama	MBPP	48	32	20
StarCoder	HumanEva	49	29	22
CodeLlama	HumanEva	43	31	23

Code Generation with Property-Based Testing. In *33rd ACM International Conference on the Foundations of Software Engineering (FSE Companion '25)*, June 23–28, 2025, Trondheim, Norway. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3696630.3728702>

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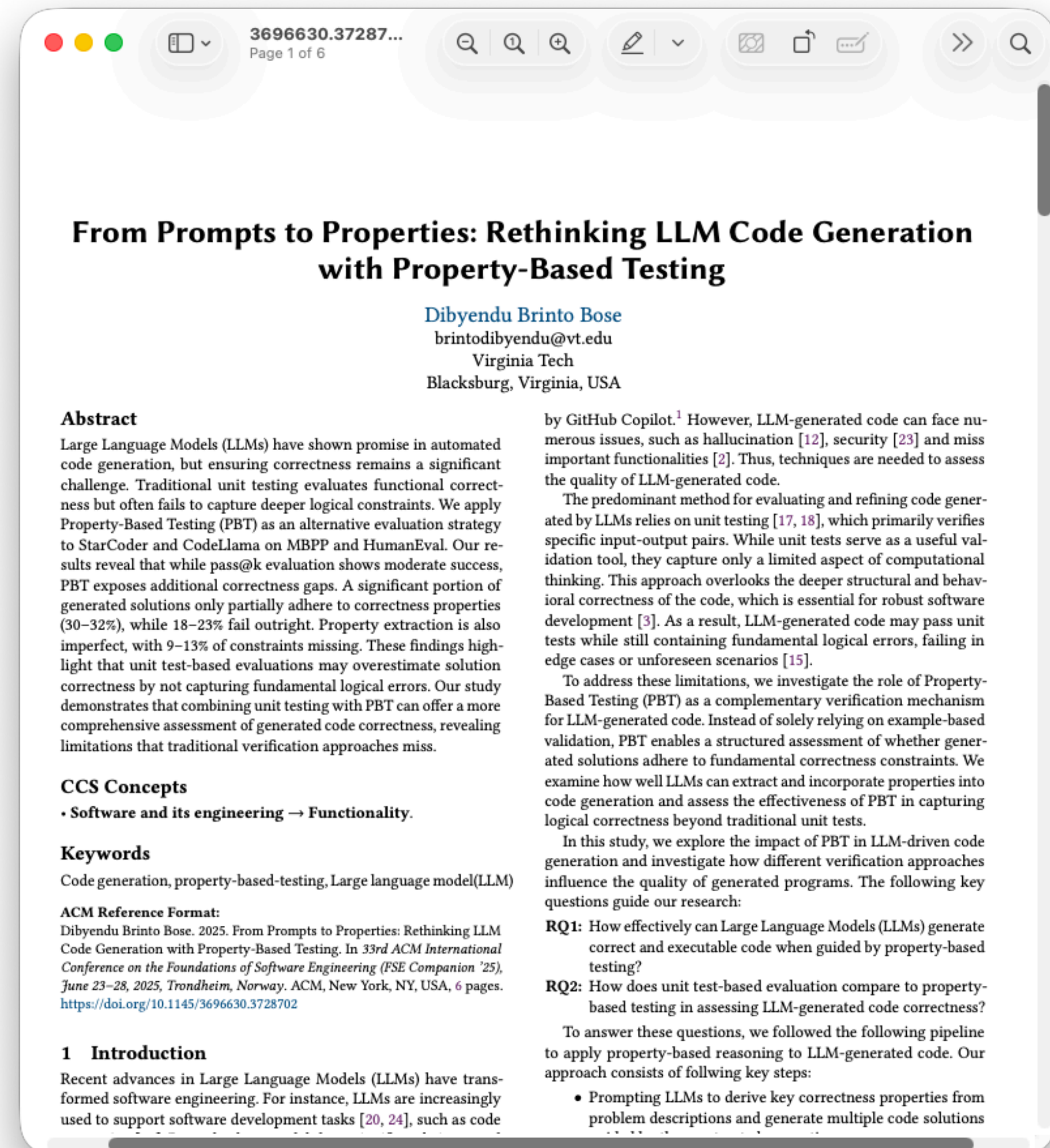
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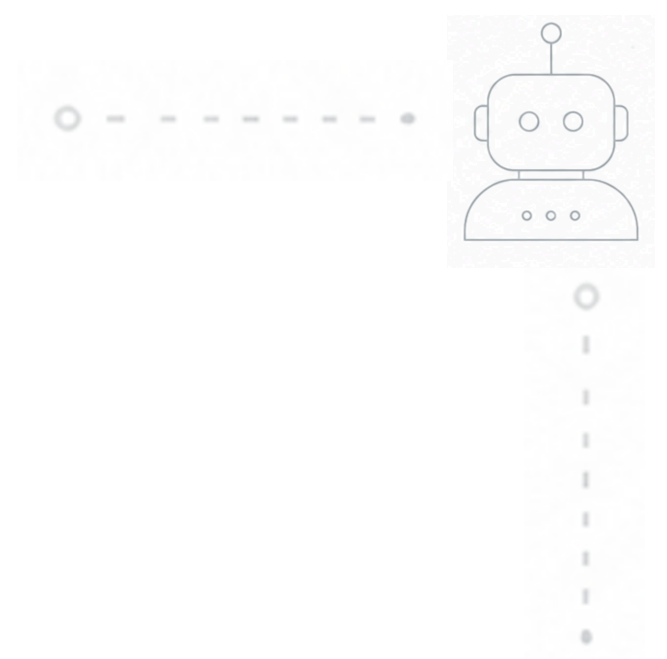
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Could LLMs Do It All?

- LLMs for property extraction
 - 76% valid property extraction
- LLMs for property compliance
 - 52% property compliance
- Quality still probabilistic
 - Human comprehension is essential

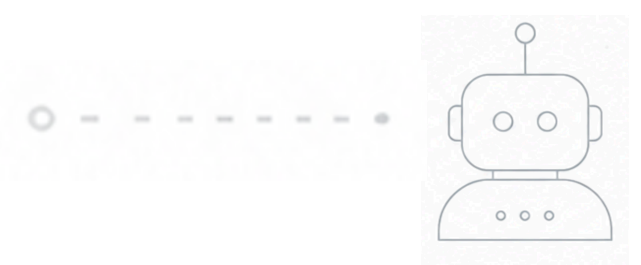
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How Did I Miss That?





Meeting Scheduling





Meeting Scheduling

Existing Meeting





Meeting Scheduling

Existing Meeting

New Meeting





Meeting Scheduling

Existing Meeting

New Meeting

New Meeting





Meeting Scheduling

Existing Meeting

New Meeting





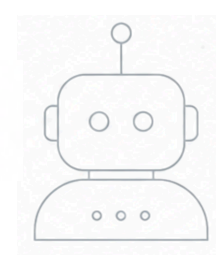
Meeting Scheduling

Existing Meeting

New Meeting

New Meeting





Meeting Scheduling

Existing Meeting

New Meeting

New Meeting

New Meeting



Meeting Scheduling

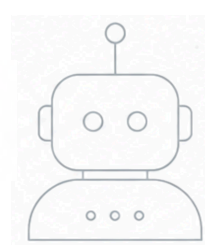
Calendar 1

Existing Meeting

New Meeting

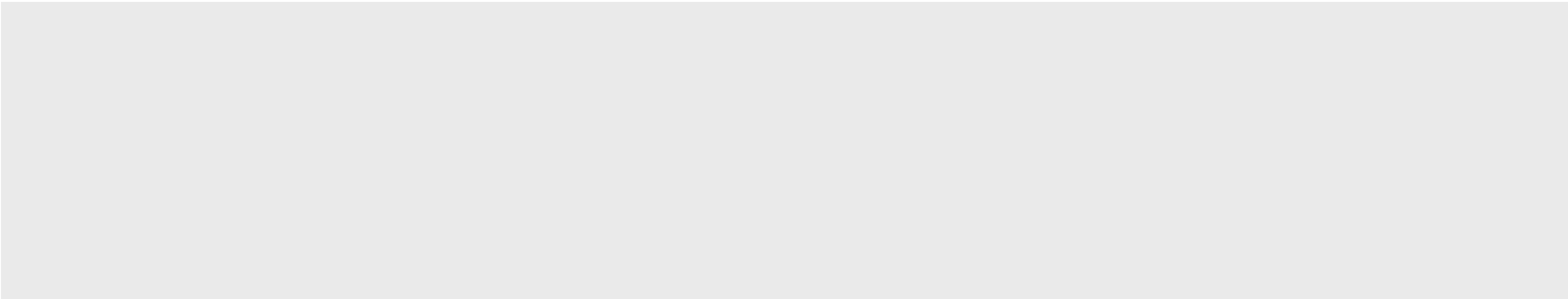
New Meeting

New Meeting

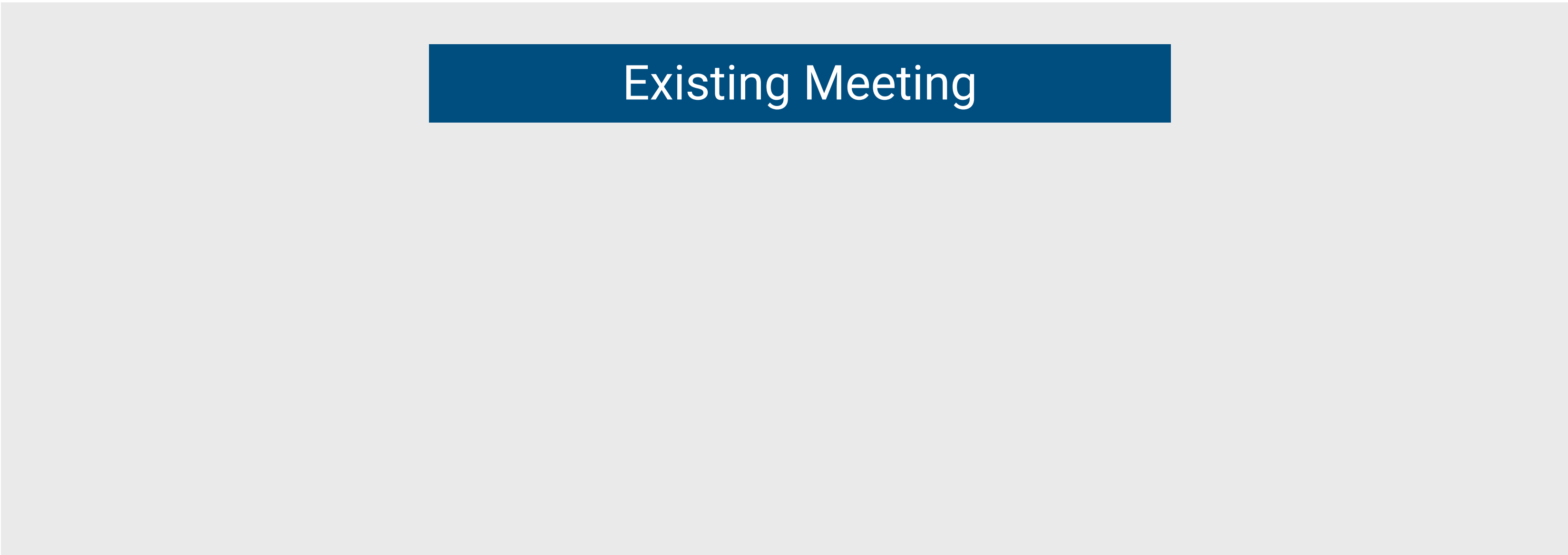


Meeting Scheduling

Calendar 2



Calendar 1



Existing Meeting



Meeting Scheduling

Calendar 2

New Meeting

Calendar 1

Existing Meeting

Meeting Scheduling

Calendar 2

New Meeting

Calendar 1

~~Existing Meeting~~

**Auto-reject lower
priority conflicts**

Meeting Scheduling, Vibe Coded



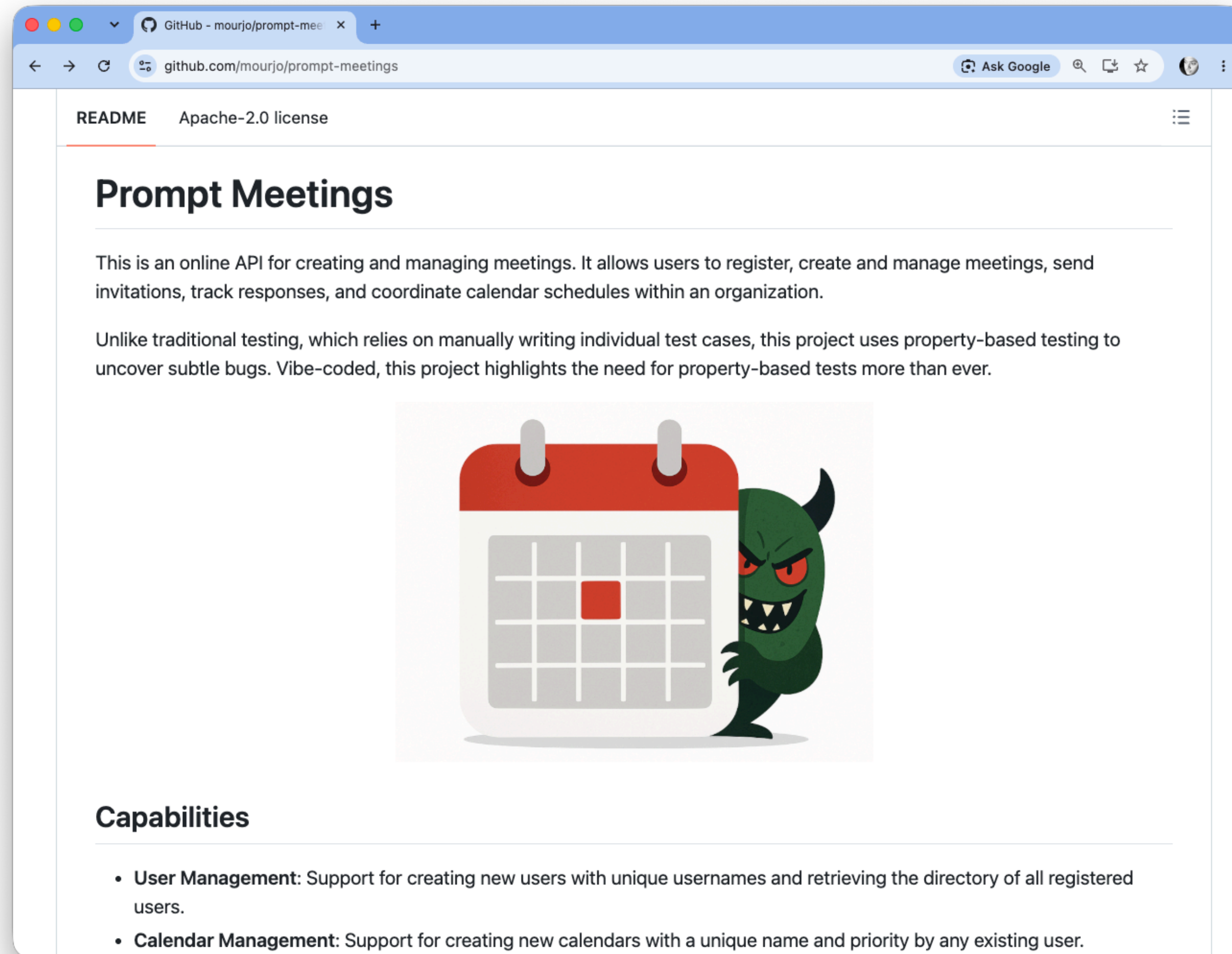
Antigravity CLI 1.1.27
~/.repos/prompt-meetings@gmail.com (Google AI Pro)
Claude Opus 4.6 (Thinking)
~/repos/prompt-meetings

> This application called "prompt-meetings" is designed to be used for creating/managing meetings. Create the following endpoints:

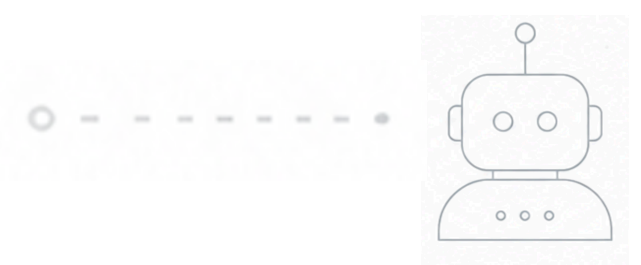
- create a user with a unique username
- view all users in the system
- create a meeting with a start and end time
- invite other users by user name to a meeting
- accept an invite
- reject and invite
- view pending invitations for a user
- view all meetings
- view all pending invitations received by a user

Claude Opus 4.6 (Thinking)

Meeting Scheduling, Vibe Coded

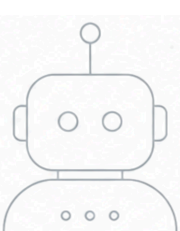


Source: github.com/mourjo/prompt-meetings



Meeting Scheduling, Vibe Coded





Meeting Scheduling, Vibe Coded

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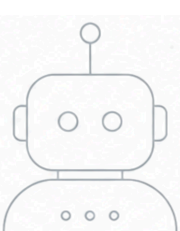
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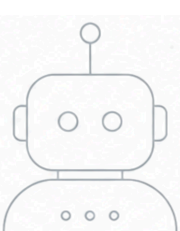
Meeting Scheduling, Vibe Coded

“

*I want to introduce now a **concept of calendars**. A calendar contains meetings. Every meeting must be part of a calendar. Calendars have priorities denoted by a floating point number. Migrate the existing meetings to a default calendar. Update the database schema. Update the create-meeting endpoints...*

”





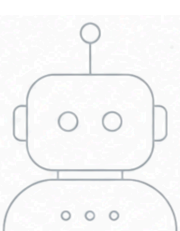
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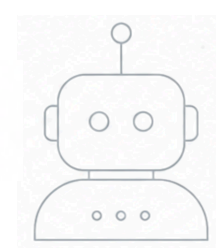
Meeting Scheduling, Vibe Coded

“

Now I want to use the priority of calendars – when a meeting is being created, check if any of my meetings are conflicting with this. If the conflicting meeting is originating from a calendar of a lower priority, reject that meeting. If the conflict is coming from a calendar of the same priority or higher, do not allow its creation...

”





Meeting Scheduling, Vibe Coded

```
$ mvn clean test
```

```
[INFO] -----
```

```
[INFO] T E S T S
```

```
[INFO] -----
```

```
...
```

```
[INFO] Results:
```

```
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0
```

```
[INFO] -----
```

```
[INFO] BUILD SUCCESS
```

```
[INFO] -----
```

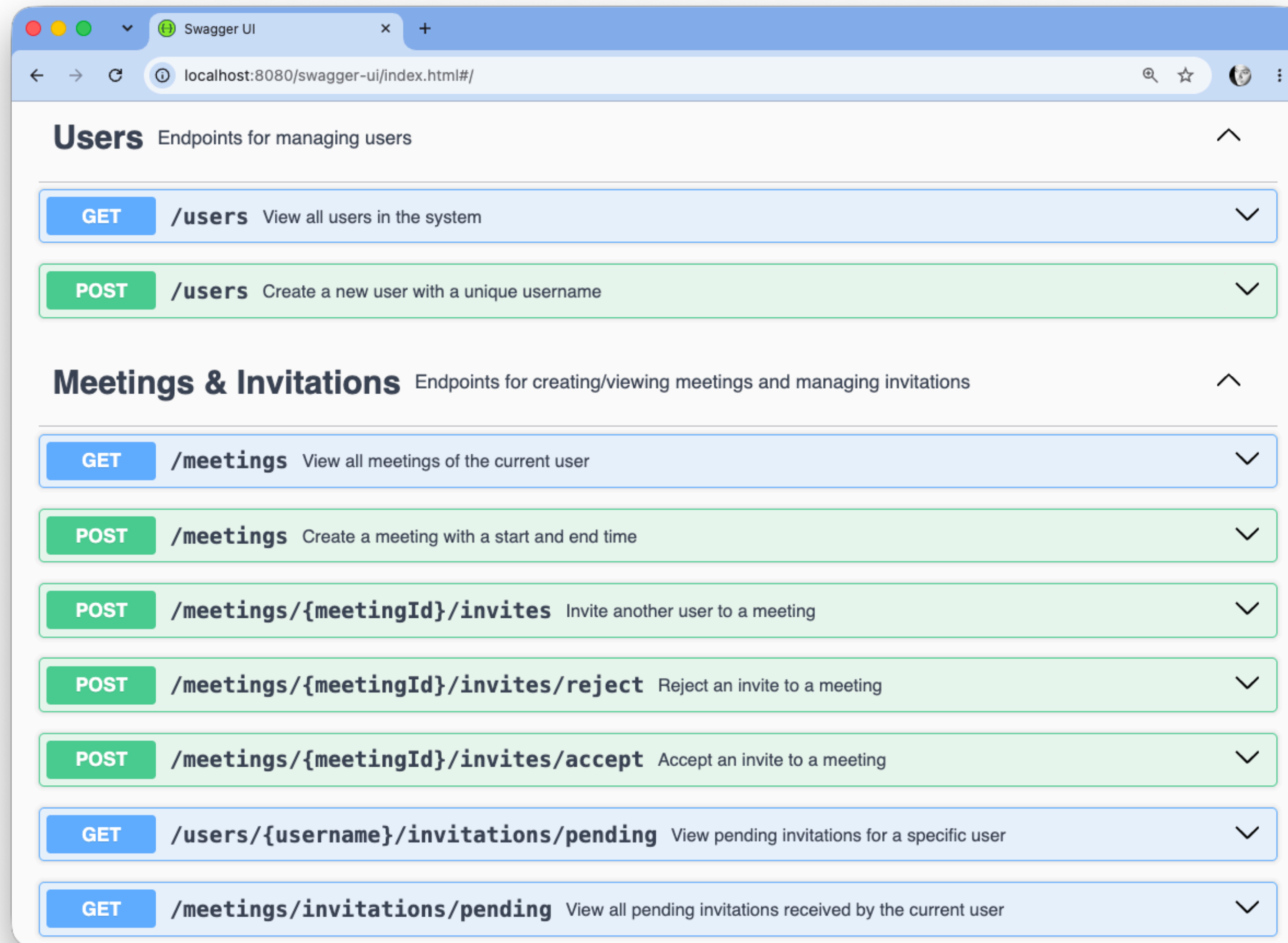
```
[INFO] Total time: 6.349 s
```

```
[INFO] Finished at: 2026-08-16T17:54:28+02:00
```

```
[INFO] -----
```



Meeting Scheduling, Vibe Coded



Source: github.com/mourjo/prompt-meetings

Meeting Scheduling, Vibe Coded





Meeting Scheduling, Vibe Coded

“

I want to add property based tests using jqwik. Use version 1.9.3. Use the spring boot extension.

- There should be 4 users in the system (this should be a constant, do this in the setup)*
- There should be 3 calendars of priorities default, medium, high, this should be a constant, do this in the setup...*

”





Meeting Scheduling, Vibe Coded

“

...the property based test should perform sequences of actions on behalf of users

- create-meeting*
- invite someone else to a meeting*
- accept an invitation*
- reject an invitation*

”



Meeting Scheduling, Vibe Coded

“

...the invariant should always hold true: no user can ever part of two meetings at the same time – no sequence of actions should ever allow this.

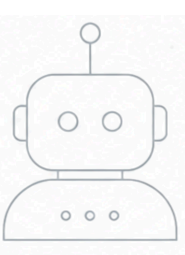
”



Meeting Scheduling, Vibe Coded

```
# ~/repos/prompt-meetings → mvn clean test
[INFO] Scanning for projects...
...
[INFO]
[INFO] -----
[INFO]  T E S T S
[INFO] -----
[INFO]
[INFO] Results:
[INFO]
[ERROR] Failures:
[ERROR]   MeetingSchedulerPropertyTests.noOverlappingMeetingsForAnyUser:90
>lambda$noOverlappingMeetingsForAnyUser$0:80 Invariant 'no-overlap' failed after the following actions: [
    alice creates Meeting-0 in calendar default (UTC) from 10:00 to 10:01
    alice creates Meeting-1 in calendar medium (UTC) from 10:00 to 10:01
    alice accepts invitation to meeting Meeting-0
]
final state: me.mourjo.prompt.meetings.MeetingSchedulerPropertyTests$CalendarMeetingsState@7fdfe501
User charlie is in overlapping meetings: Meeting-0 [10:00 - 10:01] and Meeting-1 [10:00 - 10:01]
[INFO]
[ERROR] Tests run: 3, Failures: 1, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD FAILURE
```





Meeting Scheduling, Vibe Coded

Invariant 'no-overlap' failed after the following actions: [
 alice creates Meeting-0 in calendar default (UTC) from 10:00 to 10:01
 alice creates Meeting-1 in calendar medium (UTC) from 10:00 to 10:01
 alice accepts invitation to meeting Meeting-0
]



Meeting Scheduling, Vibe Coded

Invariant 'no-overlap' failed after the following actions: [

```
charlie creates Meeting-0 in calendar high (UTC) from 10:41 to 10:49
alice creates Meeting-1 in calendar medium (UTC) from 13:49 to 13:50
bob creates Meeting-2 in calendar default (UTC) from 10:44 to 11:04
charlie creates Meeting-3 in calendar high (UTC) from 11:29 to 12:28
dan creates Meeting-4 in calendar medium (UTC) from 10:05 to 10:25
bob creates Meeting-5 in calendar default (UTC) from 10:19 to 10:37
alice rejects invitation to Meeting-2
dan accepts invitation to meeting Meeting-2
dan invites alice to Meeting-2
alice invites charlie to Meeting-1
alice rejects invitation to Meeting-0
charlie creates Meeting-6 in calendar medium (UTC) from 15:40 to 16:09
dan rejects invitation to Meeting-3
charlie accepts invitation to meeting Meeting-6
dan creates Meeting-7 in calendar high (UTC) from 10:10 to 10:14
alice invites dan to Meeting-4
charlie creates Meeting-8 in calendar high (UTC) from 13:15 to 14:01
dan invites bob to Meeting-9
charlie invites alice to Meeting-4
bob creates Meeting-9 in calendar default (UTC) from 10:07 to 10:10
dan accepts invitation to meeting Meeting-4
```

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Meeting Scheduling, Vibe Coded

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alice rejects invitation to Meeting-2

dan accepts invitation to meeting Meeting-2

dan invites alice to Meeting-2

alice invites charlie to Meeting-1

alice rejects invitation to Meeting-0

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dan creates **Meeting-4** in calendar medium (UTC) from 10:05 to 10:25

bob creates **Meeting-5** in calendar default (UTC) from 10:19 to 10:37

alice rejects invitation to **Meeting-2**

dan accepts invitation to meeting **Meeting-2**

dan invites alice to **Meeting-2**

alice invites charlie to **Meeting-1**

alice rejects invitation to **Meeting-0**

charlie creates **Meeting-6** in calendar medium (UTC) from 15:40 to 16:09

dan rejects invitation to **Meeting-3**

charlie accepts invitation to meeting **Meeting-6**

dan creates **Meeting-7** in calendar high (UTC) from 10:10 to 10:14

alice invites dan to **Meeting-4**

charlie creates **Meeting-8** in calendar high (UTC) from 13:15 to 14:01

dan invites bob to **Meeting-9**

charlie invites alice to **Meeting-4**

bob creates **Meeting-9** in calendar default (UTC) from 10:07 to 10:10

dan accepts invitation to meeting **Meeting-4**

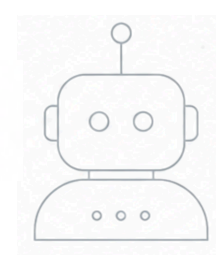
]



Meeting Scheduling, Vibe Coded

@Property





Meeting Scheduling, Vibe Coded

@Property

```
@ForAll("actions") ActionChain<CalendarState> chain
```



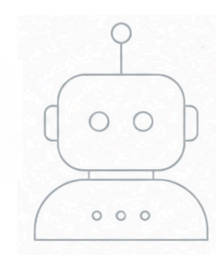
Meeting Scheduling, Vibe Coded

@Property

@ForAll("actions") ActionChain<CalendarState> chain

@Provide

```
Arbitrary<ActionChain<CalendarState>> actions() {  
    return ActionChain.startWith(CalendarMeetingsState::new)  
        .withAction(new CreateMeetingAction())  
        .withAction(new InviteUserAction())  
        .withAction(new AcceptInviteAction())  
        .withAction(new RejectMeetingAction());  
}
```



Meeting Scheduling, Vibe Coded

@Property

```
@ForAll("actions") ActionChain<CalendarState> chain
```





Meeting Scheduling, Vibe Coded

@Property

```
void noOverlappingMeetingsForAnyUser(@ForAll("actions") ActionChain<CalendarState> chain) {
```





Meeting Scheduling, Vibe Coded

@Property

```
void noOverlappingMeetingsForAnyUser(@ForAll("actions") ActionChain<CalendarState> chain) {  
    chain.withInvariant("no-overlap", sut → {
```

```
    }).run();
```





Meeting Scheduling, Vibe Coded

@Property

```
void noOverlappingMeetingsForAnyUser(@ForAll("actions") ActionChain<CalendarState> chain) {  
    chain.withInvariant("no-overlap", sut → {  
  
        for (String user : USERS) {  
  
            }  
        }).run();  
    }  
}
```





Meeting Scheduling, Vibe Coded

@Property

```
void noOverlappingMeetingsForAnyUser(@ForAll("actions") ActionChain<CalendarState> chain) {  
    chain.withInvariant("no-overlap", sut → {  
  
        for (String user : USERS) {  
            var meetings = meetingService.getMyMeetings(user).stream()  
                .filter(m → "ACCEPTED".equalsIgnoreCase(m.userStatus()))  
                .toList();  
  
        }  
    }).run();  
}
```



Meeting Scheduling, Vibe Coded

@Property

```
void noOverlappingMeetingsForAnyUser(@ForAll("actions") ActionChain<CalendarState> chain) {  
    chain.withInvariant("no-overlap", sut → {  
  
        for (String user : USERS) {  
            var meetings = meetingService.getMyMeetings(user).stream()  
                .filter(m → "ACCEPTED".equalsIgnoreCase(m.userStatus()))  
                .toList();  
  
            for (int i = 0; i < meetings.size(); i++) {  
                for (int j = i + 1; j < meetings.size(); j++) {  
  
                }  
            }  
        }  
    }).run();  
}
```


Meeting Scheduling, Vibe Coded

@Property

```
void noOverlappingMeetingsForAnyUser(@ForAll("actions") ActionChain<CalendarState> chain) {
    chain.withInvariant("no-overlap", sut → {

        for (String user : USERS) {
            var meetings = meetingService.getMyMeetings(user).stream()
                .filter(m → "ACCEPTED".equalsIgnoreCase(m.userStatus()))
                .toList();

            for (int i = 0; i < meetings.size(); i++) {
                for (int j = i + 1; j < meetings.size(); j++) {
                    if (overlaps(meetings.get(i), meetings.get(j))) {
                        throw new AssertionError(...);
                    }
                }
            }
        }
    }).run();
}
```

Meeting Scheduling, Vibe Coded

```
[ERROR]    Invariant 'no-overlap' failed after the following actions: [  
    charlie creates Meeting-0 in calendar default (UTC) from 10:00 to 10:01  
    charlie creates Meeting-1 in calendar medium (UTC) from 10:00 to 10:01  
    charlie accepts invitation to meeting Meeting-0  
]
```


Meeting Scheduling, Vibe Coded

[ERROR] Invariant 'no-overlap' failed after the following actions: [
charlie creates Meeting-0 in calendar default (UTC) from 10:00 to 10:01
charlie creates Meeting-1 in calendar medium (UTC) from 10:00 to 10:01
charlie accepts invitation to meeting Meeting-0
]

Charlie creates Meeting-0



Meeting Scheduling, Vibe Coded

[ERROR] Invariant 'no-overlap' failed after the following actions: [
charlie creates Meeting-0 in calendar default (UTC) from 10:00 to 10:01
charlie creates Meeting-1 in calendar medium (UTC) from 10:00 to 10:01
charlie accepts invitation to meeting Meeting-0
]

Charlie creates Meeting-0

Charlie creates an overlapping meeting in a higher priority calendar





Meeting Scheduling, Vibe Coded

[ERROR] Invariant 'no-overlap' failed after the following actions: [
charlie creates Meeting-0 in calendar default (UTC) from 10:00 to 10:01
charlie creates Meeting-1 in calendar medium (UTC) from 10:00 to 10:01
charlie accepts invitation to meeting Meeting-0
]

Charlie creates Meeting-0

Charlie creates an overlapping meeting in a higher priority calendar

Charlie accepts Meeting-0





Meeting Scheduling, Vibe Coded

“

When accepting a meeting ensure that the invitation is in pending state – ie, do not allow acceptance of a meeting which is already accepted or rejected.

”



Meeting Scheduling, Vibe Coded

[ERROR] Invariant 'no-overlap' failed after the following actions: [
charlie creates Meeting-0 in calendar default (UTC) from 10:00 to 10:01
alice creates Meeting-1 in calendar default (UTC) from 10:00 to 10:01
alice invites charlie to Meeting-1
charlie accepts invitation to meeting Meeting-1
]

Meeting Scheduling, Vibe Coded

[ERROR] Invariant 'no-overlap' failed after the following actions: [
charlie creates Meeting-0 in calendar default (UTC) from 10:00 to 10:01
alice creates Meeting-1 in calendar default (UTC) from 10:00 to 10:01
alice invites charlie to Meeting-1
charlie accepts invitation to meeting Meeting-1
]

Charlie creates a Meeting-0

Meeting Scheduling, Vibe Coded

[ERROR] Invariant 'no-overlap' failed after the following actions: [
charlie creates Meeting-0 in calendar default (UTC) from 10:00 to 10:01
alice creates Meeting-1 in calendar default (UTC) from 10:00 to 10:01
alice invites charlie to Meeting-1
charlie accepts invitation to meeting Meeting-1
]

Charlie creates a Meeting-0

Alice creates an overlapping meeting

Meeting Scheduling, Vibe Coded

[ERROR] Invariant 'no-overlap' failed after the following actions: [
charlie creates Meeting-0 in calendar default (UTC) from 10:00 to 10:01
alice creates Meeting-1 in calendar default (UTC) from 10:00 to 10:01
alice invites charlie to Meeting-1
charlie accepts invitation to meeting Meeting-1
]

Charlie creates a Meeting-0

Alice creates an overlapping meeting

Alice invites Charlie to their meeting

Meeting Scheduling, Vibe Coded

[ERROR] Invariant 'no-overlap' failed after the following actions: [
charlie creates Meeting-0 in calendar default (UTC) from 10:00 to 10:01
alice creates Meeting-1 in calendar default (UTC) from 10:00 to 10:01
alice invites charlie to Meeting-1
charlie accepts invitation to meeting Meeting-1
]

Charlie creates a Meeting-0

Alice creates an overlapping meeting

Alice invites Charlie to their meeting

Charlie accepts the invitation



Meeting Scheduling, Vibe Coded

[ERROR] Invariant 'no-overlap' failed after the following actions: [
charlie creates Meeting-0 in calendar default (UTC) from 10:00 to 10:01
alice creates Meeting-1 in calendar default (UTC) from 10:00 to 10:01
alice invites charlie to Meeting-1
charlie accepts invitation to meeting Meeting-1
]

[ERROR] Invariant 'no-overlap' failed after the following actions: [
charlie creates Meeting-0 in calendar default (UTC) from 10:00 to 10:01
charlie creates Meeting-1 in calendar medium (UTC) from 10:00 to 10:01
charlie accepts invitation to meeting Meeting-0
]





Meeting Scheduling, Vibe Coded

[

charlie creates Meeting-0 in calendar default (UTC) from 10:00 to 10:01
alice creates Meeting-1 in calendar default (UTC) from 10:00 to 10:01
alice invites charlie to Meeting-1
charlie accepts invitation to meeting Meeting-1

] Four actions

Three actions

[

charlie creates Meeting-0 in calendar default (UTC) from 10:00 to 10:01
charlie creates Meeting-1 in calendar medium (UTC) from 10:00 to 10:01
charlie accepts invitation to meeting Meeting-0

]



Meeting Scheduling, Vibe Coded

```
[
  charlie creates Meeting-0 in calendar default (UTC) from 10:00 to 10:01
  alice creates Meeting-1 in calendar default (UTC) from 10:00 to 10:01
  alice invites charlie to Meeting-1
  charlie accepts invitation to meeting Meeting-1
]
```

] Two users

One user

```
[
  charlie creates Meeting-0 in calendar default (UTC) from 10:00 to 10:01
  charlie creates Meeting-1 in calendar medium (UTC) from 10:00 to 10:01
  charlie accepts invitation to meeting Meeting-0
]
```

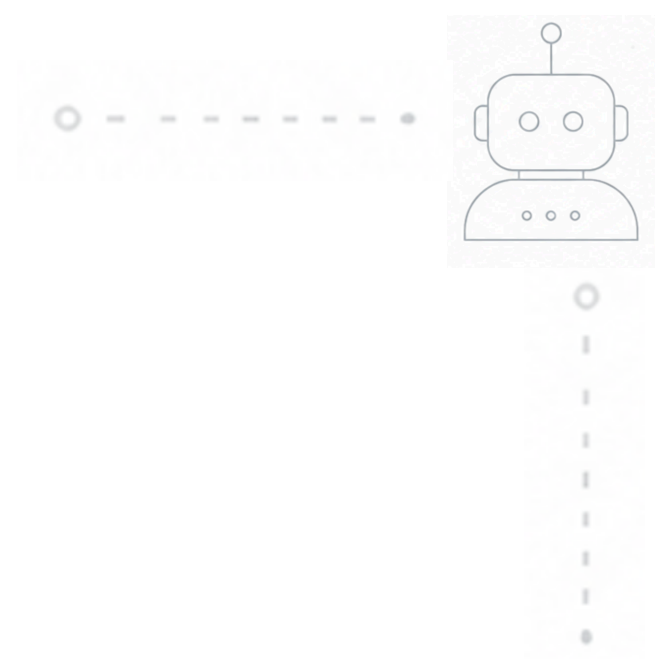
Meeting Scheduling, Vibe Coded

```
[
charlie creates Meeting-0 in calendar default (UTC) from 10:00 to 10:01
alice creates Meeting-1 in calendar default (UTC) from 10:00 to 10:01
alice invites charlie to Meeting-1
charlie accepts invitation to meeting Meeting-1
]
```

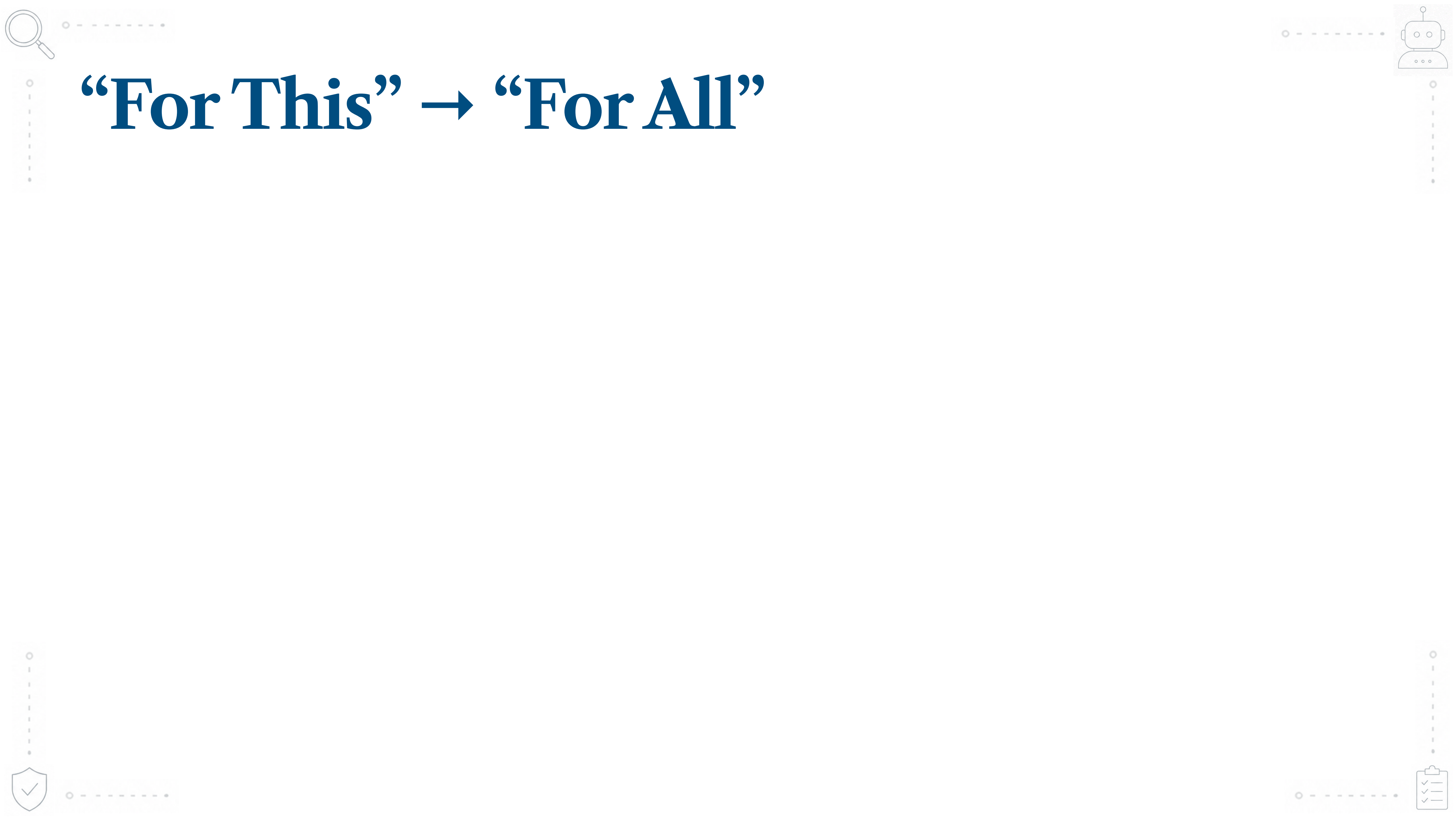
One calendar

Two calendars

```
[
charlie creates Meeting-0 in calendar default (UTC) from 10:00 to 10:01
charlie creates Meeting-1 in calendar medium (UTC) from 10:00 to 10:01
charlie accepts invitation to meeting Meeting-0
]
```

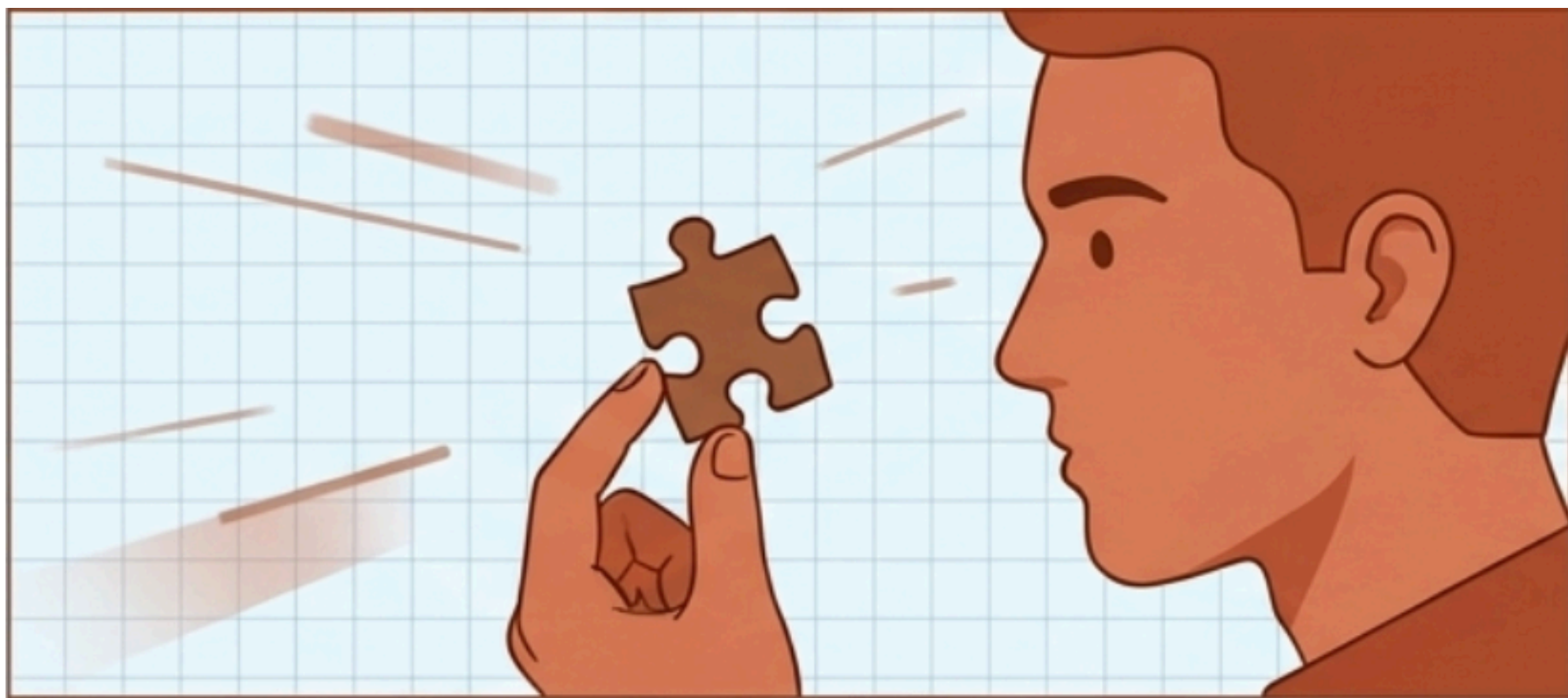



More Than Just Testing

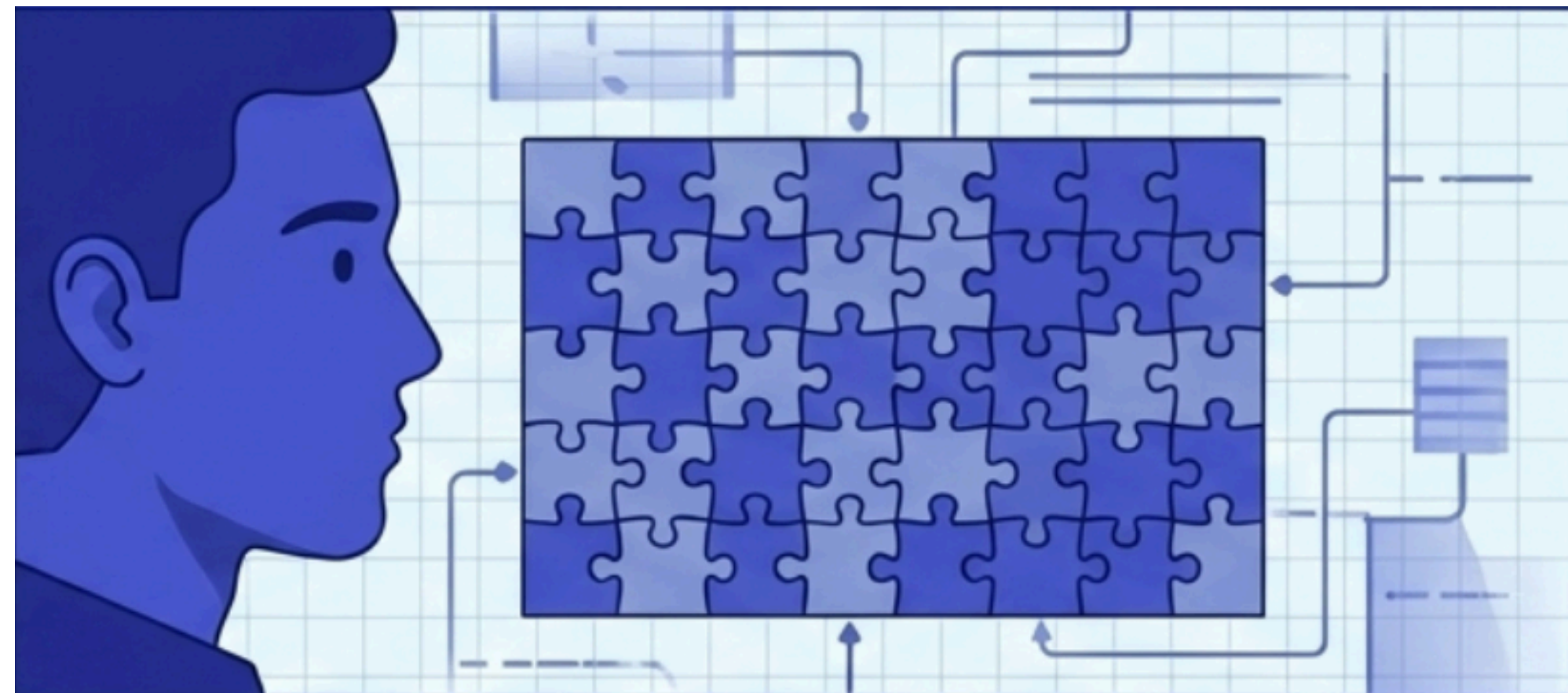
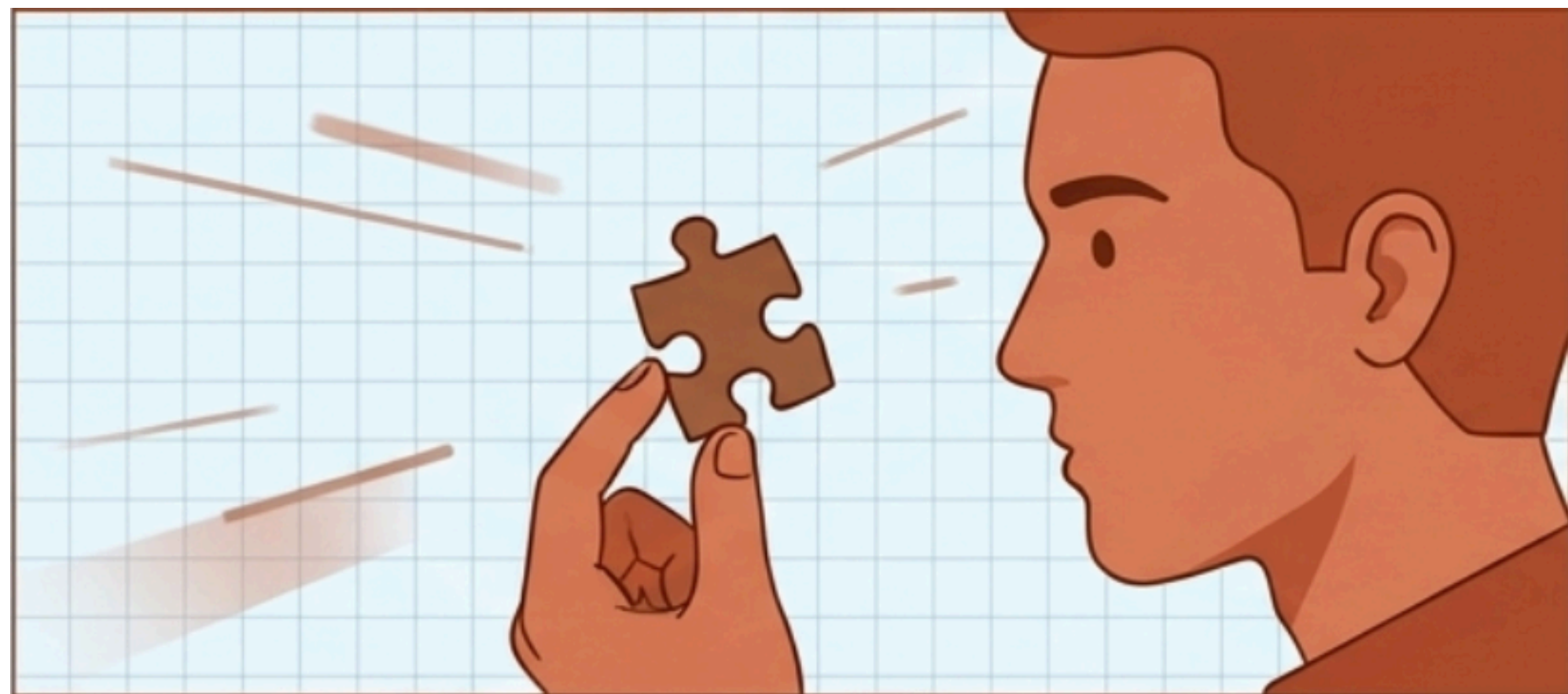


“For This” → “For All”

“For This” → “For All”

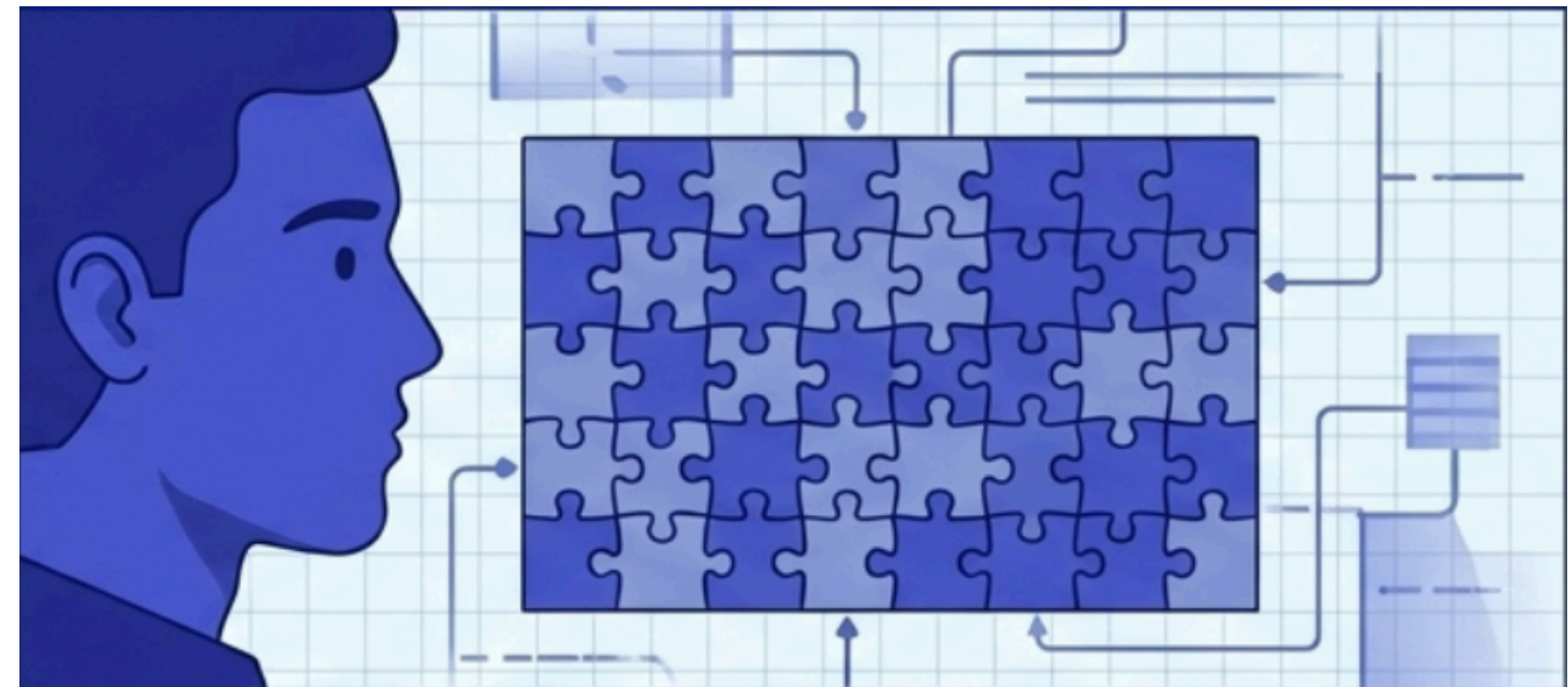
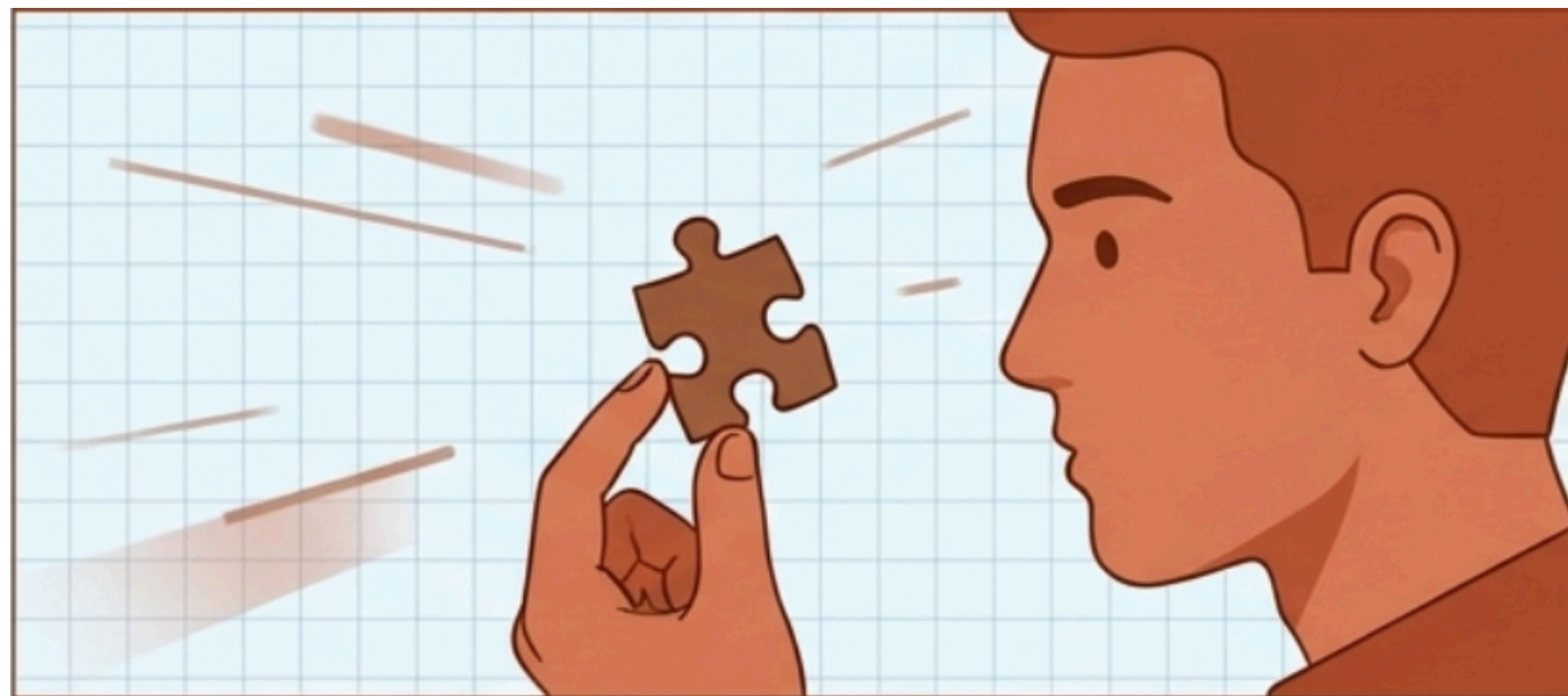


“For This” → “For All”



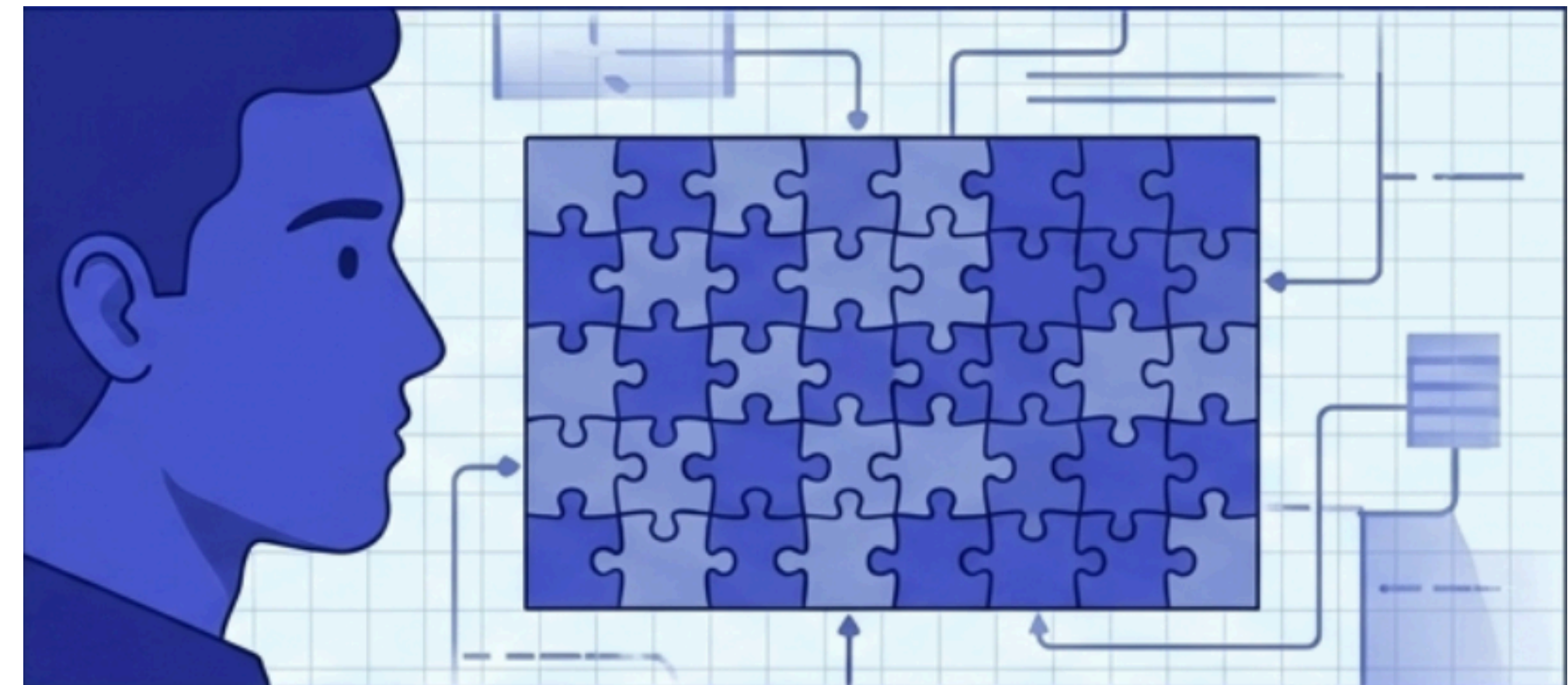
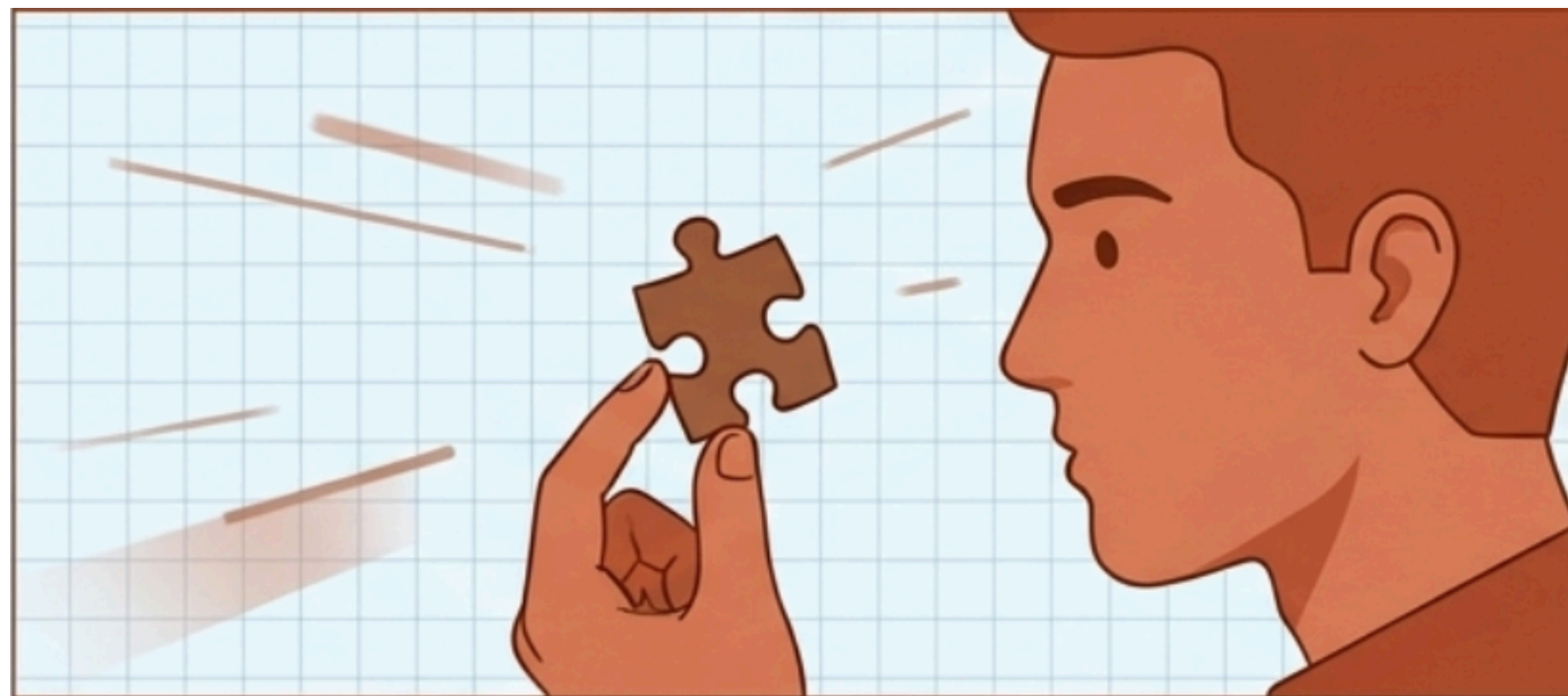
“For This” → “For All”

- Defining what must always hold true



“For This” → “For All”

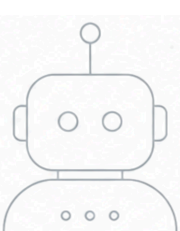
- Defining what must always hold true
- Understanding counter examples





“For This” $\rightarrow$ “For All”

- Defining what must always hold true
 - Understanding counter examples
- 
- 



“For This” → “For All”

- Defining what must always hold true
- Understanding counter examples

“Add a new property based test which ensures this invariant: **for every auto rejected meeting, there should be at least one higher priority meeting that overlaps with this meeting and is accepted** by the current user.”



“For This” → “For All”

- Defining what must always hold true
- Understanding counter examples

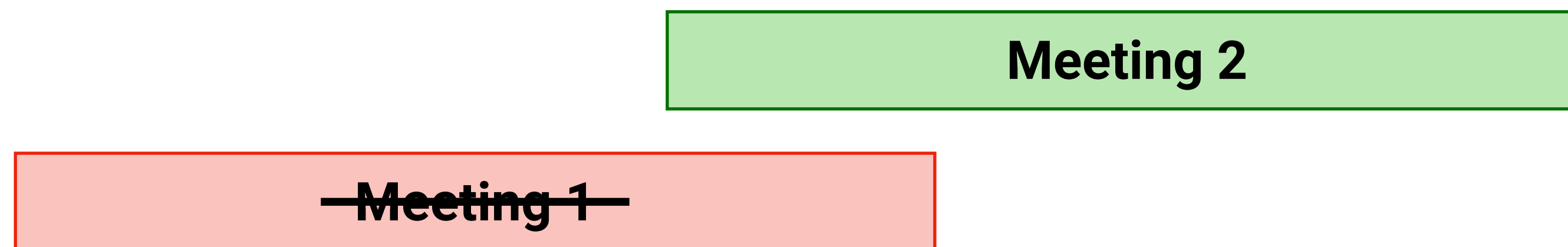
“Add a new property based test which ensures this invariant: **for every auto rejected meeting, there should be at least one higher priority meeting that overlaps with this meeting and is accepted** by the current user.”

Meeting 1

“For This” → “For All”

- Defining what must always hold true
- Understanding counter examples

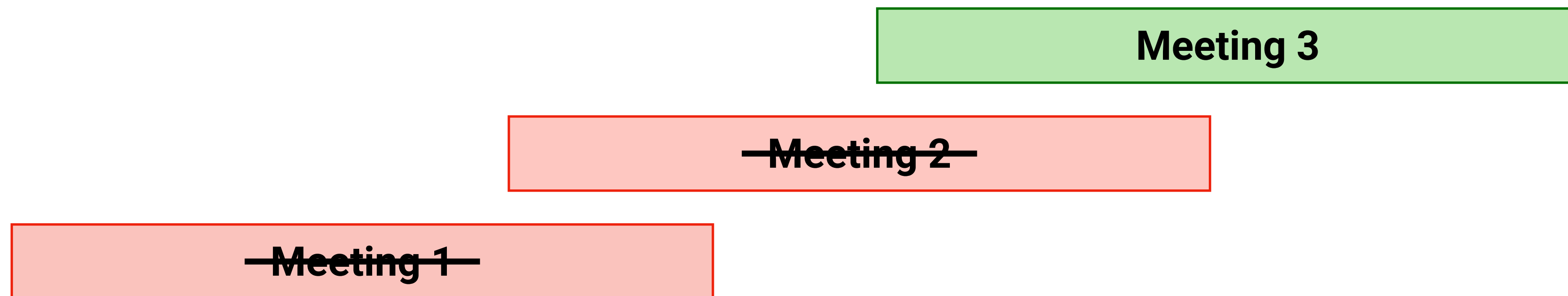
“Add a new property based test which ensures this invariant: **for every auto rejected meeting, there should be at least one higher priority meeting that overlaps with this meeting and is accepted** by the current user.”



“For This” → “For All”

- Defining what must always hold true
- Understanding counter examples

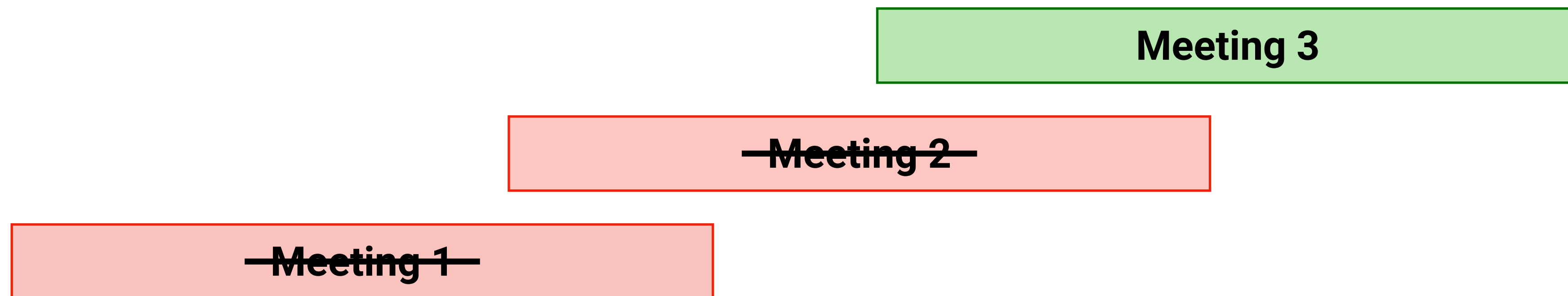
“Add a new property based test which ensures this invariant: **for every auto rejected meeting, there should be at least one higher priority meeting that overlaps with this meeting and is accepted** by the current user.”



“For This” → “For All”

- Defining what must always hold true
- Understanding counter examples

“Add a new property based test which ensures this invariant: **for every auto rejected meeting, there should be at least one higher priority meeting that overlaps with this meeting and is accepted** by the current user.”



No overlapping accepted meeting



Counter Examples are Transformative



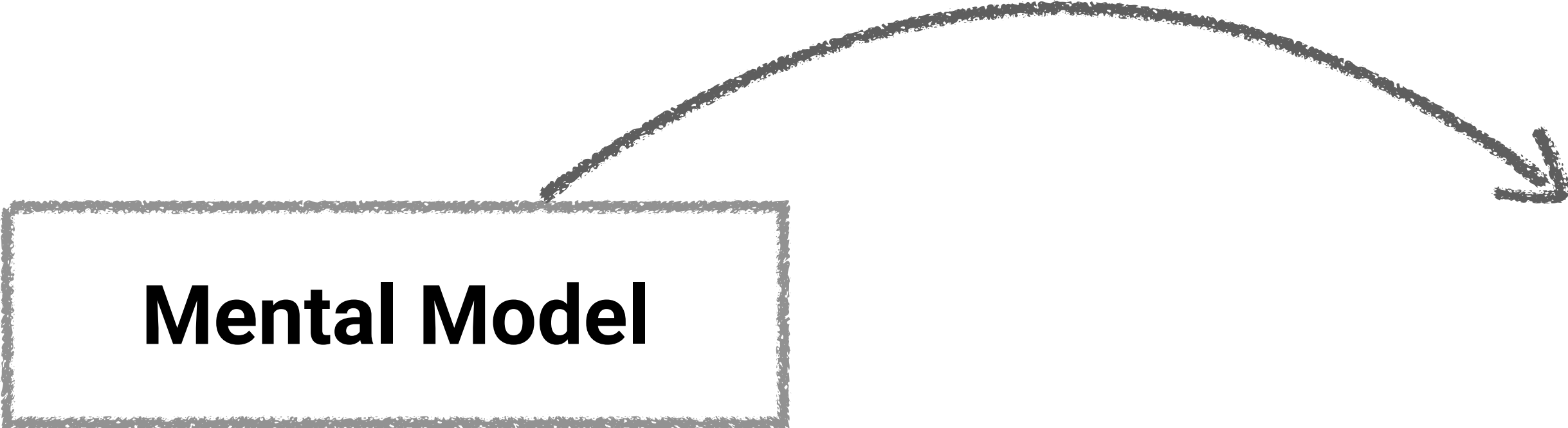
Counter Examples are Transformative

Mental Model

Counter Examples are Transformative

What must ***always*** hold true

Mental Model



Counter Examples are Transformative

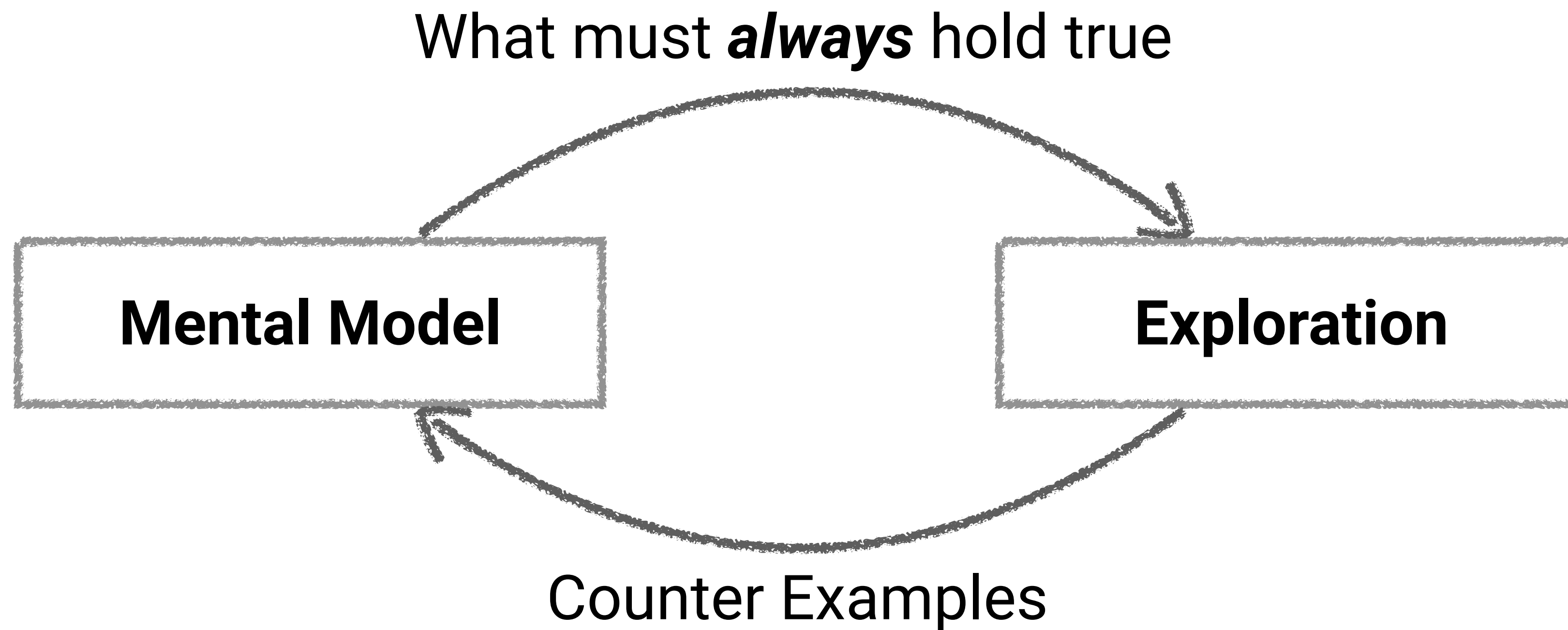
What must *always* hold true

Mental Model

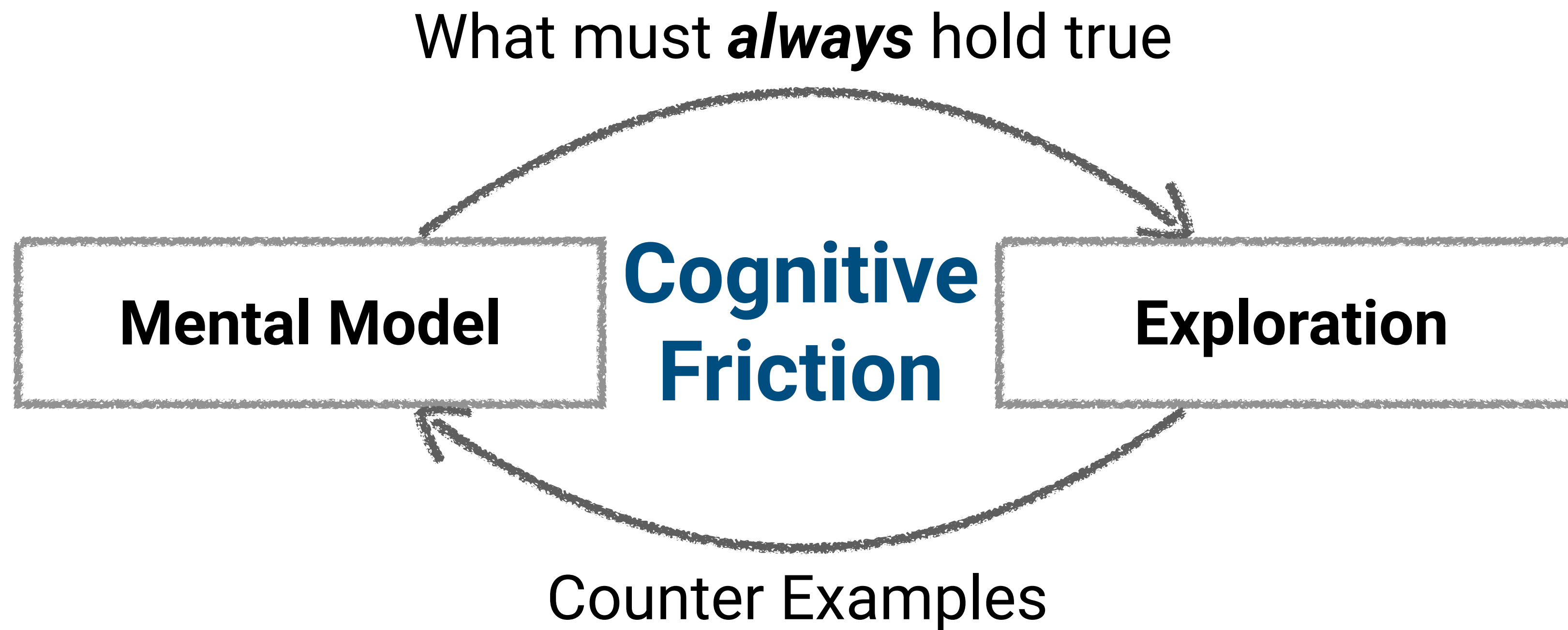
Exploration

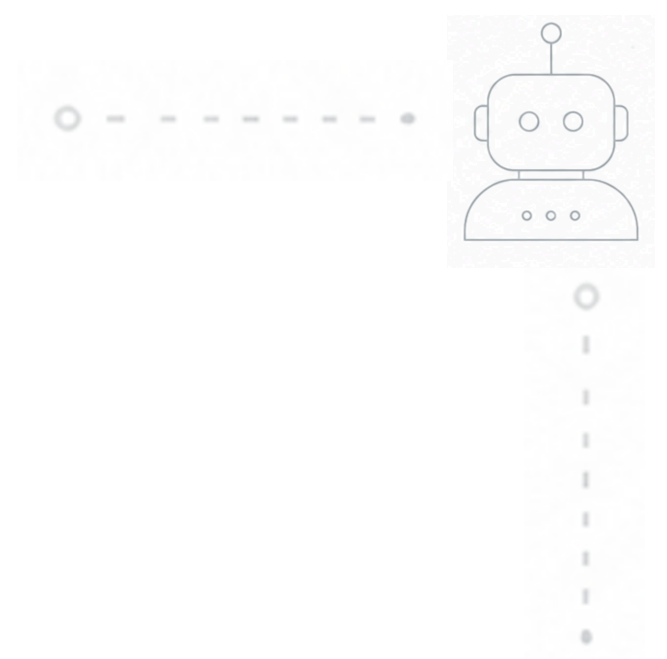


Counter Examples are Transformative



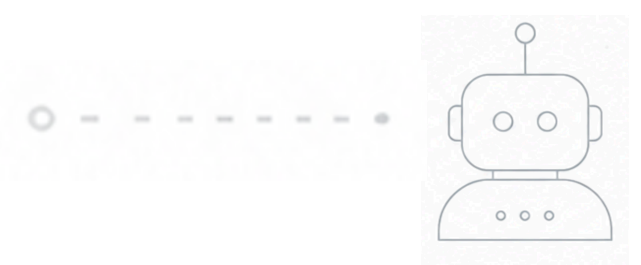
Counter Examples are Transformative





Everything's Changed





Passive Overseeing is Not Enough



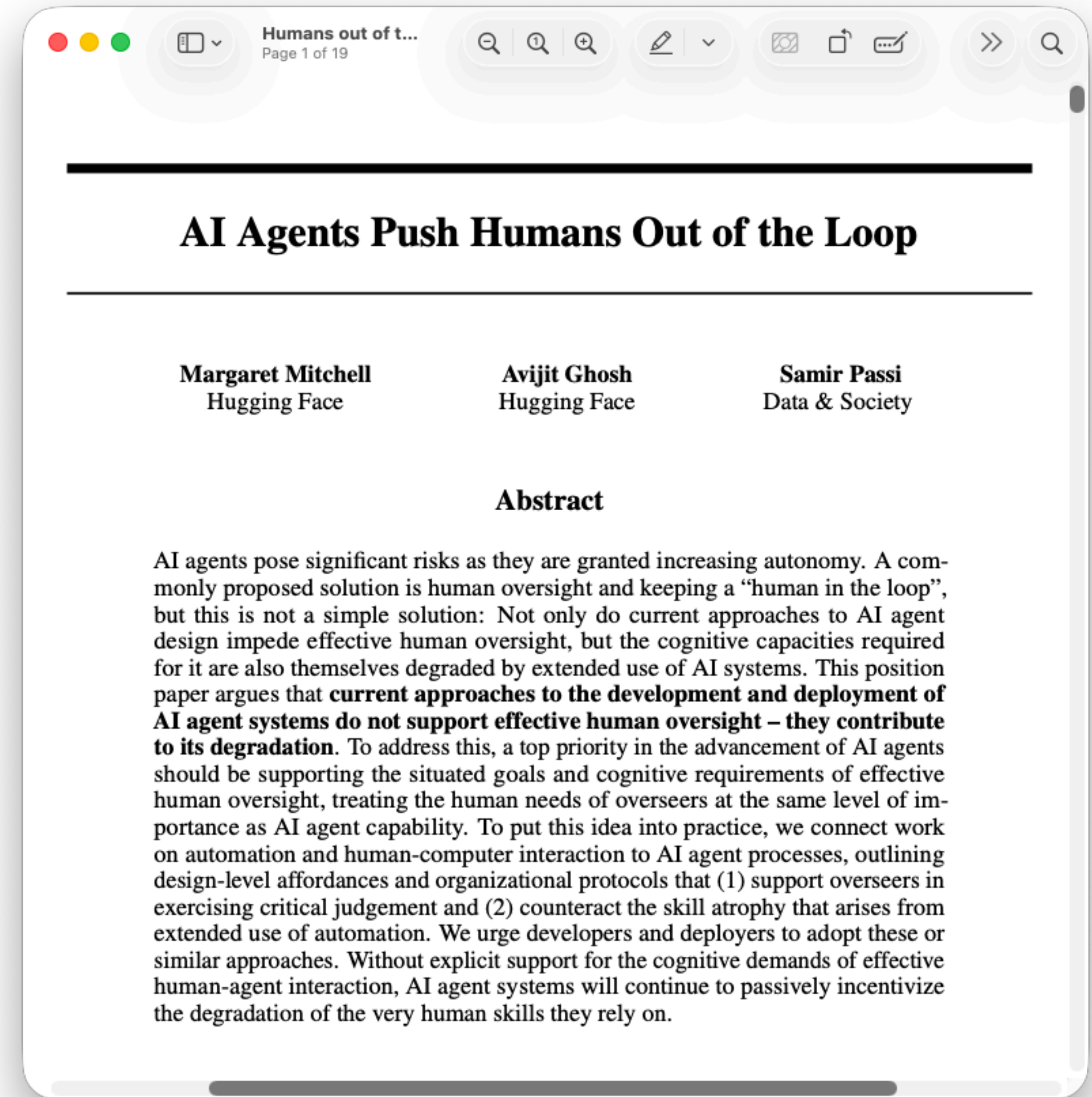


Passive Overseeing is Not Enough

- LLMs code faster than we can read
 - Quality is not guaranteed
 - Just being “in the loop” doesn’t work
- 
- 

Passive Overseeing is Not Enough

- LLMs code faster than we can read
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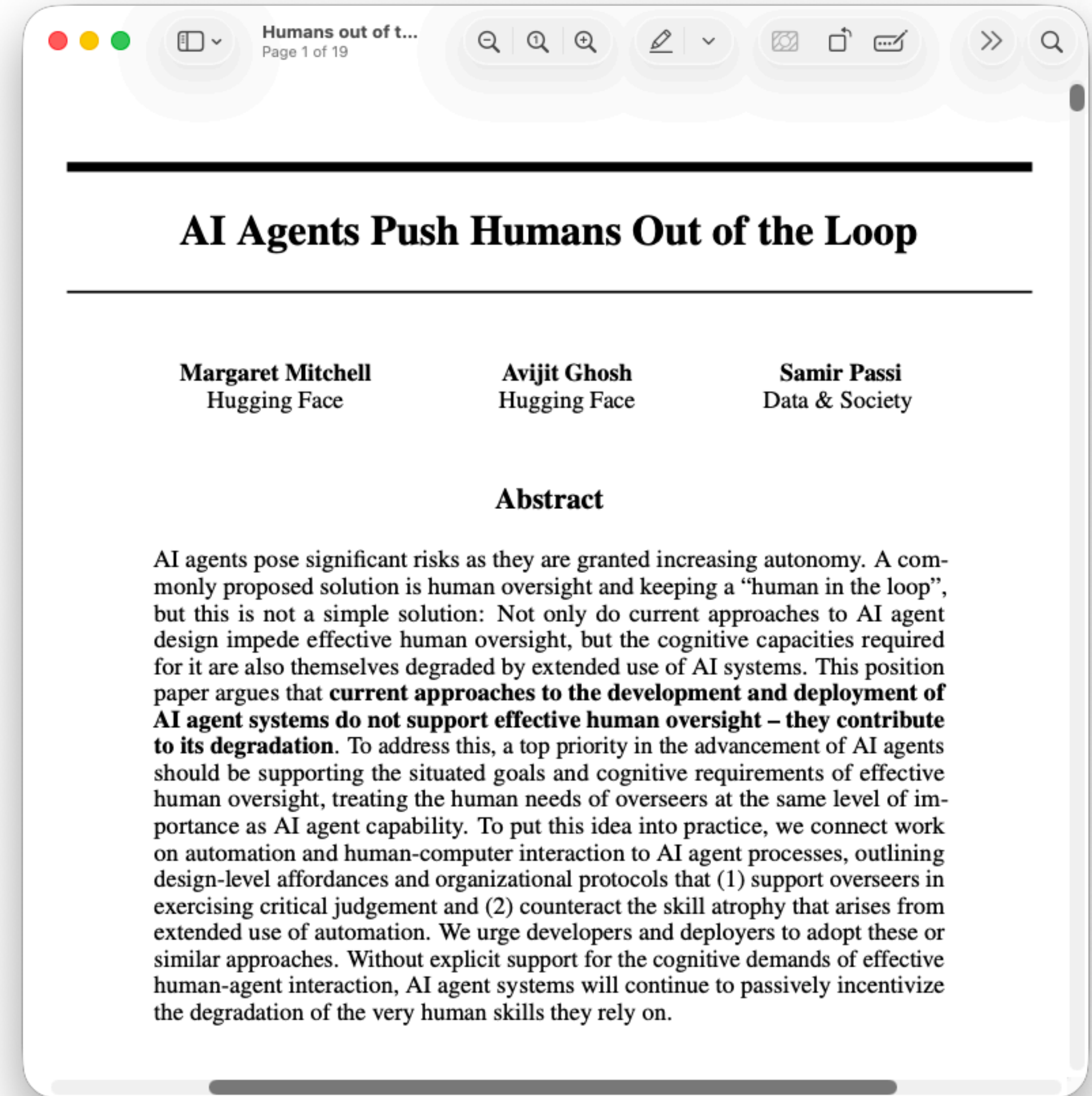


Source: <https://arxiv.org/abs/2608.23642>

Passive Overseeing is Not Enough

- LLMs code faster than we can read
 - Quality is not guaranteed
 - Just being “in the loop” doesn’t work
- Strategic cognitive friction
 - Embed human comprehension into software engineering

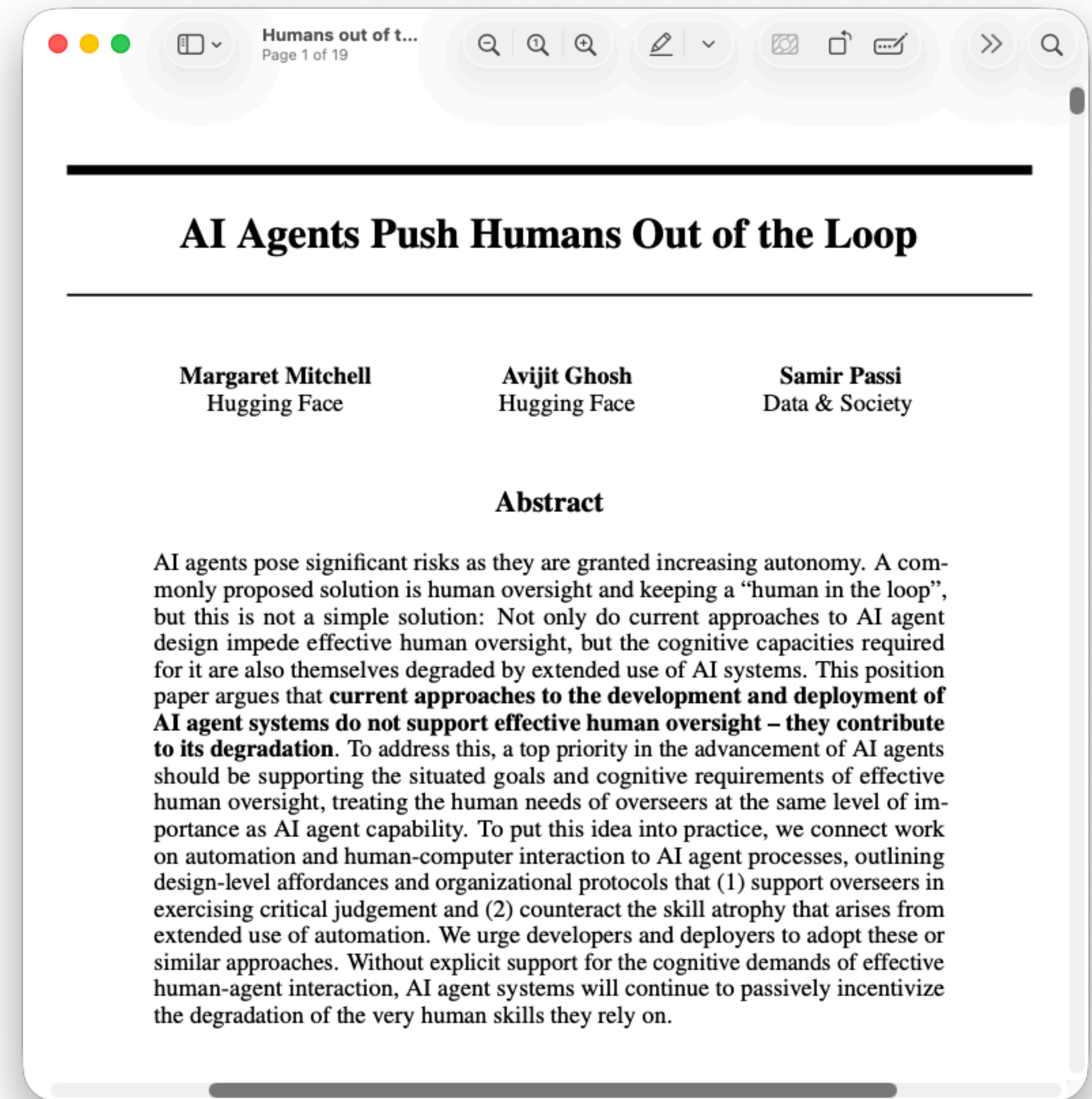
Source: <https://arxiv.org/abs/2608.23642>



Passive Overseeing is Not Enough

- LLMs code faster than we can read
 - Quality is not guaranteed
 - Just being “in the loop” doesn’t work
- Strategic cognitive friction
 - Embed human comprehension into software engineering
- Exploratory testing brings cognitive friction back

Source: <https://arxiv.org/abs/2608.23642>

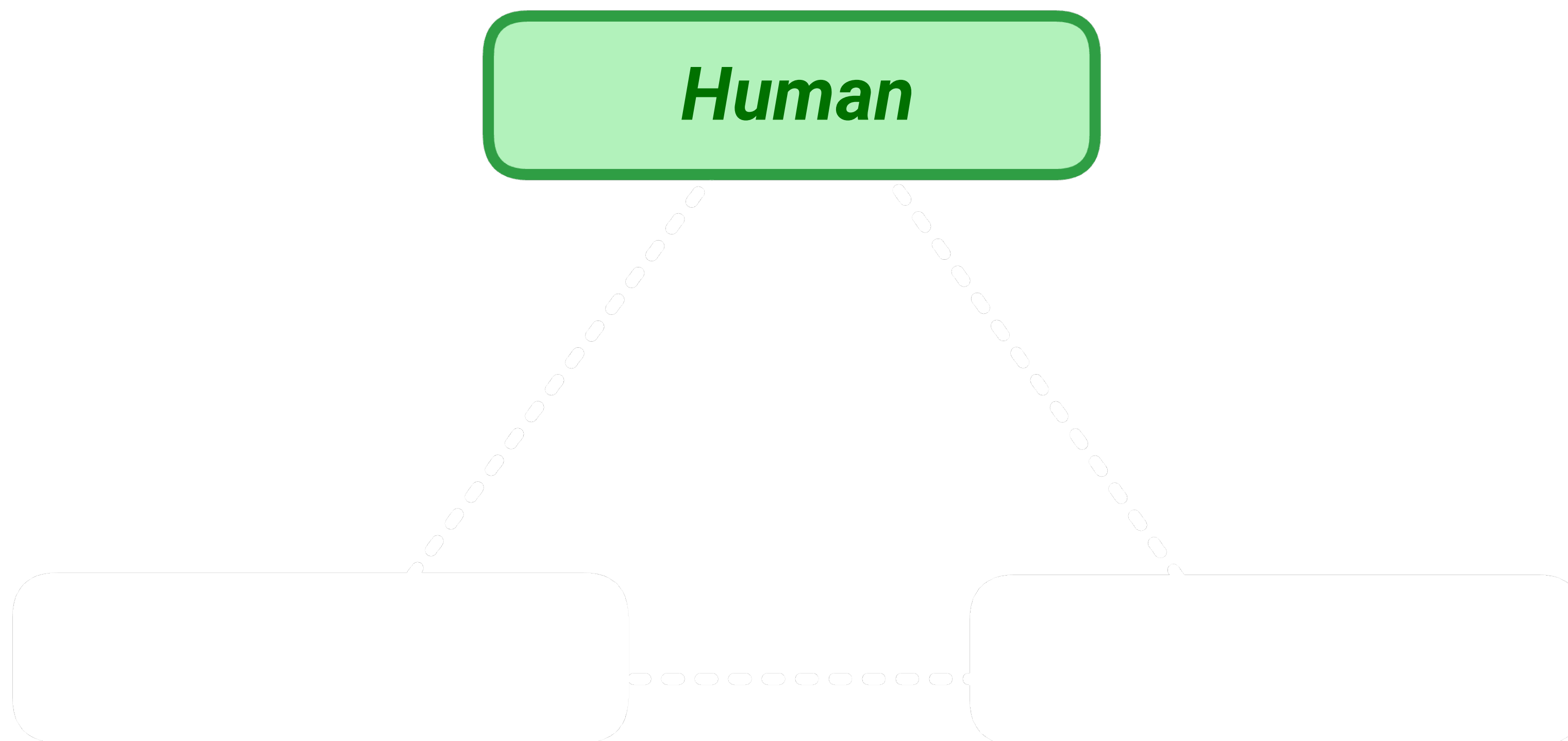




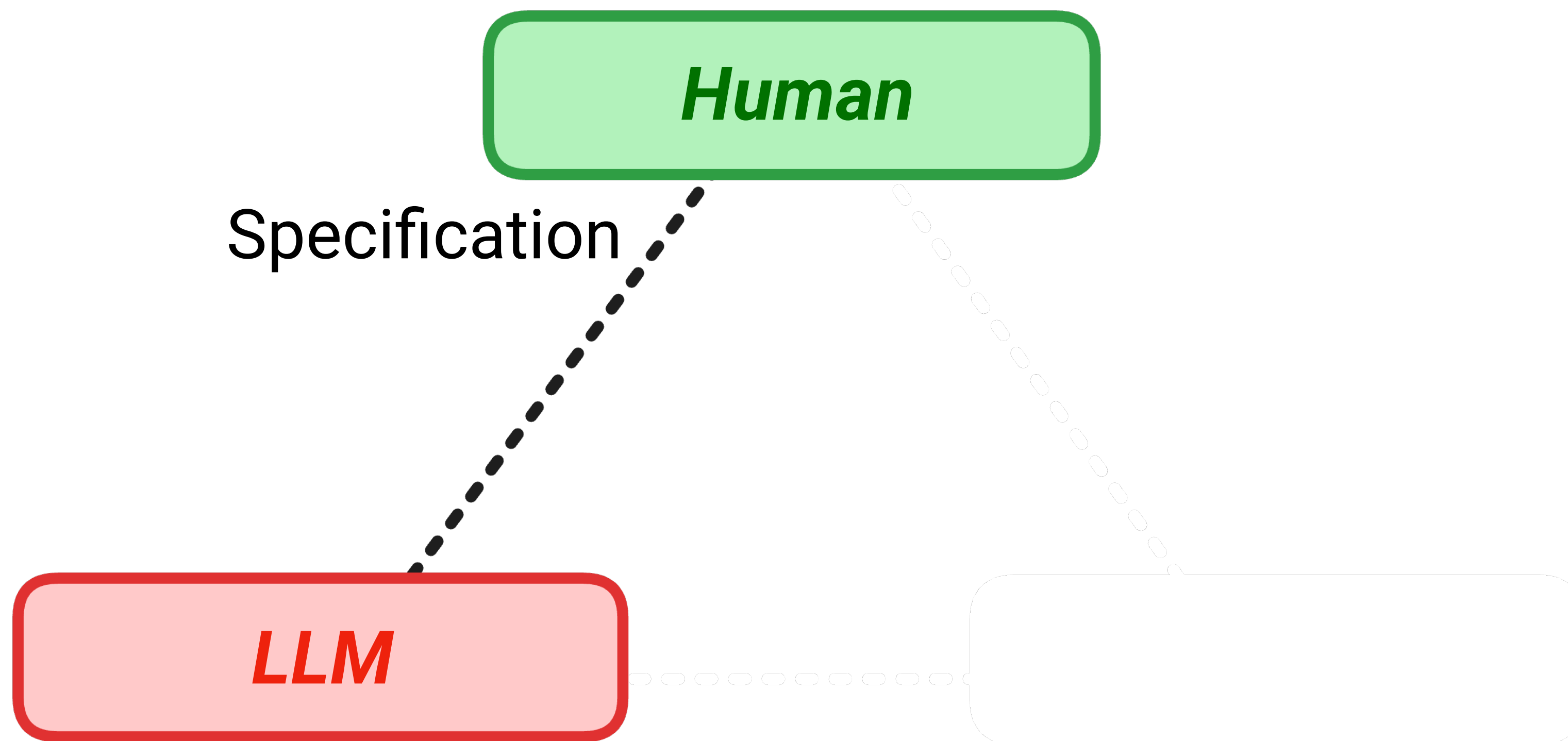
Division of Cognitive Labour



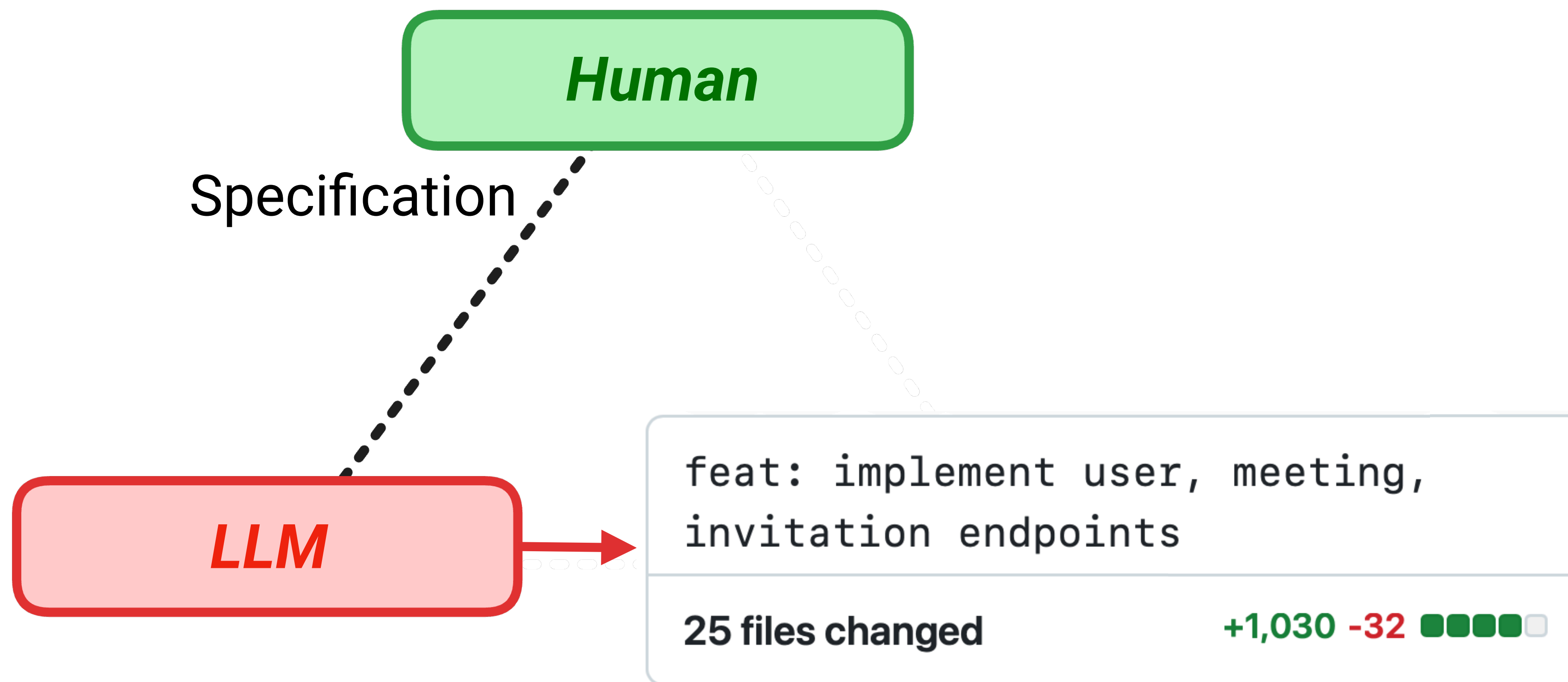
Division of Cognitive Labour



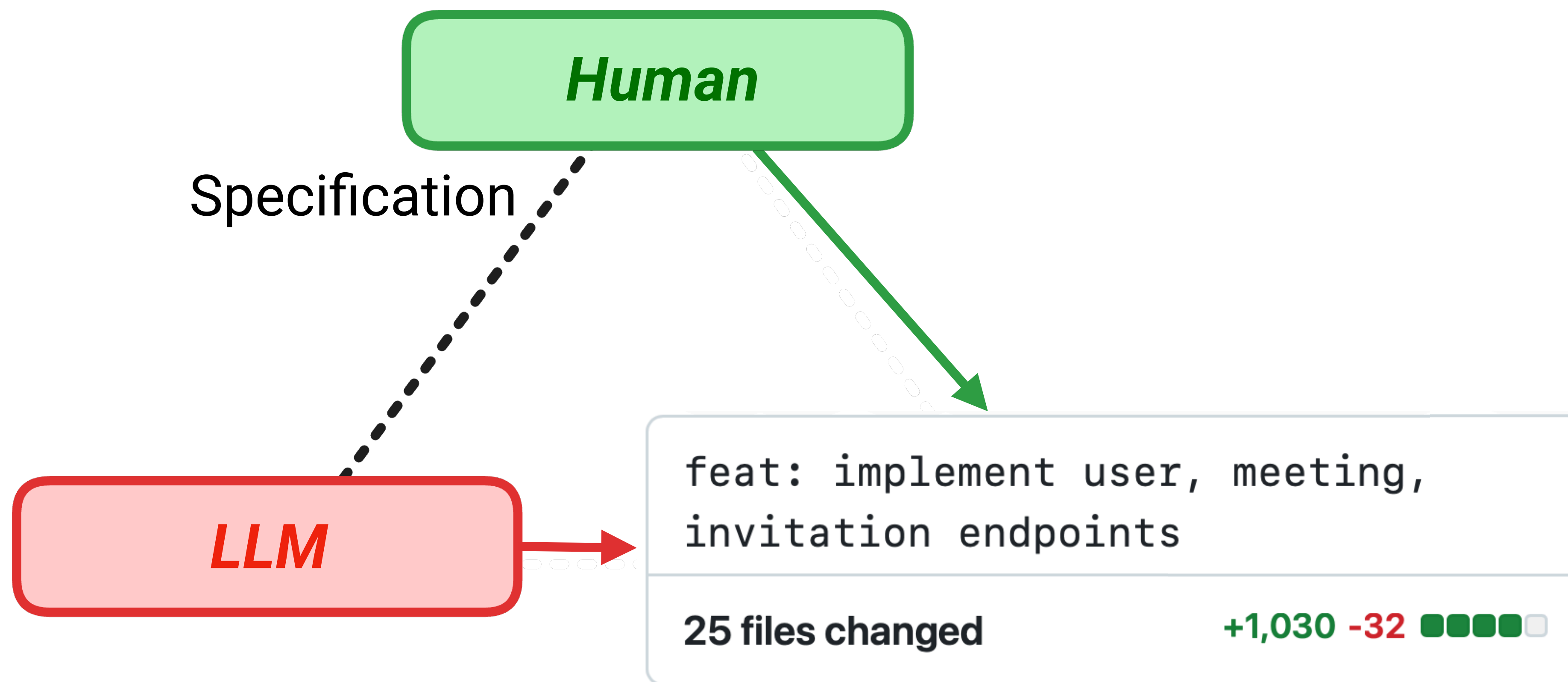
Division of Cognitive Labour



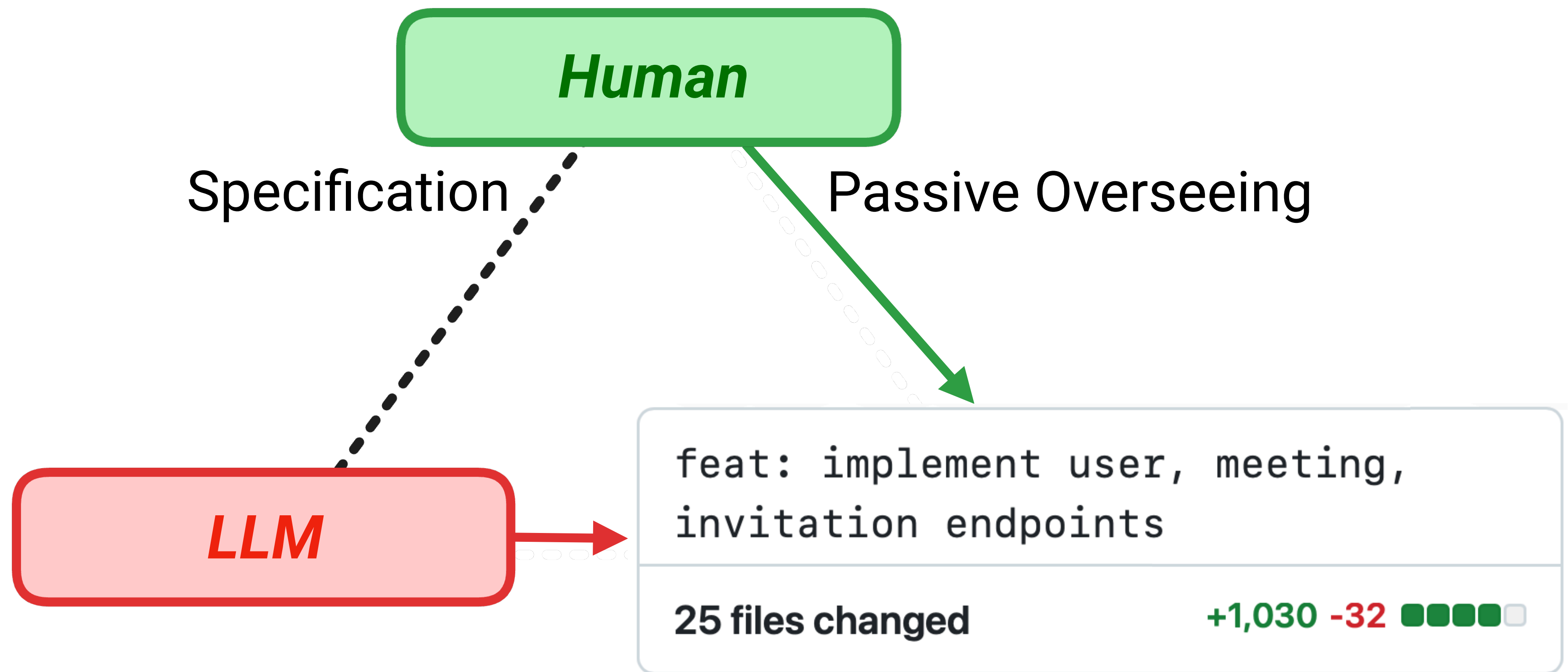
Division of Cognitive Labour



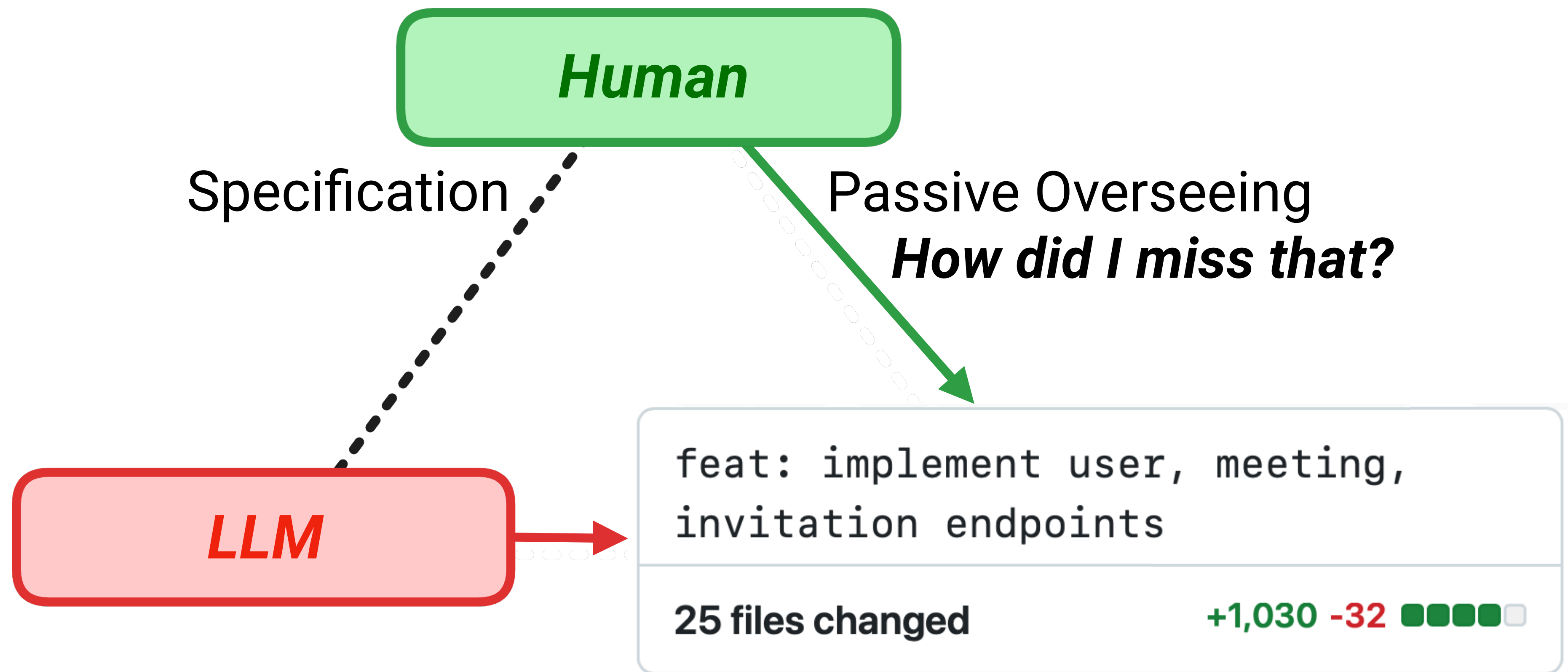
Division of Cognitive Labour



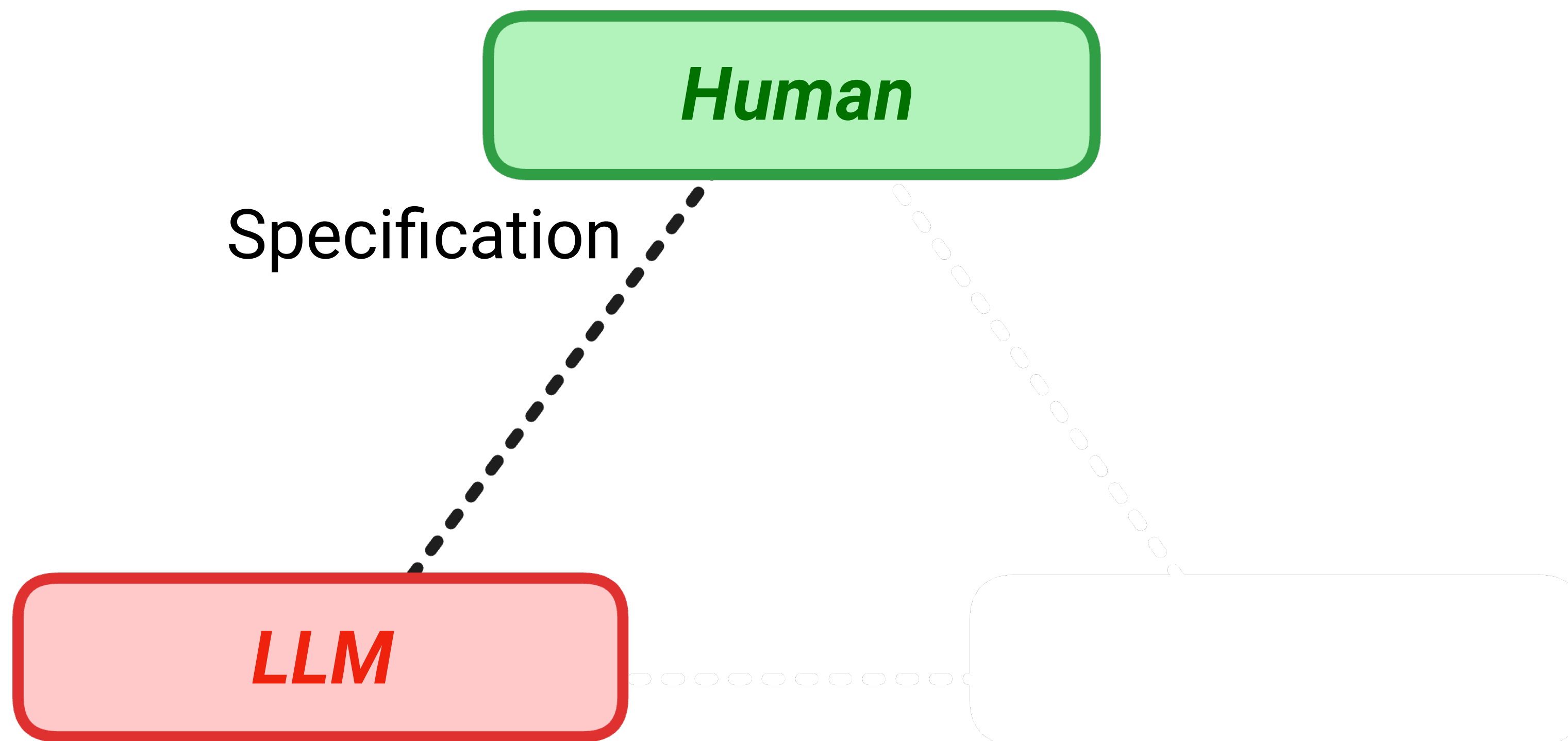
Division of Cognitive Labour



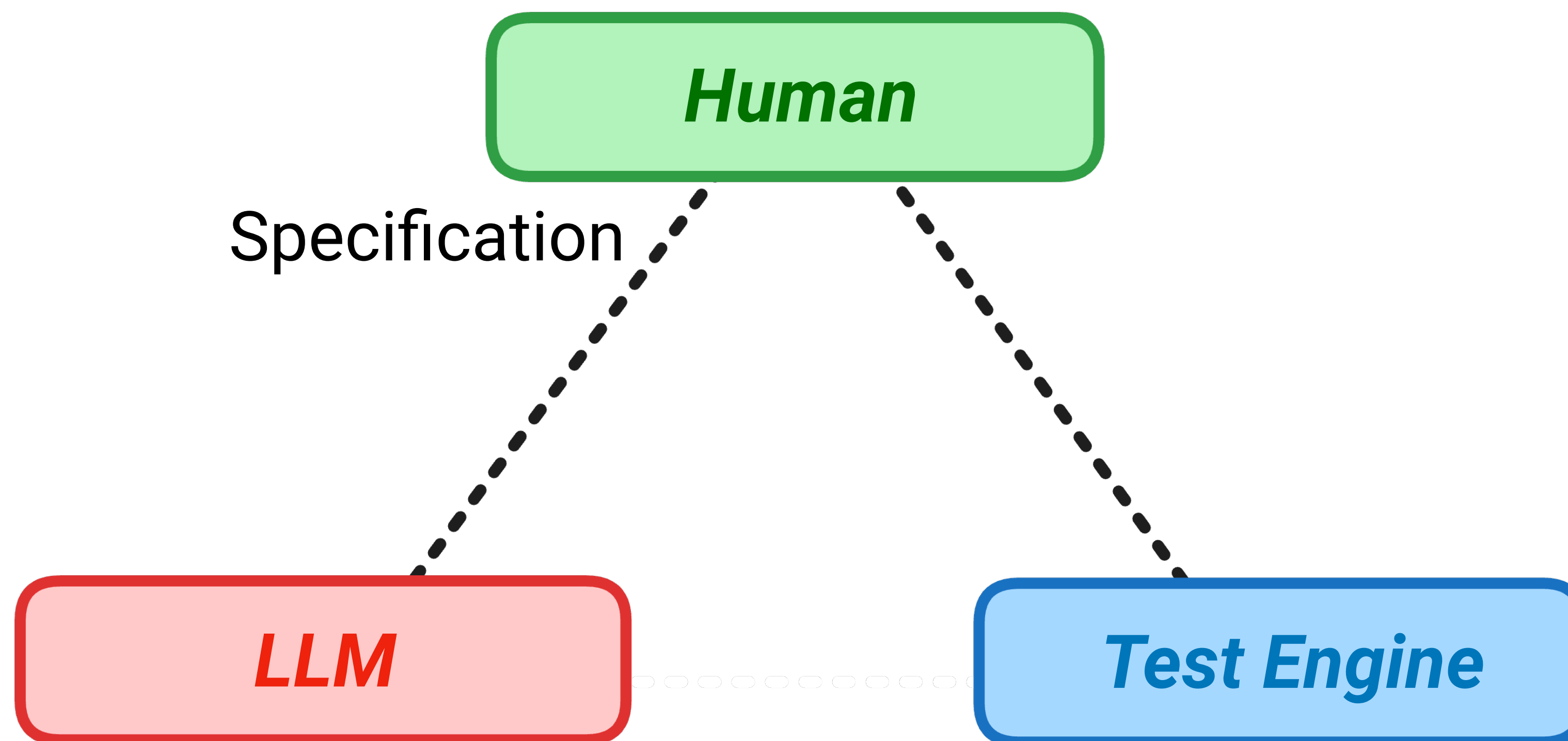
Division of Cognitive Labour



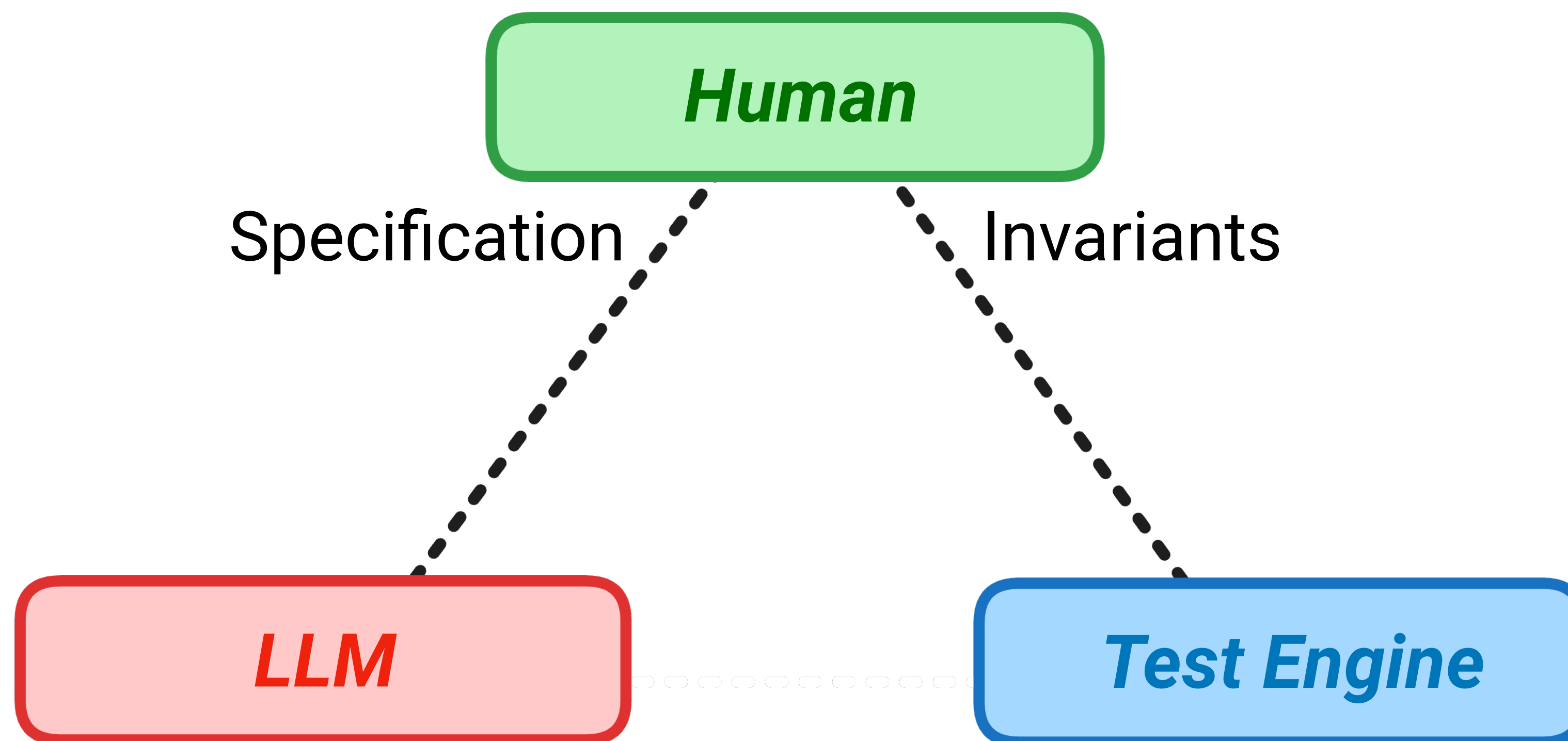
Division of Cognitive Labour



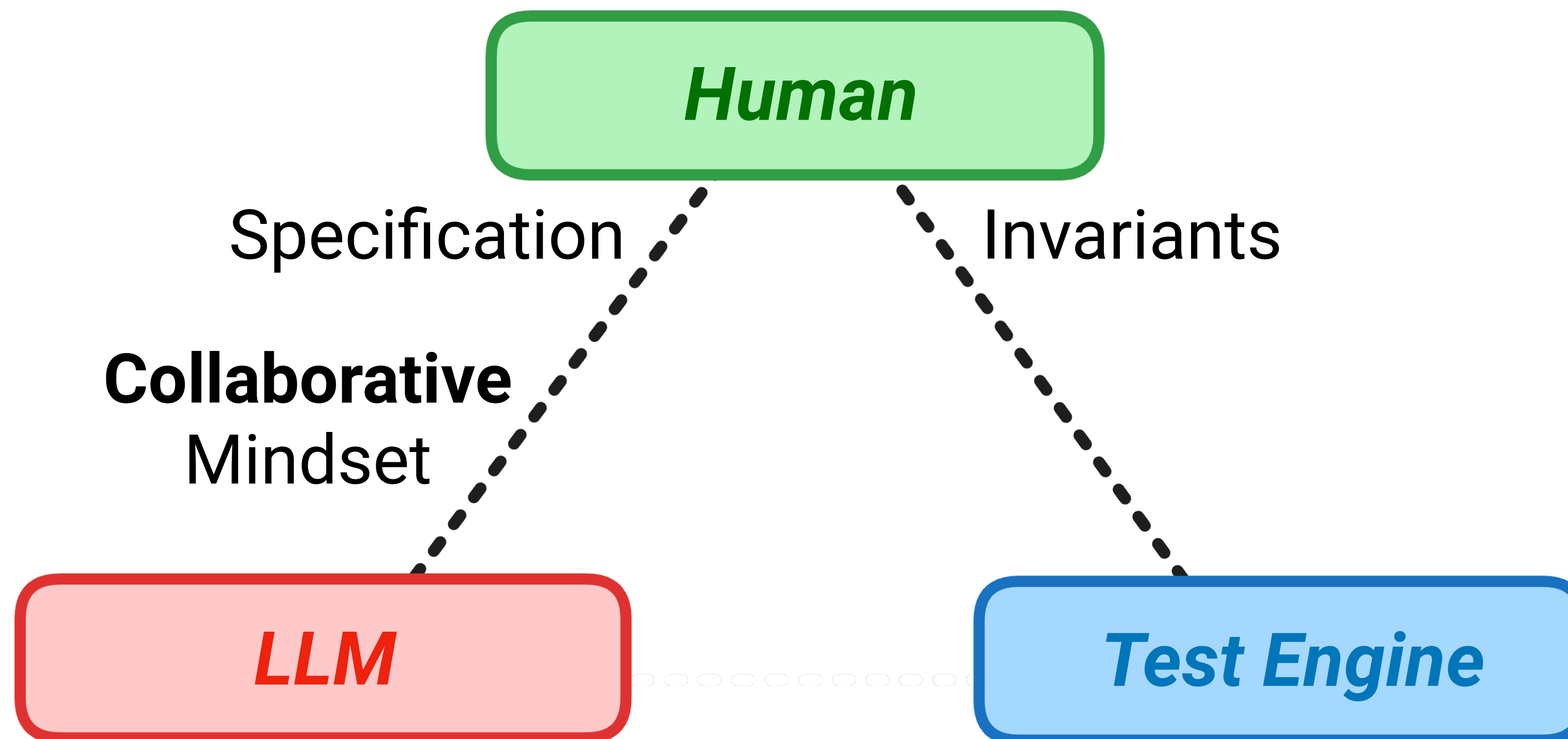
Division of Cognitive Labour



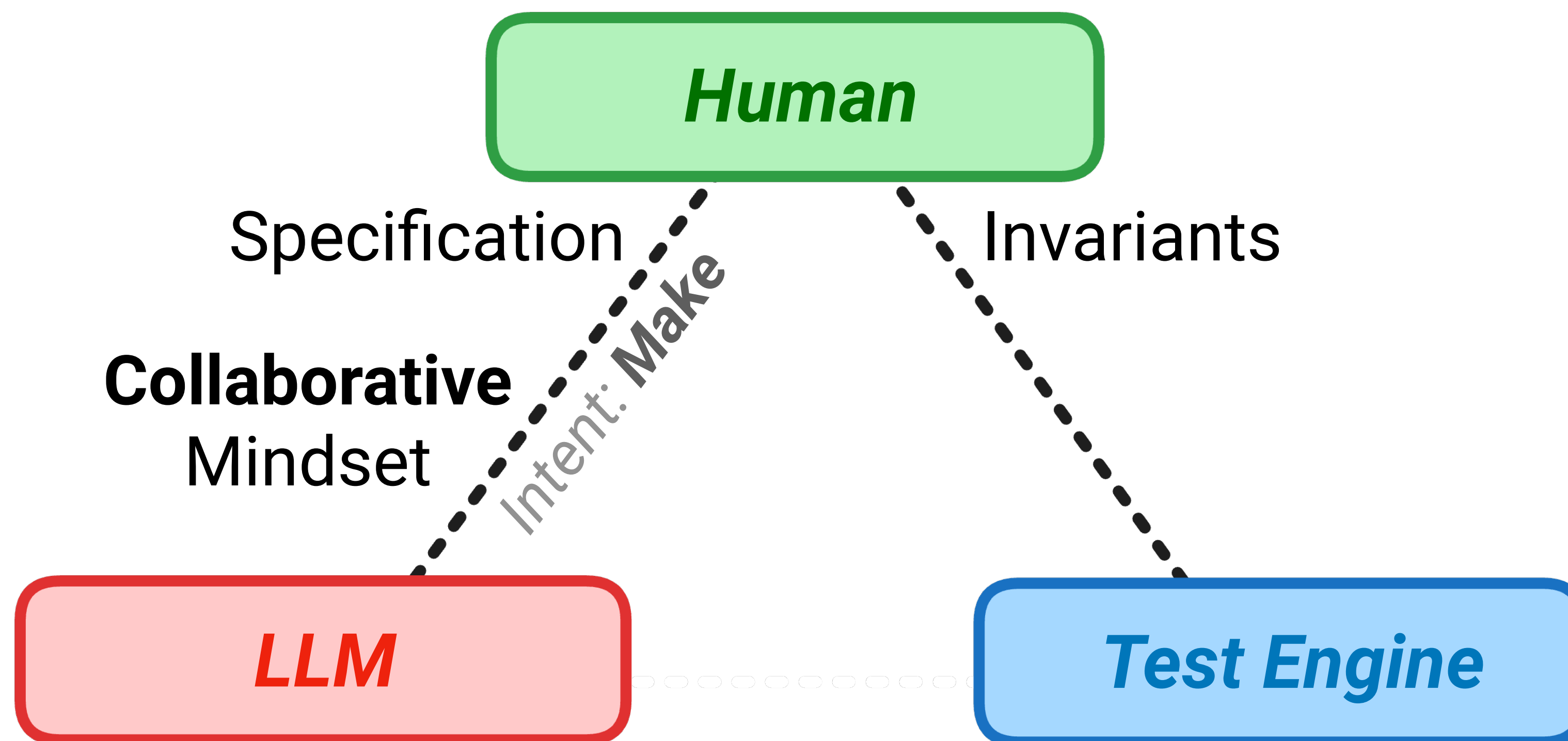
Division of Cognitive Labour



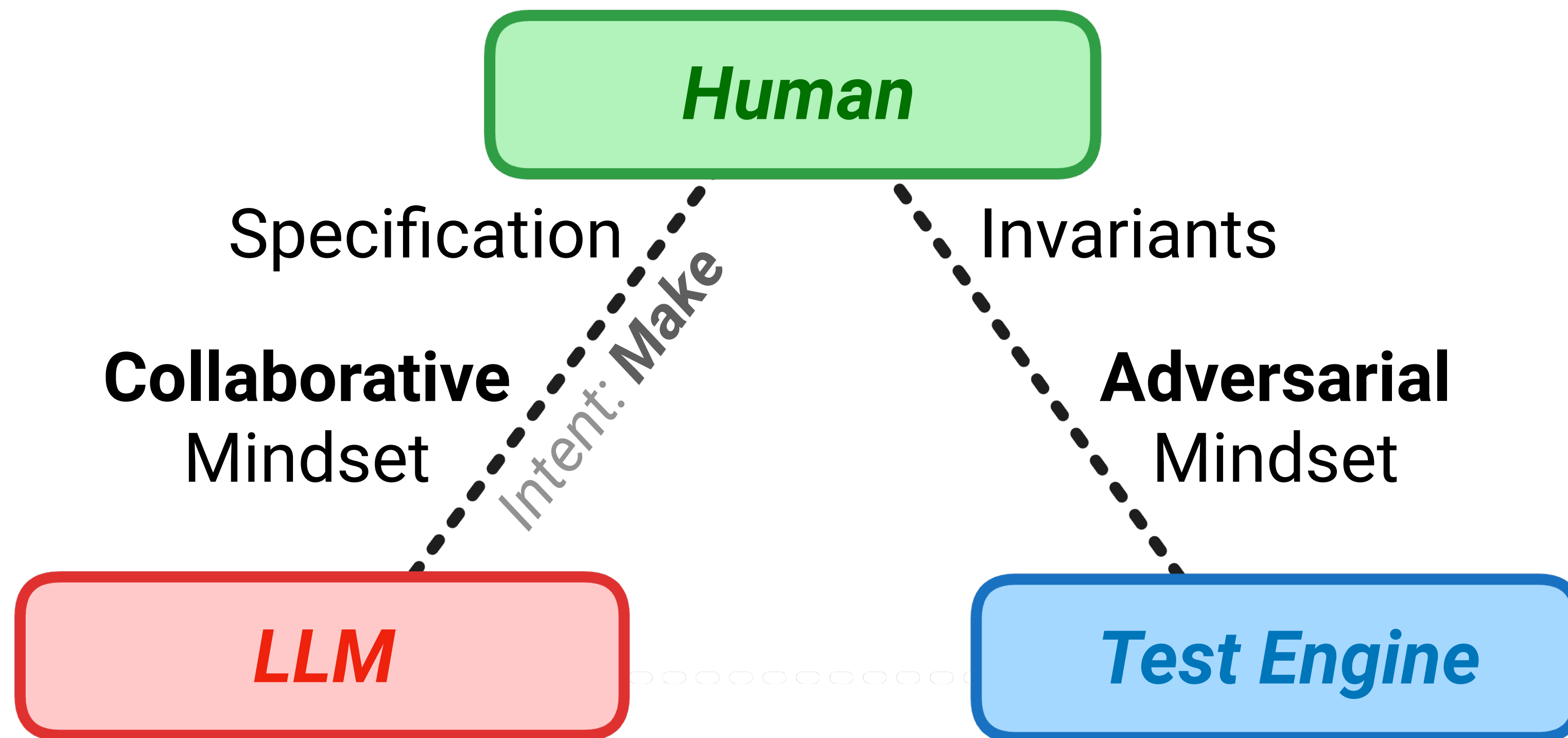
Division of Cognitive Labour



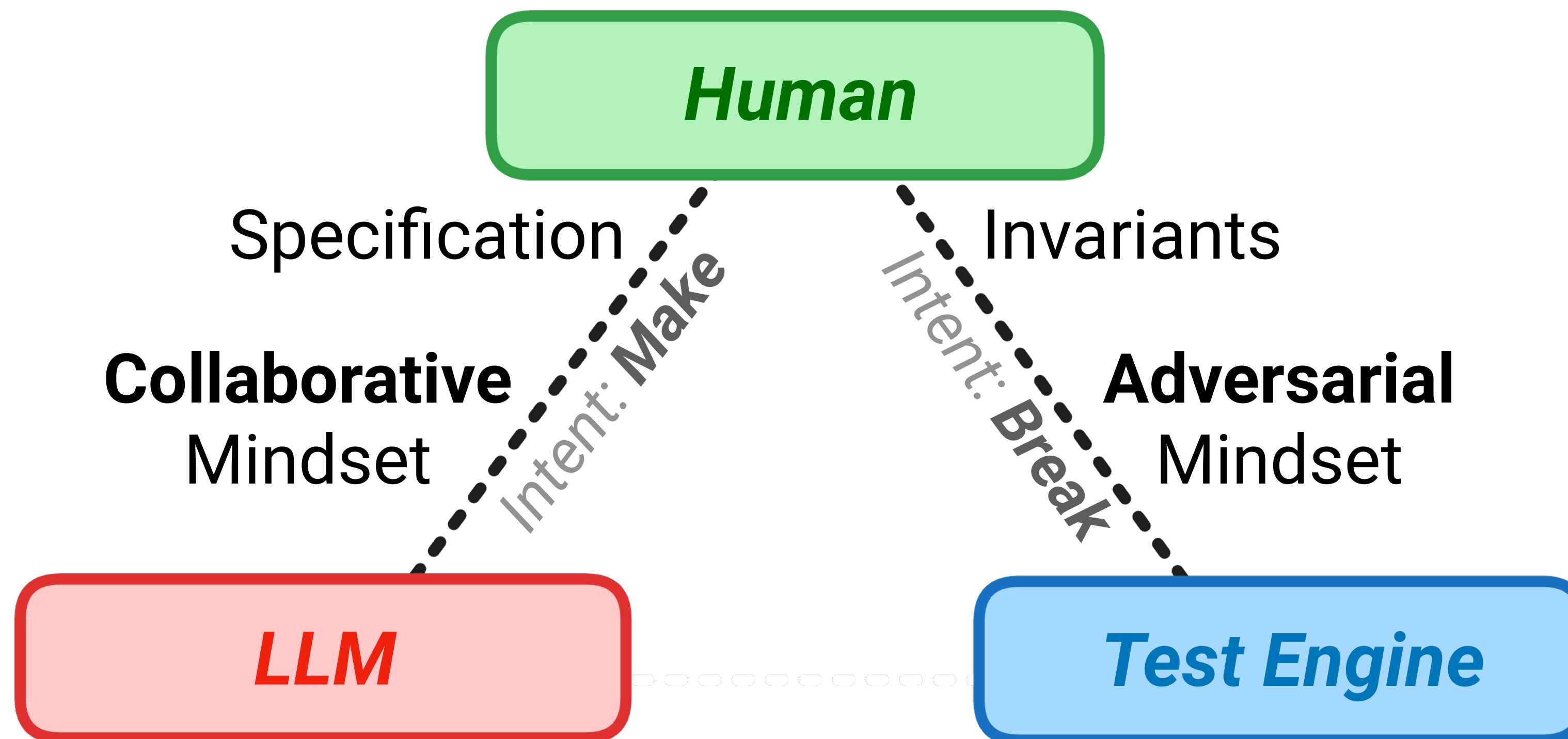
Division of Cognitive Labour



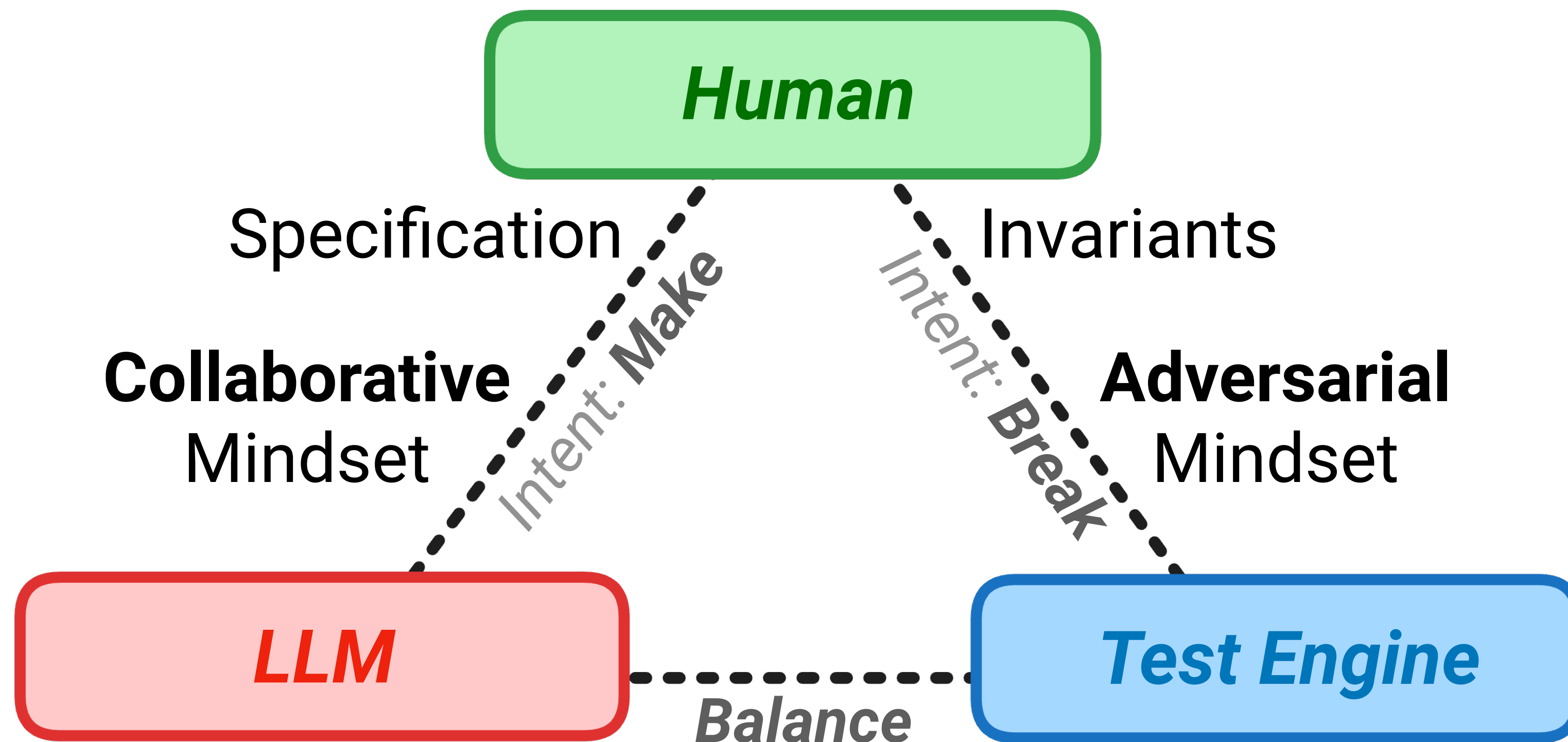
Division of Cognitive Labour



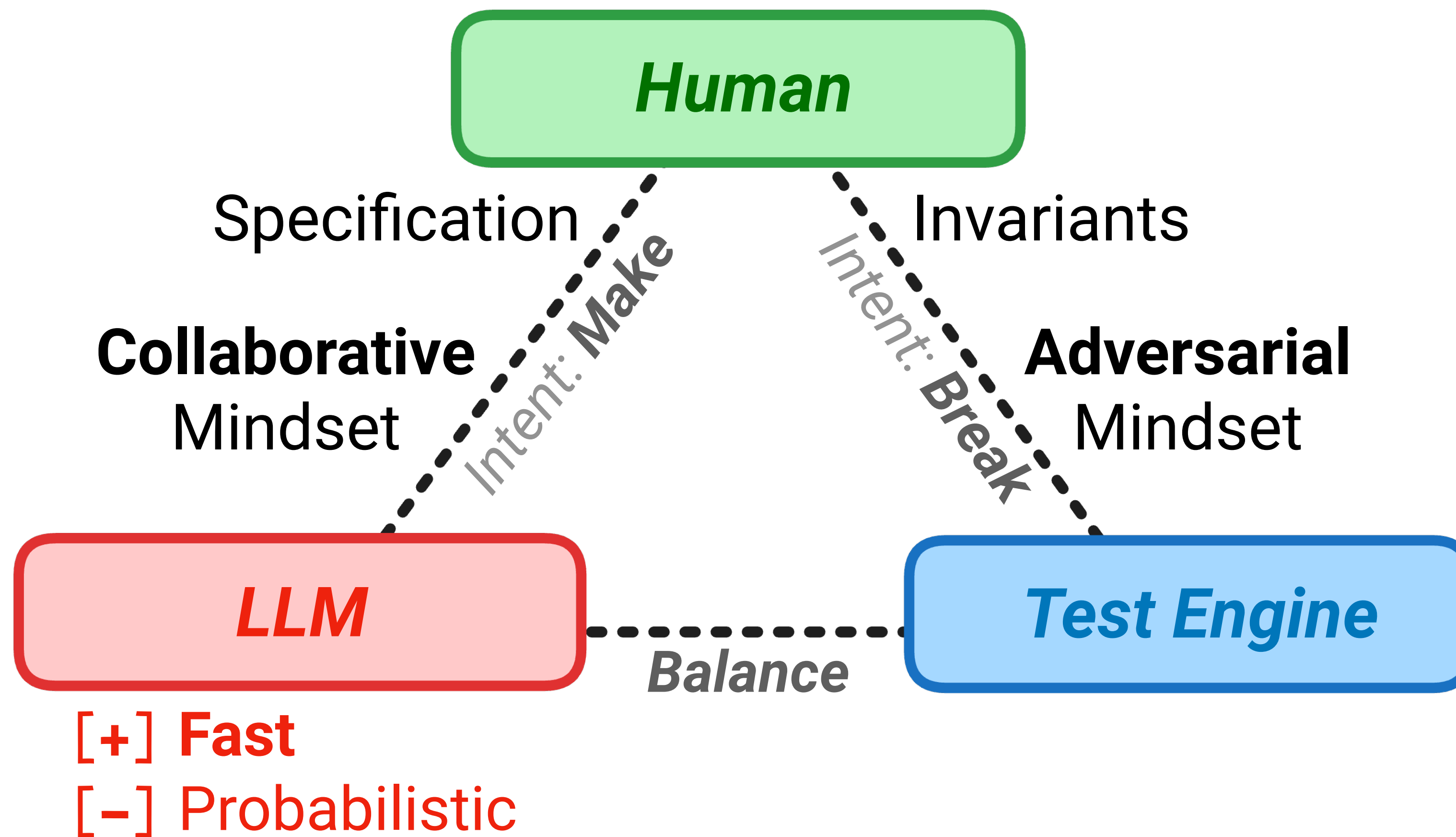
Division of Cognitive Labour



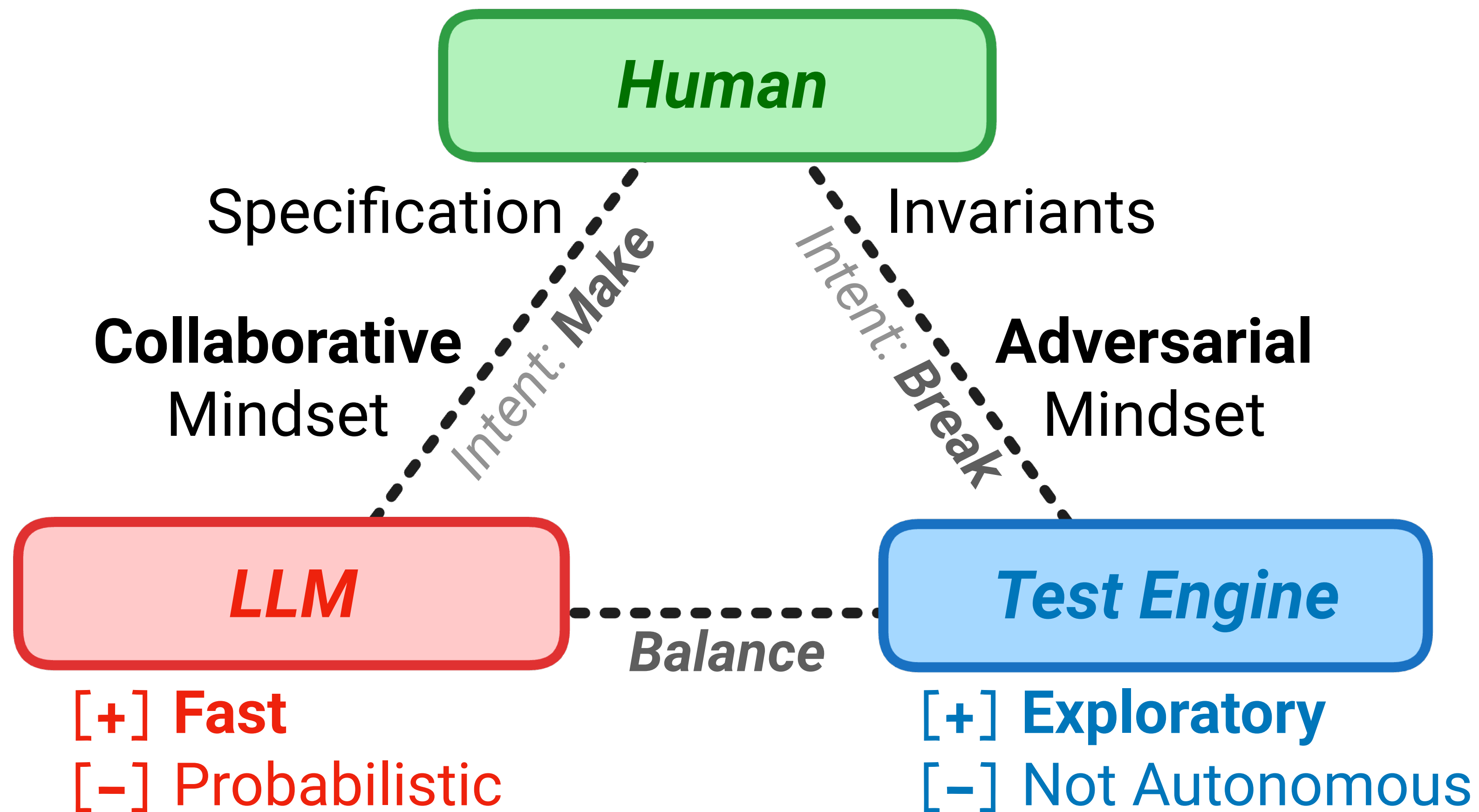
Division of Cognitive Labour



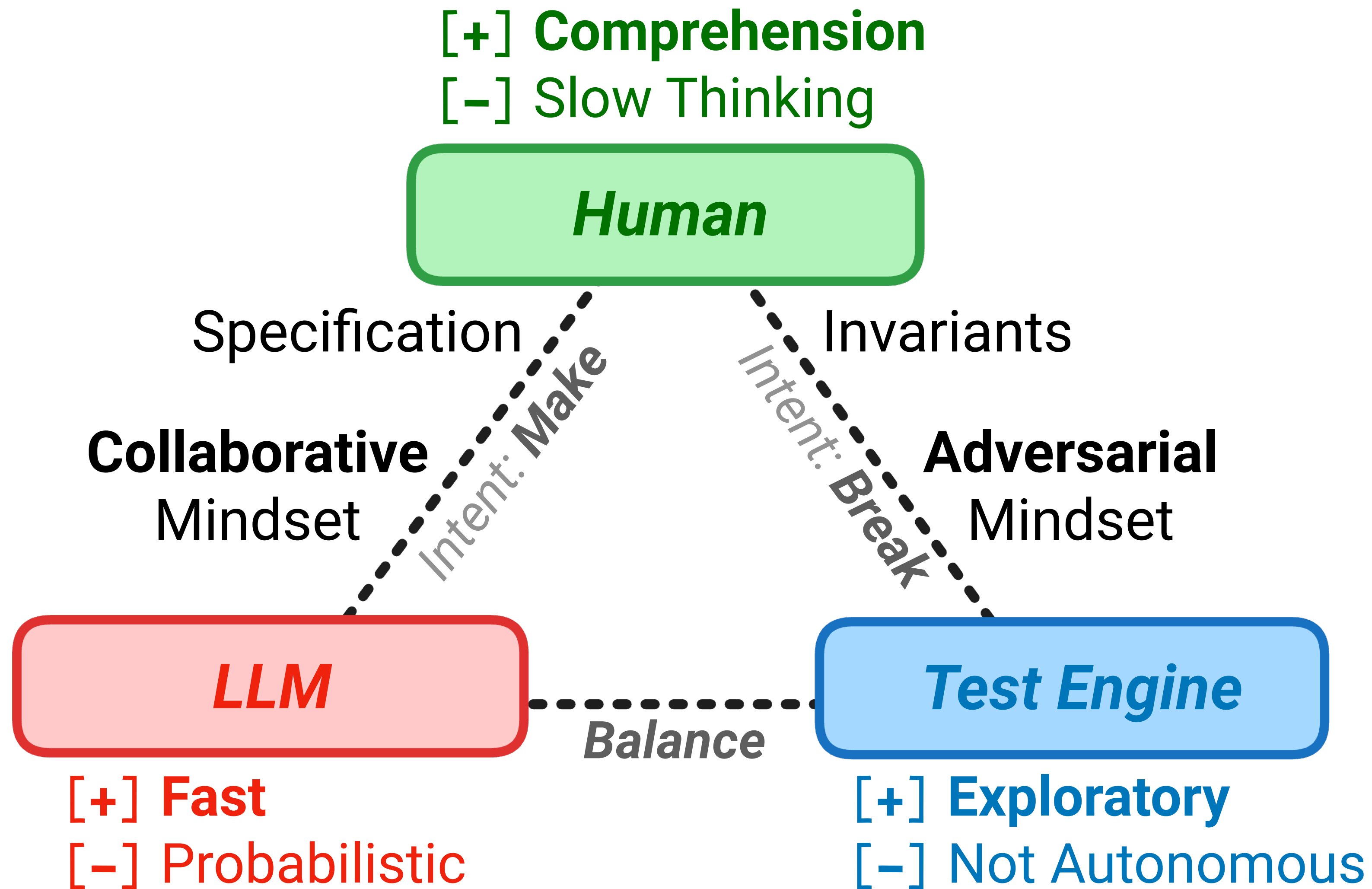
Division of Cognitive Labour



Division of Cognitive Labour



Division of Cognitive Labour



Thank You!

- mourjo.me
- github.com/mourjo/prompt-meetings
- mourjo.me/slides/2026-09-11-signals.pdf



linkedin.com/in/mourjo